

Can Community Health Workers Rebuild Trust? Evidence from India's NRHM and Emergency-Era Sterilization

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Abstract

India's 1975–77 Emergency subjected districts to a mass sterilization campaign of sharply varying intensity, and a persistent deficit in public health-care utilization followed in the most affected districts, consistent with durable distrust of the public health system. Three decades later, the National Rural Health Mission (NRHM) deployed nearly one million community health workers (ASHAs) as household-level intermediaries between communities and the formal health system. While NRHM was designed as a supply-side health system reform, its scope could have created unintended positive externalities, precisely along the dimension of (re)building trust. Using a difference-in-differences design that interacts district-level sterilization intensity with the national reform, and individual-level data from six survey rounds spanning 1998–2021 (approximately 2.1 million observations; 402 matched districts, 379 in the balanced sample), this paper asks whether the reform narrowed the deficit, and how fast. In the short run (two to three years) there is no detectable narrowing, and the design had the power to detect effects well below those that later emerged, so the null is informative. Over a decade to fifteen years the deficit closes: by 2019–21, a one-standard-deviation increase in sterilization intensity is associated with differential gains of 3.0 percentage points in DPT vaccination and 3.2 points in antenatal care, outcomes with clean pre-reform trends, and comparable gains in institutional and public-facility delivery. The convergence matches or exceeds the measured pre-reform deficit, worse-off districts shift differentially toward public providers, and the path of recovery implies a half-life of roughly eight to ten years. Direct trust measures from 2012–13 and the timing of recovery are consistent with trust rebuilt through sustained community-embedded intermediation, plausibly via ASHAs; a differential rise in voluntary sterilization adoption provides corroborating but not dispositive evidence. Historically rooted distrust, these results suggest, is reversible, but recovery may take a decade or more.

Keywords: institutional trust, community health workers, NRHM, ASHA, sterilization, India Emergency, difference-in-differences, persistence, family planning

JEL Codes: I12, I18, N35, O15, Z13

1. Introduction

In India, households systematically bypass government health providers in favour of private practitioners who are typically less qualified and more expensive (Das, Holla, Mohpal, and Muralidharan 2016; Das, Daniels, Ashok, Shim, and Muralidharan 2022). The gap in formal qualifications is stark: in the average rural village, 86% of primary-care providers are private and 68% have no formal medical training, yet these providers capture the majority of patient visits (Das et al. 2022). What draws patients is effort rather than credentials. Audit studies in which *standardised patients* present identical, scripted clinical cases find that private providers complete more of the diagnostic checklist and are at least as likely to recommend the correct treatment as their public counterparts, and among doctors who practise in both sectors, quality is uniformly higher in their private clinics (Das et al. 2016).¹ This pattern presents a puzzle: the private share of care is higher even in markets where qualified doctors offer the same services free of charge. In this paper, I explore two shocks that occurred during the last 50 years to investigate this puzzle further.

While supply-side explanations for it (weak provider incentives, understaffing, absenteeism) are well established, the demand side, why households continue to shun free public care, is less understood. One under-explored demand-side channel is historically rooted distrust. A growing literature documents that historical episodes of state coercion persistently damage citizens' trust in public institutions, with consequences for economic behaviour decades or even centuries later (Nunn and Wantchekon 2011; Lowes and Montero 2021; Alsan and Wanamaker 2018; Martinez-Bravo and Stegmann 2022). Several of these episodes operated specifically through health: forced colonial medical campaigns later depressed vaccination rates and trust in medicine (Lowes and Montero 2021), the public disclosure of the Tuskegee study reduced healthcare utilization and raised mortality among older Black men (Alsan and Wanamaker 2018), and a covert vaccination ruse lowered immunization rates in Pakistan (Martinez-Bravo and Stegmann 2022), outcomes closely related to those I study. A separate literature establishes the converse for the present day: who delivers care matters for whether it is taken up. Alsan, Garrick, and Graziani (2019) find that Black men in Oakland randomly assigned to a racially concordant physician take up substantially more preventive care, and Alsan and Eichmeyer (2024) show that a socially proximate non-expert outperforms a credentialed expert in raising vaccine demand, particularly among the least willing, precisely because credibility rather than expertise binds where trust is low. Yet neither literature speaks to the question of repair: the first documents persistence, the second identifies the effect of a trusted messenger at a point in time. Whether and how quickly trust that has been destroyed can be rebuilt remains an open question, one with direct implications for the design of public health systems in settings where historical trauma depresses demand for government services.

The Indian case offers a sharp instance of such a coercive shock: the forced sterilization campaign during the Emergency period (1975–77), which varied dramatically across districts. Pelras and

¹These studies measure provider quality using *standardised patients*: trained fieldworkers who present identical, scripted clinical cases to providers, holding patient case-mix fixed so that observed differences in diagnosis, treatment, and effort are attributable to the provider rather than to the patient.

Renk (2023) show that this campaign depressed demand for public health services, with children in higher-coercion areas less likely to be vaccinated or delivered in an institution; the distrust attaches not to the government in the abstract but to the specific clinics, sub-centres, and health workers through which rural citizens encounter the state. Their evidence, however, comes from a household survey fielded only a few years after the Emergency (the 1982 wave of the Rural Economic and Demographic Survey), so it is not clear to what extent this under-utilization extended beyond that early window. Nor is it clear whether such a deficit persists over the longer run, or how it evolves as health policy changes over the intervening decades.²

I ask whether the National Rural Health Mission (NRHM), launched in April 2005, succeeded in reducing this deficit in healthcare usage. The NRHM’s centrepiece was the Accredited Social Health Activist (ASHA): a female community health worker recruited from the village she serves, compensated through performance-based payments rather than a government salary, and tasked with facilitating institutional delivery, immunization, and antenatal care. I exploit the interaction of within-state variation in Emergency-era sterilization intensity with the temporal variation from the NRHM rollout in a difference-in-differences framework, using individual-level data from six survey rounds (DLHS-1 through DLHS-4 and NFHS-4/5) spanning 1998 to 2021.

Cross-sectional persistence regressions, which dominate the existing literature on historical distrust, suffer from a well-documented inference problem: spatial autocorrelation and unmodelled spatial trends inflate reported t -statistics in OLS specifications, making much of the published significance fragile.³ My difference-in-differences design mitigates the level component of this problem by exploiting within-unit temporal variation rather than purely cross-sectional spatial variation, though spatially correlated differential trends remain a concern, which I address through Conley spatial standard errors and state-level clustering. If sterilization intensity captures genuine health-specific distrust rather than spatially correlated development differences, then a trust-relevant intervention should differentially narrow the utilization gap in high-sterilization districts while leaving non-health outcomes unaffected. The conceptual framework is straightforward: trust is an inherited prior, transmitted across generations through cultural learning (Tabellini 2008; Bisin and Verdier 2001), that is updated through direct experience with institutions. Negative experiences (coercive sterilization) shift the prior downward; sustained positive interactions (an ASHA who is intrinsically and extrinsically motivated to facilitate primary caregiving) could gradually shift it back. The key empirical prediction is that trust recovery, if it occurs, should be slow: inherited priors are sticky, and updating requires repeated positive signals. The speed of recovery is itself an economically important parameter that the literature has not previously estimated. Appendix J formalizes this logic in a minimal signal-based updating model whose predictions match the shape, timing, and cross-outcome ordering of the estimated recovery paths: no jump at the reform date, recovery that saturates in cumulative contacts yet can accelerate in calendar time as contact rates rise, and earliest convergence

²Complementary evidence of a longer-lived deficit is provided by Sur (2022) and Singh and Vincent (2024); relative to the short-run health-utilization deficit examined here, both trace effects over a longer horizon, and Singh and Vincent (2024) in particular focus on non-health outcomes such as crime and intimate-partner violence rather than healthcare use.

³Conley and Kelly (2024) show that once spatial correlation is properly accounted for, the conventional standard errors and t -statistics reported throughout the historical-persistence literature are systematically overstated.

for the services the community health worker delivers in person.

The results confirm this prediction. In the short run (2005–08), the sterilization-outcome gradient is essentially unchanged: the DiD coefficients on the interaction of sterilization intensity with the post-NRHM indicator are small and statistically insignificant across ten health outcomes (six primary and four secondary maternal- and child-health measures, defined in Section 3.2). A within-survey mother fixed effects specification, which absorbs all time-invariant maternal characteristics including inherited distrust by comparing outcomes for children born to the same mother before and after the NRHM cutoff, confirms the null. In the medium run (by 2015–16), vaccination outcomes begin to converge. In the long run (by 2019–21), convergence becomes broad-based and highly significant: a one-standard-deviation increase in sterilization intensity is associated with 1.8 percentage-point greater gains in institutional delivery ($p = 0.008$), 2.6 pp in public facility delivery ($p = 0.006$), 3.2 pp in antenatal care ($p = 0.020$), and 3.0 pp in DPT vaccination ($p = 0.020$). These magnitudes are economically meaningful: measured against the pre-NRHM deficit itself, tabulated in Table 7, the NFHS-5 convergence amounts to full closure of the earliest measured (1998–99) deficit for institutional delivery, roughly 1.4 times it for antenatal care, and a differential shift toward public providers for facility delivery, an outcome whose pre-reform gradient was statistically indistinguishable from zero. By contrast, non-health outcomes (literacy, urbanization, employment) show no relationship with sterilization intensity in any period, and male health outcomes are unaffected, confirming that the convergence operates through the demand-side trust channel rather than through general development or supply-side improvements. A direct test using DLHS-4 (2012–13) trust measures corroborates: sterilization intensity no longer predicts stated distrust of government health facilities.

Two mechanism results sharpen the interpretation. First, unbundling the NRHM into its cash transfer component (the Janani Suraksha Yojana, JSY) and its community health worker component (ASHAs) shows that convergence is concentrated in the period after ASHA deployment reached operational scale, not in the initial years when only the JSY financial incentive was available. Second, high-sterilization districts experienced differentially larger increases in voluntary female sterilization adoption post-NRHM, suggesting that trust had recovered sufficiently for women to choose the very procedure that was coercively imposed during the Emergency, a pattern I argue reflects voluntary trust recovery rather than renewed coercive pressure (Section 6.2).

This paper makes two kinds of contribution. Broadly, it bears on how state coercion shapes long-run institutional trust, state capacity, and the demand for public services. Work on the historical persistence of distrust has shown that coercive episodes depress trust and behaviour for decades; this paper shifts the question from *whether* distrust persists to *whether, and how fast, it can be rebuilt*, a speed-of-recovery parameter the literature has not previously estimated. The closest antecedent is Christensen, Dube, Haushofer, Siddiqi, and Voors (2021), who show that community health-facility monitoring built resilient trust that sustained utilization through the 2014 Ebola crisis; I ask whether a comparable dynamic operates at national scale over two decades. Closer to this setting, prior work on the Emergency’s health legacy has documented the utilization deficit but stopped at persistence, high-coercion districts still under-use public

health services decades on; this paper is the first to show that a later reform actively reversed that deficit, and to identify the female community health worker (the ASHA) as the repair mechanism, distinct from the supply-side effort deficit emphasized in the public–private health-care literature (Das et al. 2016, 2022). In doing so it joins three literatures, on the persistence of institutional distrust, on the public–private gap in developing-country healthcare, and on the evaluation of the NRHM, by identifying demand-side trust recovery as a channel common to all three.

Section 2 provides historical background. Section 3 describes the data. Section 4 develops the empirical strategy. Section 5 reports results, robustness, and heterogeneity. Section 6 examines mechanisms and Section 7 concludes.

2. Historical Background

2.1 The Emergency and Forced Sterilization (1975–77)

On June 25, 1975, Indian Prime Minister Indira Gandhi declared a state of internal emergency, suspending civil liberties and concentrating power in the central government. Among the most consequential policies implemented during the Emergency was an aggressive population control programme centred on sterilization. While India’s family planning programme had existed since the 1950s and had included sterilization as a component, the Emergency period transformed what had been a nominally voluntary programme into a de facto coercive campaign⁴. This coercive turn had deep roots and a distinct ideological framing: Connelly (2006) traces the substantial influence of the international population-control movement on Indian family planning both before and during the Emergency, while Williams (2014) shows that the sterilization drive was justified as an instrument of poverty eradication.

The scale of the campaign was unprecedented. Sterilization performance targets were set at all levels of government, and administrators faced professional consequences for failing to meet quotas. Incentives and disincentives were deployed to encourage compliance: public employees who did not undergo sterilization or produce “motivated cases” faced salary deductions or loss of promotions; access to government services such as ration cards, housing plots, and irrigation water was conditioned on proof of sterilization; and in some areas, coercion extended to physical force (Shah Commission of Inquiry 1978; Gwatkin 1979; Singh and Vincent 2024). The campaign predominantly targeted men through vasectomy in its initial phase, but also included tubectomy (female sterilization). Between April 1976 and March 1977, over 8 million sterilizations were performed nationally, compared to approximately 2.7 million in the preceding year.

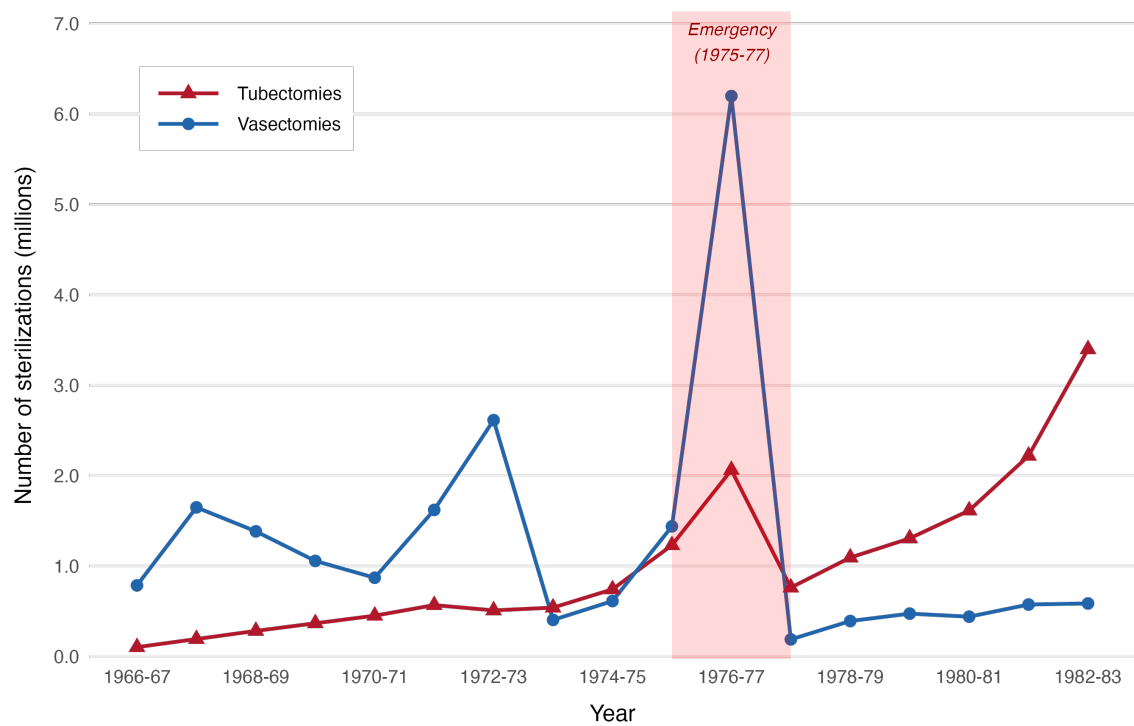
Figure 1 illustrates the scale and nature of the Emergency sterilization campaign using annual national data from the Ministry of Health and Family Welfare yearbooks. The vasectomy spike during 1976–77 is stark: vasectomies surged from approximately 1.4 million in 1975–76 to over 6.2 million during the Emergency year, before collapsing to under 200,000 in 1977–78 as the programme was dismantled. The figure also reveals a structural shift in India’s sterilization

⁴Refer to Singh and Vincent (2024) and Pelras and Renk (2023) for a detailed background.

programme: prior to the Emergency, vasectomies constituted the majority of sterilizations; after the backlash, tubectomies permanently replaced vasectomies as the dominant method, reflecting both men’s refusal to engage with the state’s family planning apparatus and a policy pivot toward female sterilization that persists to this day.

A crucial dimension of the campaign was its reliance on the existing community health infrastructure. Since the early 1960s, the government had trained extension workers, predominantly male paramedics serving catchment areas of 10,000–25,000 people, to conduct home visits and deliver targeted health interventions including family planning services (Bhatia 2014). During the Emergency, these extension workers were mobilised alongside government teachers, revenue officials, and other public-sector employees to identify and “motivate” sterilization cases (Gwatkin 1979). Female multipurpose health workers earmarked specifically for family welfare services became direct instruments of the coercive programme (Das Gupta 2005; Bhatia 2014). This implication of the community health workforce in the sterilization apparatus is central to understanding why distrust attached not merely to the government in the abstract but to the specific institutions, health workers, clinics, sub-centres, through which rural citizens interfaced with the state’s health system.

Crucially for identification, the intensity of the sterilization campaign varied dramatically across districts. This variation was driven by the zealousness and administrative capacity of local officials rather than by underlying demand for family planning services. Some districts experienced sterilization rates several times higher than the national average particularly in northern India, where Sanjay Gandhi (the Prime Minister’s son, who spearheaded the campaign) wielded the most influence while others, particularly in southern states with already-established voluntary family planning programs, were comparatively less affected.



Source: Ministry of Health and Family Welfare Yearbooks, digitized by the author.

Figure 1

Vasectomies and Tubectomies Performed in India, 1966–1983. The shaded region marks the Emergency period (1975–77). The dramatic spike in vasectomies during 1976–77 reflects the coercive sterilization campaign. After the Emergency, vasectomies collapsed and tubectomies permanently became the dominant method. Source: Ministry of Health and Family Welfare Yearbooks, digitized by [Pelras and Renk \(2023\)](#); figures plotting the same series appear in [Pelras and Renk \(2023\)](#) and [Singh and Vincent \(2024\)](#).

The Emergency ended in March 1977 when Indira Gandhi called general elections and was decisively defeated, largely due to public backlash against the sterilization campaign and other Emergency excesses, including roughly 8 million sterilizations performed in 1976–77 (nearly triple the previous year’s volume of roughly 2.7 million), 9,566 documented complaints of compulsion or use of force, 1,690 reported sterilization-related deaths, and 110,701 cases of forcible sterilization of non-eligible persons (Shah Commission of Inquiry 1978; Tarlo 2003; Jaffrelot and Anil 2021). The new Janata Party government immediately dismantled the coercive family planning apparatus. The political fallout was so severe that India’s family planning programme was renamed “Family Welfare” and sterilization targets were officially abolished. These were largely formal repudiations, however. A subsequent literature documents that performance targets persisted in practice long after the Emergency: notwithstanding the 1996 “Target-Free Approach,” state and district authorities continued to set de facto “Expected Levels of Achievement” and to press frontline health workers to meet them, with documented threats to salary and tenure (Basu 1981; Human Rights Watch 2012). This gap between the official abolition of targets and their de facto persistence is what later reconciles the apparent tension between the post-Emergency disavowal of sterilization targets and the sterilization-linked incentives still embedded in the ASHA compensation schedule, which I examine in Section 6.2.

2.2 The Anatomy of Coercion: Archival Evidence

The treatment variable used throughout this paper, the district-level surge in sterilizations during 1976–77, is a summary statistic for a campaign whose qualitative character is essential to the trust mechanism it is meant to capture. Contemporary records from the Shah Commission of Inquiry, the National Archives, and the contemporaneous press allow that character to be reconstructed, and four features stand out, each of which maps directly onto the empirical design.⁵

First, coercion was administratively manufactured through a target cascade that operated district by district. A national target (4.3 million sterilizations for 1976–77, against 7.5 million acceptors the prior year) was pushed down to states, which then *revised their own quotas sharply upward*: Uttar Pradesh raised its sterilization target from the 4 lakh assigned by the Centre to 15 lakh, and West Bengal from 3.25 to 10 lakh.⁶ States in turn allocated quotas to individual districts and even to government departments. Far from restraining this escalation, the Health Ministry’s communications “reveal the anxiety on the part of the Minister to keep up the tempo of sterilisation,” and the Commission found nothing in the files “to show that the Central Health Ministry tried to dissuade the State Governments from upward revision of the targets.” Because the binding quotas were set and pursued by district administrations, the coercion “dose” varied at the district level within states, precisely the variation the difference-in-differences design exploits.

⁵The district- and state-level family-planning records used here were collected by the author at the National Archives of India, with the assistance of Abhishek Paliwal and Ritesh Singh.

⁶Shah Commission of Inquiry, *Third and Final Report* (1978), ch. XXI; the figures and the quoted ministerial communications below are reproduced in that chapter. State and district target schedules are corroborated by the family-planning questionnaire files consulted at the National Archives of India.

Second, the instruments of the campaign made access to the state itself conditional on submission to its health apparatus. Alongside graduated cash and in-kind incentives,⁷ the state deployed *disincentives* that struck at the citizen–state relationship directly. In Varanasi, the District Magistrate ordered the treasury to withhold the salary of every government employee who failed to “motivate at least two people each for vasectomy or tubectomy operations”;⁸ elsewhere, ration cards, housing, loans, and irrigation water were conditioned on sterilization certificates, and Maharashtra went so far as to introduce a bill making sterilization compulsory and non-compliance a cognisable offence. The campaign was delivered through the ordinary machinery of public health, the sub-centres, primary health centres, and the predominantly male extension workers and female multipurpose workers who staffed them, so that the coercion was experienced not as an abstraction but at the very point of contact through which rural households would later be asked to seek antenatal care, institutional delivery, and immunization. This is the channel through which a sterilization shock becomes *health-specific* distrust. Contemporary accounts capture the mechanism directly: in Delhi, rumours that the school vaccination drive was a covert means of sterilizing children set off panic and refusal of routine immunization, and women began avoiding government hospitals to deliver for fear of being sterilized after childbirth (Bose and Dayal 2018).

Third, the human cost was real, was recorded, and was understood at the time to be damaging the programme. The government acknowledged in Parliament that sterilization-related deaths rose from 85 in 1974–75 and 219 in 1975–76 to 914 in 1976–77.⁹ The Shah Commission’s own chapter records that the fixation of high targets in a short time had “led to certain administrative steps resulting in undesirable and in some instances disastrous consequences,” and that “hostile reactions ha[d] come to notice in some [States], particularly U.P. and to a lesser extent in Bihar, West Bengal and Maharashtra.” The most notorious episode was the police action at Uttawar, a village of some 8,000 people in Gurgaon district, Haryana, “inhabited mostly by the Muslims belonging to the Meo community,” which was raided in November 1976 by a large police force that “carried away in trucks about 550 villagers” for interrogation, scores of whom were sterilized.¹⁰ The episode points to a community dimension of the campaign, visible also in the religion-wise acceptance statistics of the period, that the trust mechanism plausibly inherited.

Finally, the campaign was overwhelmingly a male, vasectomy-driven drive whose political backlash permanently re-gendered Indian family planning.¹¹ As the vasectomy share of sterilizations climbed from 54 to 75 percent over the course of 1976, it kindled fears of “emasculatation” that, in a patriarchal society, hardened into open resistance and compelled the state to deploy the police to enforce the drive (Prakash 2019). Female sterilizations were only a quarter of the

⁷Compensation was graded by parity (e.g. Rs. 150/100/70 for acceptors with two-or-fewer/three/four-or-more children), and was supplemented locally by in-kind awards and private-sector payments; see, e.g., “Saris, dhotis as incentives for sterilisation,” *Times of India*, 2 October 1976.

⁸“Sterilisation, or no pay,” *Times of India*, 8 March 1976.

⁹“Rs. 37 lakhs paid to sterilisation victims,” *Times of India*, 10 March 1978 (Lok Sabha replies of the Health Minister).

¹⁰Shah Commission of Inquiry (1978), ch. XVII.

¹¹The male character was pronounced but not uniform across regions: in parts of western and southern India women were already the majority of those sterilized even during the Emergency, and the national female share rose decisively only after 1977 (Scott 2018).

1976–77 total but jumped to four-fifths the following year and remained there; this reversal, as [Basu \(1985\)](#) argues, was “not so much the vagaries of consumer demand as the changing preferences of the official family planning programme itself,” which after the backlash could not promote vasectomy “without [further] destroy[ing] its credibility” and so turned to women, “which lacked the individual and group capability to protest.” The coercion of the Emergency was thus inflicted disproportionately on men, while the public-health system that had to win back trust became, and remains, organized around women’s reproductive health, the setting in which the female, community-recruited ASHA operates. The asymmetry between the gender of the original harm and the gender of the later repair is not incidental to this paper’s interpretation; it is, as Section 6.2 argues, central to it. One directional caveat applies from the outset: when Section 6.6 tests this asymmetry at the district level, recovery is slower, not faster, where the Emergency campaign was most heavily male, so the asymmetry shapes the interpretation of the repair without predicting that it accelerates it.

2.3 The Legacy: Distrust in Public Health

The Emergency-era sterilization campaign, particularly in high-intensity districts, has been linked with reduced utilization of public health facilities for services entirely unrelated to sterilization including childhood vaccination, antenatal care, and institutional delivery. The mechanism through which this underutilization propagates has been hypothesised as low demand rather than supply of health services, driven by a loss of trust. If this is indeed the case, it is likely a combination of direct experience (among older generations) and intergenerational cultural transmission of negative attitudes toward the state.

[Pelras and Renk \(2023\)](#) provide the most systematic evidence of this persistent utilization gap. This recent body of work is the first to document the link between Emergency-era sterilization and persistent health distrust systematically: although the Emergency’s coercion has long been examined by historians and political scientists, its measurable, district-level legacy for present-day health-seeking behaviour has been quantified only in the last few years. Using district-level variation in Emergency-era sterilization intensity and data from the National Family Health Surveys, they demonstrate that a doubling of sterilization intensity is associated with decreases of 1.4, 2.5, and 3.2 percentage points in DPT vaccination, any vaccination, and hospital births, respectively. These effects persist across multiple survey rounds spanning decades, consistent with the intergenerational transmission of distrust. [Singh and Vincent \(2024\)](#) provide complementary evidence on the long shadow of the Emergency on crime and intimate partner violence as a result of intergenerational trauma. [Sur \(2022\)](#) offers further corroboration, showing that Emergency-exposed districts exhibit higher private and lower public healthcare utilization consistent with eroded confidence in government providers. This paper builds directly on this body of evidence, but shifts the question from *whether* Emergency-era distrust persists to *whether a later policy reform can actively reverse it*.

This pattern connects to a broader literature on the persistence of institutional distrust following state coercion. [Nunn and Wantchekon \(2011\)](#) show that ethnic groups more heavily affected by the transatlantic and Indian Ocean slave trades exhibit lower levels of trust in rel-

atives, neighbors, and local government today. [Lowes and Montero \(2021\)](#) demonstrate that colonial-era medical campaigns in Central Africa which employed coercive methods including forced treatment and involuntary medical examinations reduced trust in modern medicine, with communities more exposed to historical campaigns showing lower vaccination rates and greater reliance on traditional healers. [Alsan and Wanamaker \(2018\)](#) find that the revelation of the Tuskegee syphilis experiment in 1972 reduced Black men’s interactions with physicians and led to increased mortality. [Martinez-Bravo and Stegmann \(2022\)](#) document an analogous mechanism in Pakistan, where the CIA’s covert use of a hepatitis B vaccination campaign to locate Osama bin Laden led to a 23–39% decline in childhood immunization after disclosure, a case in which the state’s misuse of health infrastructure for non-health purposes directly mirrors the Emergency’s co-optation of community health workers for sterilization. [Archibong and Annan \(2023\)](#) find a similar pattern following Pfizer’s controversial 1996 drug trial on children in Kano, Nigeria: disclosure of the deaths caused Muslim mothers to reduce routine vaccination of their children. Even non-coercive negative health events can erode trust: [Giulietti, Vlassopoulos, and Zenou \(2023\)](#) show that local vaccine-related deaths reduce take-up, while [Aassve, Alfani, Gandolfi, and Le Moglie \(2021\)](#) demonstrate that the 1918 Spanish Flu persistently reduced social trust among affected populations and their descendants. The positive counterpart also holds: [Calvi and Mantovanelli \(2018\)](#) show that Protestant medical missions in colonial India produced persistent improvements in health practices, suggesting that the sign and durability of trust effects depend on the nature of the initial health encounter. [Christensen, Dube, Haushofer, Siddiqi, and Voors \(2021\)](#) provide the closest precedent for the present paper: community-driven development programmes that included health facility monitoring in Sierra Leone built resilient trust that sustained health utilization even through the 2014 Ebola crisis, demonstrating that institutional trust can be actively rebuilt through community engagement. Across these settings, the common finding is that institutional misuse of health infrastructure creates durable deficits in trust that persist for decades through intergenerational cultural transmission ([Tabellini 2008](#)), though the evidence from [Christensen et al. \(2021\)](#) and [Calvi and Mantovanelli \(2018\)](#) suggests that positive institutional encounters can reverse these deficits.

2.4 Community Health Workers and the NRHM

India’s attempts to bring health services to rural communities through village-level workers have a long but troubled history. As early as 1940, the Sokhey Sub-Committee of the National Planning Committee recommended one health worker per 1,000 villagers for basic curative and preventive services; the Bhore Committee echoed this in 1943 ([Bhatia 2014](#)). Neither recommendation was implemented at scale. In the 1960s, the Chadha Committee established basic health workers for malaria eradication at a ratio of one per 10,000 population. In 1975, the Integrated Child Development Services (ICDS) scheme created the Anganwadi worker: a female village-level worker providing nutrition, preschool education, and maternal and child health services to children under six. The Anganwadi programme survives to this day, but its mandate was limited to child development rather than general primary health care ([Bhatia 2014](#)).

In 1974, the Srivastava Committee laid the groundwork for a dedicated cadre of community health workers for basic health services (Srivastava Committee 1975). The Committee’s vision was explicit: these workers should be neither remunerated nor supervised by the state, and should instead work with communities “on the basis of the trust and confidence they can generate” (Strodel and Perry 2019). When the Janata Party defeated Indira Gandhi in 1977, the new government, under intense pressure to regain the confidence of rural populations after the sterilization backlash launched the Village Health Guide (VHG) scheme on October 2, 1977, less than seven months after taking office. The scheme drew on the Srivastava Committee’s recommendations but deviated from them in ways that proved fatal.

The VHG scheme aimed to train one health guide per 1,000 population, and by 1987 approximately 400,000 VHGs had been trained across India (Strodel and Perry 2019). However, the programme suffered from three fundamental design flaws. First, 75–94% of VHGs selected were male, despite the Committee’s recommendation for female workers; the selection process was captured by political patronage rather than driven by community needs (Strodel and Perry 2019). Second, VHGs received a government stipend of Rs. 50 per month, though the Srivastava Committee had explicitly warned against remuneration, which caused communities, health personnel, and the VHGs themselves to regard them as government employees rather than community advocates (Strodel and Perry 2019). This identification with the state was particularly damaging in communities where the state had just perpetrated coercive sterilizations through its health apparatus. Third, training was perfunctory: three months at local Primary Health Centres where trainers were often unaware of the programme’s goals, centres were overcrowded, and only half of trainees received training manuals (Strodel and Perry 2019). Evaluations revealed that VHGs could correctly answer only 20–30% of basic health questions, and communities reported that government-provided primary healthcare “lacked responsive and caring health care workers” (Strodel and Perry 2019). The government formally terminated financial support for the VHG scheme in April 2002, though it had effectively collapsed by the mid-1980s.

For the next two decades, India’s rural communities were served by the same overstretched workforce: Auxiliary Nurse-Midwives (ANMs) nominally covering 5,000 people but in practice often serving 20,000 or more, with 22% of sub-centres lacking an ANM entirely (Rao, Bhatnagar, and Berman 2012). In 30% of districts, sub-centres served double their intended population. The doctor-to-population ratio in rural India stood at 0.03 per 1,000 (Bajpai, Sachs, and Dholakia 2010). State-level CHW programmes emerged during the 1990s almost all using female workers, but no national scheme replaced the VHGs until 2005 (Bhatia 2014).

Launched in April 2005, the National Rural Health Mission (NRHM) represented India’s most ambitious health system strengthening effort. The mission was motivated by India’s persistently poor maternal and child health indicators, particularly in rural areas and the “Empowered Action Group” (EAG) states of northern India (Uttar Pradesh, Bihar, Madhya Pradesh, Rajasthan, Odisha, Chhattisgarh, Jharkhand, and Uttarakhand, along with Assam and Jammu & Kashmir). The NRHM’s objectives were explicitly supply-side: to improve access to and quality of public health services through infrastructure upgrades, human resource expansion, and demand-side financing.

The NRHM’s most innovative component was the creation of Accredited Social Health Activists (ASHAs). ASHAs are village-level community health workers, one per village or per approximately 1,000 rural inhabitants. Crucially, ASHAs are recruited from the communities they serve: they must be female residents of the village, aged 25–45, with at least eighth-grade education (relaxed in some areas), and are selected through a community process involving the *gram sabha* (village assembly), self-help groups, the village health and sanitation committee, and block- and district-level nodal officers. This local recruitment creates a personal, trust-based relationship between the ASHA and the households she serves, a feature consistent with the broader evidence on community-based development (Mansuri and Rao 2004) and one that distinguishes the ASHA model from the male, externally-appointed, stipended Village Health Guides described above, and from any cadre of health worker imposed on the community by the state.

The ASHA’s mandated activities span the full maternal-and-child-health continuum. She identifies and registers pregnant women in her area; arranges and accompanies the schedule of antenatal check-ups (including tetanus immunisation and iron-folic acid supplementation); counsels women on safe pregnancy practices and the case for institutional delivery; *escorts* the woman to the facility for delivery and remains with her until discharge; ensures the newborn is immunised (the vaccines themselves are administered by the Auxiliary Nurse-Midwife); and conducts six postnatal home visits (days 3, 7, 14, 21, 28, and 42) in the case of institutional deliveries. She additionally reports births and deaths to the local health system, mobilises children for childhood-immunisation drives, distributes basic medicines (oral rehydration salts, iron-folic acid tablets, condoms, oral contraceptive pills), and serves as the village-level interface for national programmes such as JSY, the Janani Shishu Suraksha Karyakaram (JSSK), and Mission Indradhanush.

Unlike the salaried Auxiliary Nurse-Midwives or the earlier stipended VHGs, ASHAs receive no fixed wage. They are compensated through a piece-rate schedule of *performance-based incentives* (Gopalan and Varatharajan 2012) defined in the NRHM Framework for Implementation and subsequently revised by Ministry of Health and Family Welfare circulars. The largest single item is the JSY institutional-delivery package, approximately Rs. 600 in rural Low-Performing States (Rs. 300 for the antenatal-care component plus Rs. 300 for the delivery itself), followed by smaller per-case payments for full immunisation (around Rs. 100), IUD or postpartum-IUD insertions (Rs. 150), and motivating sterilisation acceptors under the Family Planning Compensation Scheme (Rs. 150 per tubectomy and Rs. 200 per vasectomy; raised in the 146 high-fertility districts covered by Mission Parivar Vikas from 2016 onward). Incentives for pills and condoms are essentially negligible, while the IUD incentive (Rs. 150) matches the tubectomy rate, so the schedule’s tilt toward permanent methods holds against the short-acting spacing methods rather than against the IUD. The schedule therefore embeds a non-trivial financial pull toward permanent contraception relative to pills and condoms; in practice this pull operates overwhelmingly through female sterilisation, which accounts for the vast majority of sterilisation uptake, even though the per-case rate is nominally higher for vasectomy. This structural feature is relevant to the family-planning mechanism analysed in Section 6.2.

The ASHA programme operates alongside the Janani Suraksha Yojana (JSY), a conditional cash transfer that provides financial incentives directly to women for delivering at a health facility. Together, these programmes create both supply-side (ASHA facilitation and information) and demand-side (JSY cash transfers) incentives for institutional health-seeking.

The JSY’s incentive structure varies by state. In the ten Low-Performing States (the EAG states plus Assam and Jammu & Kashmir), all women delivering in a public facility are eligible regardless of income or caste; in the High-Performing States, eligibility is restricted to women aged 19 and above who are below the poverty line or from Scheduled Caste or Scheduled Tribe households, and only for their first two births (Debnath 2021; Powell-Jackson, Mazumdar, and Mills 2015). Cash payments to mothers range from approximately Rs. 1,400 (rural) and Rs. 1,000 (urban) in Low-Performing States to Rs. 700 and Rs. 600 in High-Performing States. The ASHA is the operational link in this scheme: she identifies and registers the pregnant woman, arranges her antenatal care, escorts her to the facility for delivery, and brings the newborn for immunisation, herself receiving a performance payment (approximately Rs. 600 in rural Low-Performing States) for each institutional delivery she facilitates (Debnath 2021; Andrew and Vera-Hernández 2024). The JSY became the most-evaluated component of the NRHM and is credited with a large national shift away from home delivery (Lim et al. 2010; Powell-Jackson, Mazumdar, and Mills 2015). Over the same period the public sector’s share of maternal care rose sharply while out-of-pocket and catastrophic health spending in public facilities fell, even as maternal care in private facilities remained several times more costly (Mohanty and Kastor 2017). Its rule-based eligibility variation provides the identification for the analysis in Section 4.3.

3. Data

3.1 Treatment: Sterilization Intensity

The treatment variable measures district-level exposure to the Emergency-era sterilization campaign. I use data on district-level sterilization counts from 1969–70 through 1979–80, mapped to contemporary district boundaries.¹² The treatment variable is defined as the proportional surge in sterilizations during the Emergency year:

$$c_{int,d} = \frac{S_{d,76-77} - S_{d,75-76}}{S_{d,75-76}} \quad (1)$$

where $S_{d,t}$ denotes the sterilization rate, measured as the share of eligible couples sterilized, in district d during year t . This proportional surge in sterilizations is the *coercion intensity* index, hence the subscript *int* on $c_{int,d}$, and captures the excess sterilization intensity attributable to the Emergency: districts where the campaign was most aggressively implemented show the

¹²The series was originally compiled by Gwatkin (1979). This extensive sterilization dataset was collated through joint efforts with fellow researchers Charlotte Pelras, Andréa Renk, Aditi Singh, and Sarah Vincent; the dataset, alongside the one from Jolly (1986), was digitized by Aditi Singh, and the annual yearbooks of the Ministry of Health and Family Welfare were digitized by Pelras and Renk (2023). I am grateful for their efforts and scholarship.

largest proportional surge relative to their 1975–76 baseline. I retain the 1975–76 base in the headline measure for comparability with the literature built on the same digitized records. Because Indian fiscal year 1975–76 runs from April 1975 to March 1976, however, it contains roughly nine months of Emergency-era mobilization and is not strictly pre-Emergency. I therefore also construct an alternative index that is identical in form but uses the fully pre-Emergency 1974–75 rate as the base year. The alternative has a rank correlation of 0.74 with the headline measure and, unlike it, is nearly uncorrelated with its own denominator; Section 5.7.3 uses it as an internal-validity check on the long-run results, which prove as strong or stronger on the cleaner base.¹³ The case for the proportional surge over target-based alternatives is *ex ante*, grounded in what the trust mechanism requires rather than in empirical performance. The mechanism concerns what households experienced: a sudden, time-compressed escalation of sterilization activity in their district, which is what the surge measures. Constructions built from over-achievement relative to centrally assigned targets instead measure administrative performance against plans that households never observed, an object relevant to bureaucratic compliance but not to the lived shock that plausibly destroyed trust. The archival record of district-level implementation in Section 2.2 documents precisely this escalation logic, and Section 6.6 revisits the measurement question against archival alternatives; validating the surge district-by-district against surviving target files remains a task for future archival work.

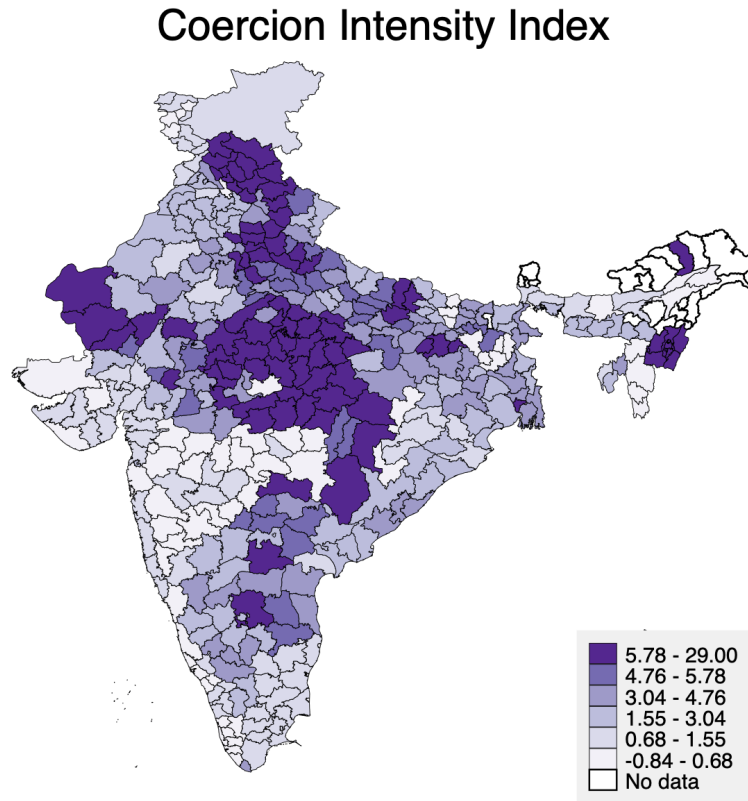
The proportional measure is mechanically sensitive to low denominators: a district with a near-zero 1975–76 sterilization base produces an extreme or undefined ratio. Districts whose ratio exceeds 100 are therefore capped at 29, just above the maximum value observed elsewhere in the distribution (28.7). Six districts are affected: five districts of Nagaland, whose recorded 1975–76 base is exactly zero and none of which enters the balanced estimation sample, and Lahaul and Spiti in Himachal Pradesh, the single capped district that does.¹⁴ The mean sterilization intensity across the 379 districts in the balanced treatment sample is approximately 4.5, with substantial variation (standard deviation approximately 5.3).

Figure 2 maps the coercion intensity index across Indian districts. The geographic pattern is striking: the highest-intensity districts are concentrated in the northern Hindi-speaking belt, particularly in Uttar Pradesh, Madhya Pradesh, Rajasthan, and Haryana, where Sanjay Gandhi’s political influence was strongest. Southern states, which had established voluntary family planning programs prior to the Emergency, generally exhibit lower coercion intensity. This north-south gradient is a recurring feature of India’s Emergency experience and motivates the inclusion of state-level fixed effects and EAG-state heterogeneity analyses in the empirical

¹³The measure is the district-level coercion-intensity index of Singh and Vincent (2024), defined as the growth rate of the share of sterilized couples between 1975–76 and 1976–77, and is constructed from the shared digitized dataset described above. Pelras and Renk (2023), who use the same underlying data, instead build their preferred measure as a state-level over-achievement of the centrally assigned sterilization targets; that alternative construction is examined in Section 5.7.

¹⁴Results are robust to the log transformation $\log(1 + c_{int})$ (Appendix Table A9, Panel A). They are also not driven by the capped district itself, whose assigned value of 29 sits roughly 4 standard deviations above the treatment mean: dropping Lahaul and Spiti entirely (equivalent to coding it as missing) leaves the NFHS-5 event-study coefficients essentially unchanged (institutional delivery 1.82 to 1.78 percentage points per standard deviation, public-facility delivery 2.64 to 2.62, antenatal care 3.21 to 3.19, DPT unchanged at 2.96), with no change in statistical significance; nor does the capped value distort the standardization scale, since excluding it changes the treatment standard deviation by only 2% (5.89 to 5.77).

strategy.



Source: Jolly, KG, Family Planning in India, 1969-1984: A District Level Study

Figure 2

Coercion Intensity Index (c_{int}) Across Indian Districts. The map displays the proportional increase in sterilizations during 1976–77 relative to 1975–76 at the district level, using quantile breaks. Darker shading indicates higher coercion intensity. The geographic concentration in northern India reflects the political geography of the Emergency campaign. Source: [Jolly \(1986\)](#), digitized by Aditi Singh.

The primary treatment variable in all DID specifications is c_{int}^{std} , the z-score of c_{int} (mean zero, unit standard deviation), so that coefficients are interpretable as the effect of a one-standard-deviation increase in sterilization intensity. A range of alternative treatment constructions, the raw unstandardised c_{int} , a binary above-median indicator, the log transformation, the raw 1976–77 sterilization level, the absolute excess, quintile dummies, the state-level over-target measure, and two archival target-based measures, is examined separately in the robustness analysis (Section 5.7).¹⁵

3.2 Outcome Data

The analysis draws on six rounds of nationally representative household surveys:

¹⁵The two archival measures are constructed from original Ministry of Health and Family Welfare records: a *target revision ratio* (revised over original 1976–77 target; 41 districts) and a *target overshoot* (achievement over revised target; 36 districts). The over-target construction corresponds to the preferred coercion measure of [Pelras and Renk \(2023\)](#). States are additionally classified by whether they offered enhanced positive incentives (payments to motivators, institutional prizes, or additional payments to acceptors beyond the central baseline) versus relying primarily on coercive implementation, based on the Family Planning yearbooks.

Short-run DiD panel: The primary analysis compares DLHS-2 (2002–04, pre-NRHM baseline) with DLHS-3 (2007–08, post-NRHM). Both rounds survey ever-married women and collect detailed information on antenatal care, institutional delivery, and childhood immunization. Variable construction follows the established DLHS-based programme-evaluation literature exactly, ensuring comparability.¹⁶ The balanced treatment sample includes approximately 257,000 delivery-outcome observations across 379 districts observed in both rounds (the full sample totals approximately 730,000 observations, with variable-specific N depending on module availability).

Long-run event-study panel: To trace the trajectory of trust recovery over a longer horizon, I construct a long-run panel incorporating four additional survey rounds: DLHS-1 (1998–99, earliest pre-NRHM baseline), DLHS-4 (2012–13, late post-NRHM), NFHS-4 (2015–16, late post-NRHM), and NFHS-5 (2019–21, latest post-NRHM). This six-round panel, spanning over two decades, enables an event study analysis with DLHS-2 as the reference period and a pre-trend check using DLHS-1. The long-run panel combines surveys with different sampling designs and question wording. To mitigate concerns that the convergence pattern is driven by cross-survey measurement differences, the event study uses round-specific coefficients rather than pooled pre/post comparisons, and the convergence pattern is visible within survey families: the NFHS-4 to NFHS-5 progression alone shows strengthening positive coefficients for institutional delivery, public delivery, and ANC. The long-run panel contains approximately 2.1 million observations; 402 distinct districts appear in at least one round, with per-round matched counts ranging from 272 (DLHS-1) to 396 (DLHS-3) and 276–279 in the NFHS rounds, and 379 districts forming the balanced DLHS-2/3 estimation sample (per-round counts are reported in Table 2).

Trust outcomes (DLHS-4 only). DLHS-4 uniquely contains direct measures of trust in government health facilities, ASHA engagement, and government scheme participation, enabling a direct cross-sectional test of persistent distrust seven years after NRHM. The direct-trust measures are drawn from the DLHS-4 household questionnaire (items Q49–Q52), which records where household members usually seek treatment and, when they avoid government facilities, why, with explicit response options for “poor quality of care” and “no trust in government facility”, alongside health-scheme and RSBY coverage. Because these items are answered by the household respondent rather than the sampled woman, they capture household-level rather than strictly individual attitudes. The ASHA-engagement and scheme-participation measures (pregnancy registration, ASHA-facilitated antenatal care and delivery, JSY receipt) are instead taken from the ever-married-woman questionnaire and are therefore reported by the woman herself. Exact question wording is reproduced in Appendix E.

Family planning outcomes. To test mechanisms related to family planning methods, I extract family planning variables from both DLHS-2 (W-prefix variables: W403 for current FP use, W404 for method type) and DLHS-3 (V-module: V405 for current FP use, V406 for method type). I classify methods into spacing (IUD, pills, condoms, injectables), limiting (female sterilization, male sterilization), and traditional (rhythm, withdrawal).

All outcome variables are binary indicators coded 0/1; coefficients in tables are therefore in proportion units (e.g., 0.018 corresponds to 1.8 percentage points). The primary outcomes

¹⁶Variable definitions follow [Debnath \(2021\)](#).

are institutional delivery, any vaccination (BCG, DPT, Polio, or Measles), DPT vaccination,¹⁷ BCG, measles, and full immunization.¹⁸ Secondary outcomes include: any antenatal care, iron-folic acid supplementation, tetanus injection during pregnancy, and public facility delivery.

3.3 District Harmonization

A significant data challenge is that district boundaries change between survey rounds due to administrative reorganization (districts are split or merged). I construct a master crosswalk that harmonizes district identifiers across all six survey rounds and maps them to the historical sterilization data. The crosswalk combines exact matching on district codes with fuzzy string matching on district names, supplemented by manual verification of ambiguous cases. When districts have been split between survey rounds, I assign the parent district’s sterilization intensity to all daughter districts. The magnitude of this challenge is non-trivial: of the district identifiers coded in each survey, the share that could be matched to the 1976–77 sterilization baseline and retained with non-missing c_{int} is roughly two-thirds in the DLHS rounds (381 of 593 in DLHS-2; 396 of 601 in DLHS-3) and about 40% in the NFHS rounds (276–279 of 640–707), with the master crosswalk resolving to 409 usable districts; the unmatched residual is disproportionately composed of newly created districts in the smaller north-eastern states. Per-round counts at each harmonization step are reported in Appendix Table A1. Because c_{int} is fixed at the 1976–77 district boundary, it cannot itself respond to post-1976–77 administrative reorganization, so the harmonization induces no anticipatory relationship between boundary changes and treatment. The residual measurement error from assigning parent intensities to daughter districts is of the Berkson type rather than classical: the assigned parent value is less variable than the unobserved daughter-specific intensity, and the daughter-level deviations are uncorrelated with the assigned value by construction. Berkson error leaves the point estimate approximately unbiased but reduces the precision with which β is estimated. The assignment has an inference counterpart: daughter districts carrying a common parent value do not contribute independent treatment variation. Grouping districts that share an identical intensity value within the same state, which identifies daughters that inherited a common parent value, the 409 usable districts embody 353 distinct parent-boundary treatment values, and the 379 balanced-sample districts embody 335 (of which 303 are single-district groups). Re-estimating the long-run event study with standard errors clustered on these parent groups rather than on districts leaves inference essentially unchanged: the largest movements among the NFHS-5 coefficients are for DPT vaccination ($p = 0.020$ to $p = 0.025$) and any vaccination ($p = 0.056$ to $p = 0.080$), with the remaining outcomes unaffected.

¹⁷Raw DPT coverage in NFHS-5 (2019–21) appears artificially depressed relative to NFHS-4 because of a change in the survey’s vaccination-recall window: NFHS-4 collected child vaccination for all children under five, whereas NFHS-5 collected it only for children under three. The variable is constructed over children under five, so the roughly two-fifths of NFHS-5 children aged three or four, for whom vaccination was never recorded, enter as unvaccinated and deflate the unconditional mean to about 0.50. Restricting to the comparable vaccination-eligible (under-three) window, DPT coverage is stable to slightly rising across the two rounds (0.814 in NFHS-4 to 0.869 in NFHS-5, the weighted means reported in Table 2), so the apparent decline is an artifact of the denominator rather than a real change. The DPT outcome used throughout, including in Table 2, is constructed on this corrected under-three denominator; the resulting NFHS-5 event-study coefficient is 3.0 pp per standard deviation, slightly larger than on the uncorrected denominator (2.3 pp).

¹⁸The choice of primary outcomes follows [Pelras and Renk \(2023\)](#) and [Debnath \(2021\)](#).

A related concern is that the roughly three-fifths of NFHS district identifiers that cannot be matched to the 1976–77 baseline (and roughly one-third of DLHS identifiers) constitute a selected subsample. The dropped districts cannot be compared on c_{int} itself, which is undefined for them, but they can be compared on outcomes and demographics: in NFHS-5, weighted means for matched versus unmatched districts are nearly identical for institutional delivery (0.888 versus 0.888), public-facility delivery (0.617 versus 0.635), antenatal care (0.928 versus 0.932), DPT (0.869 versus 0.879), education, and rural share, with only tetanus showing a statistically detectable but negligible gap (0.954 versus 0.948). NFHS-4 shows the same pattern, with matched districts 2.5 percentage points higher on institutional delivery ($p = 0.048$) as the largest difference; the larger raw gaps for public-facility delivery and DPT are statistically imprecise ($p = 0.67$ and $p = 0.86$ in NFHS-5; $p = 0.54$ and $p = 0.44$ in NFHS-4). The unmatched residual, concentrated in newly created districts of the smaller north-eastern states, therefore shows no economically large selection on outcome levels, though institutional delivery in NFHS-4 and tetanus in NFHS-5 differ at conventional significance; and level balance is in any case only suggestive, since the design requires the dropped districts not to differ in treatment intensity (undefined for them) or in trends, neither of which is testable.

3.4 Sample and Summary Statistics

Table 1
Summary Statistics by Survey Round (Short-Run DiD Sample)

Variable	DLHS-2 (2002–04)		DLHS-3 (2007–08)		Difference	
	Mean	SD	Mean	SD	DLHS-3 – DLHS-2	<i>p</i> -value
<i>Panel A: Primary Outcomes</i>						
Institutional delivery	0.375	0.484	0.434	0.496	0.067***	0.000
Any vaccination ^a	0.756	0.430	0.993	0.081	0.232***	0.000
DPT vaccination	0.679	0.467	0.746	0.435	0.069***	0.000
BCG	0.699	0.459	0.834	0.372	0.140***	0.000
Measles	0.425	0.494	0.580	0.494	0.158***	0.000
Full immunization	0.339	0.473	0.350	0.477	0.006	0.537
<i>Panel B: Secondary Outcomes</i>						
Any ANC	0.654	0.476	0.732	0.443	0.070***	0.000
Iron-folic acid	0.608	0.488	0.561	0.496	-0.048***	0.000
Tetanus injection	0.783	0.412	0.713	0.452	-0.071***	0.000
Public delivery	0.193	0.395	0.149	0.356	-0.044***	0.000
<i>Panel C: Controls</i>						
Woman’s age	29.864	7.471	31.844	8.551	2.126***	0.000
Woman’s education (yrs)	4.371	4.939	4.272	4.762	-0.015	0.757
Husband’s education (yrs)	6.822	5.193	6.591	4.986	-0.167***	0.000
Hindu	0.774	0.418	0.774	0.419	-0.014**	0.047
Muslim	0.112	0.315	0.115	0.319	-0.002	0.646
Scheduled Caste	0.171	0.376	0.179	0.383	0.005	0.138
Scheduled Tribe	0.156	0.363	0.177	0.382	0.042***	0.000
OBC	0.378	0.485	0.397	0.489	0.010	0.155
Rural	0.682	0.466	0.783	0.412	0.093***	0.000
<i>Panel D: Treatment</i>						
Sterilization intensity ^b	4.542	5.399	4.504	5.194	-0.038	0.670

Notes: $N = 1,151,566$ (DLHS-2: 507,622; DLHS-3: 643,944). 379 districts in the balanced short-run DiD sample. The “Difference” column reports the DLHS-3 minus DLHS-2 mean (weighted) and the p -value from a univariate regression of the variable on a post indicator with errors clustered at the district level. ^aThe near-ceiling mean (0.994) for any vaccination in DLHS-3 reflects the inclusive definition (at least one of BCG, DPT, OPV, or measles); this leaves minimal room for differential improvement and limits the informativeness of this outcome in the DID analysis. ^bSterilization intensity is the proportional surge in sterilizations during the Emergency year, $c_{int,d} = (S_{d,76-77} - S_{d,75-76})/S_{d,75-76}$; following Pelras and Renk (2023) and Singh and Vincent (2024), a single district with a near-zero pre-Emergency base is recoded to 29 (just above the maximum observed elsewhere). A predetermined-covariate balance test by high vs. low c_{int} is reported in Appendix Table A2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2
Summary Statistics: Long-Run Event-Study Sample

	DLHS-1 1998–99		DLHS-2 2002–04		DLHS-3 2007–08		NFHS-4 2015–16		NFHS-5 2019–21	
Variable	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Institutional delivery	–	–	0.375	0.484	0.434	0.496	0.791	0.406	0.888	0.316
Any vaccination	–	–	0.756	0.430	0.993	0.081	0.917	0.275	0.955	0.207
DPT vaccination ^c	–	–	0.679	0.467	0.746	0.435	0.814	0.392	0.869	0.337
Any ANC	–	–	0.654	0.476	0.732	0.443	0.815	0.389	0.929	0.258
Tetanus injection	–	–	0.783	0.412	0.713	0.452	0.919	0.274	0.953	0.212
Public delivery	–	–	0.193	0.395	0.149	0.356	0.522	0.500	0.620	0.485
Woman’s age	–	–	29.864	7.471	31.844	8.551	26.798	4.974	26.956	4.845
Woman’s education (yrs)	–	–	4.371	4.939	4.272	4.762	6.388	5.197	7.732	5.210
Rural	–	–	0.682	0.466	0.783	0.412	0.719	0.449	0.733	0.442
Sterilization intensity	–	–	4.542	5.399	4.504	5.194	4.520	5.315	4.472	5.326
<i>N</i>	474,986		507,622		643,944		259,627		232,920	
Districts	272		381		396		279		276	

Notes: Long-run event-study sample, pooling DLHS-1 through NFHS-5. Means are weighted using survey weights; SDs are within-round dispersions (not survey-design corrected). Number of districts is the count of distinct districts with non-missing c_{int} in each round. Round-by-round mean differences relative to DLHS-2 (the omitted reference round) are reflected in the event-study coefficients in Table 5; a predetermined-covariate balance check by high vs. low c_{int} (within each round) is reported in Appendix Table A2. ^cDPT vaccination is constructed on the corrected under-three denominator described in the text: NFHS-5 collects vaccination histories only for children under three, so children aged 3–4, for whom the outcome is uncollected, are excluded rather than coded as unvaccinated. The same corrected series is used in all regressions.

The balanced treatment sample for the short-run DiD analysis includes observations from DLHS-2 and DLHS-3 across 379 districts with non-missing sterilization intensity. Table 1 presents summary statistics for these two rounds. Institutional delivery increased from approximately 40% in DLHS-2 to 46% in DLHS-3; any vaccination from 76% to 99%; and DPT vaccination from 69% to 75%. These secular improvements provide the backdrop against which I test for differential gains in high-sterilization districts. One outcome, public delivery, declines from 19.5% to 14.8% despite the overall increase in institutional delivery, reflecting a shift toward private facilities during this period; the DID tests whether this decline was moderated in high-sterilization districts as trust in public facilities recovered. The long-run panel (DLHS-1 through NFHS-5) contains approximately 2.1 million observations across 402 matched districts appearing in at least one round; summary statistics for the additional rounds, reported in Table 2, are qualitatively consistent with the core panel and show broadly rising utilization over the 1998–2021 period, with non-monotonic paths for tetanus and public delivery between DLHS-2 and DLHS-3 and for the near-ceiling any-vaccination measure between DLHS-3 and NFHS-4.

4. Empirical Strategy

This section develops the empirical framework in four subsections. Section 4.1 establishes that the healthcare utilization deficit in districts with higher sterilization intensity persists in the pre-NRHM data, providing the baseline against which subsequent analyses are benchmarked. Sec-

tion 4.2 estimates the impact of NRHM on this deficit using a short-run difference-in-differences design (DLHS-2 vs DLHS-3) together with a long-run event study and a within-survey mother-child panel. Section 4.3 unbundles the NRHM into its constituent programme components, specifically the Janani Suraksha Yojana (JSY), adding the historical sterilization dimension as a triple-difference. Section 4.4 discusses identification assumptions, the relationship to the recent DiD literature, and threats to inference.

4.1 Healthcare Utilization Deficit: Pre-NRHM Evidence

The first empirical task is to establish that Emergency-era sterilization intensity continues to predict worse health outcomes in the period before the NRHM was introduced. This evidence is essential for two reasons. First, it validates in my data the previously documented utilization deficit: districts with more intense Emergency-era sterilization campaigns exhibit lower vaccination rates and fewer institutional deliveries decades after the Emergency ended.¹⁹ Second, it provides the baseline against which any post-NRHM convergence can be assessed, verifying that there is indeed a sterilization-outcome gradient.

Specifically, I estimate:

$$Y_{id} = \alpha + \beta \cdot c_{int,d} + \mathbf{X}_i' \gamma + \theta_s + \varepsilon_{id} \quad (2)$$

where Y_{id} is the health outcome for individual i in district d , $c_{int,d}$ is the district-level sterilization intensity, $\mathbf{X}_i$ is a vector of individual controls, and θ_s are state fixed effects. A significant negative β indicates that reduced child vaccination and institutional deliveries are associated with intensive/coercive sterilization historically, 20–25 years after the Emergency, conditional on state-level unobservables. I estimate equation (2) separately within each pre-NRHM round (DLHS-1 and DLHS-2); because each regression draws on a single survey round, no survey-round fixed effect is required here. Standard errors are clustered at the district level throughout, since the treatment variable varies at the district level.²⁰

I also estimate a collapsed specification in which outcomes are aggregated to district means, weighted by the number of underlying observations. This specification connects to a district-year panel approach and serves as a robustness check on the individual-level estimates. In addition, I construct a state-level treatment variable by aggregating district-level sterilization intensity to the state mean and estimate equation (2) with region fixed effects.²¹

¹⁹This deficit is documented by Pelras and Renk (2023).

²⁰This follows the clustering convention of Singh and Vincent (2024), whose coercion intensity index is defined at the district level. Pelras and Renk (2023) also define their treatment (excess sterilization rate) at the district level, but cluster standard errors at the state level because the sterilization campaign was directed through state-level policy targets, inducing within-state correlation in errors; they use wild cluster bootstrap p -values to address the small number of state-level clusters (Cameron, Gelbach, and Miller 2008). I report state-level clustering as a robustness check in Appendix C.

²¹State fixed effects cannot be included alongside a state-level treatment variable because the treatment is collinear with the state dummies. Region fixed effects (e.g., North, South, East, West, Northeast) absorb broad geographic confounders while leaving sufficient within-region, across-state variation in the state-level treatment to identify β .

Pooling rounds enables a richer fixed-effect structure than the per-round cross-section: district fixed effects δ_d (which absorb the time-invariant $c_{int,d}$ and so preclude estimating its level effect separately, unlike eq. (2), where $c_{int,d}$ enters alongside the state fixed effects because it varies across districts within a state), and state $\times$ round fixed effects μ_{st} (which absorb any common level shift between the rounds). This is the fixed-effect structure carried into the post-NRHM panel designs below. To test whether the deficit was stable or evolving across the two pre-NRHM rounds, I then pool DLHS-1 and DLHS-2 and add an interaction between sterilization intensity and a round indicator:

$$Y_{idt} = \alpha + \beta_3(c_{int,d} \times \text{Round2}_t) + \mathbf{X}'_i \gamma + \delta_d + \mu_{st} + \varepsilon_{idt} \quad (3)$$

where Round2_t equals one for DLHS-2 observations. A null β_3 with a significant β in (2) would show that there was no pre-existing differential trend, setting the stage for interpreting any post-NRHM change as attributable to the intervention. Both pre-NRHM rounds therefore enter this test: district fixed effects δ_d absorb the time-invariant level of $c_{int,d}$, so β_3 is identified purely from the change in the sterilization-outcome gradient between DLHS-1 and DLHS-2, while state $\times$ round fixed effects μ_{st} absorb any common level shift between the two surveys. In the long-run event study below, DLHS-2 likewise serves as the omitted reference round against which the DLHS-1 pre-trend coefficient is measured.

All specifications use the standardised treatment c_{int}^{std} defined in Section 3.1; the main text reports this preferred district-level construction, and the full set of alternative coercion measures is examined in the robustness analysis (Section 5.7).

4.2 The Effect of NRHM on the Utilization Deficit

4.2.1 Short-Run Difference-in-Differences Specification

The NRHM was rolled out nationally beginning in 2005. There is no staggered adoption across states or districts: all areas received the intervention at approximately the same time, following the national launch of the NRHM in April 2005 (NRHM 2005; Debnath 2021). This simultaneous rollout means that variation in treatment comes entirely from the cross-sectional dimension (historical sterilization intensity), not from differences in adoption timing. The core identification strategy exploits the interaction between cross-sectional variation in historical sterilization intensity and the temporal variation induced by the national NRHM rollout in a continuous-treatment difference-in-differences framework:

$$Y_{idt} = \alpha + \beta_1(c_{int,d}^{std} \times \text{Post}_t) + \mathbf{X}'_i \gamma + \delta_d + \mu_{st} + \varepsilon_{idt} \quad (4)$$

where Y_{idt} is the health outcome for woman i surveyed in district d in round t . Because the surveys are repeated cross-sections, i indexes an individual woman *within* a round rather than a woman tracked over time; the panel dimension is at the district level, and the only within-individual design is the mother fixed-effects panel below ((6)). The term $c_{int,d}^{std}$ is the standardized

(z-score) district-level sterilization intensity, facilitating interpretation in standard-deviation units; $Post_t$ indicates the post-NRHM period (DLHS-3, 2007–08); $\mathbf{X}_i$ is a vector of individual controls (woman’s age, woman’s education, husband’s education, religion, caste, and rural residence); δ_d are district fixed effects; and μ_{st} are state $\times$ time fixed effects. Standard errors are clustered at the district level. District fixed effects absorb all time-invariant district characteristics, including the sterilization intensity itself. State $\times$ time fixed effects absorb all state-level secular trends and policy changes, so that identification of β_1 comes purely from within-state, across-district variation in sterilization intensity interacted with the NRHM rollout, the cross-sectional dose of historical coercion, projected onto a national-level temporal shock. Note that the post-NRHM specifications use the standardized treatment $c_{int,d}^{std}$ (the z-score of the raw $c_{int,d}$ defined in Section 3.1, used in the pre-NRHM regressions of Section 4.1) so that coefficients read directly as the effect of a one-standard-deviation increase in sterilization intensity; results under the raw and other alternative scalings are reported in Section 5.7.

The coefficient β_1 captures whether districts with higher historical sterilization intensity experienced differential changes in health outcomes after NRHM. If NRHM narrows the utilization deficit, $\beta_1 > 0$: districts with larger historical utilization gaps show larger improvements in health-seeking behaviour, narrowing the gap. A one-standard-deviation increase in sterilization intensity corresponds to approximately 5.3 units of c_{int} , or roughly the interquartile range of the treatment distribution. The binary above-median indicator provides an alternative specification for comparability with the persistence literature; additional treatment variants are examined in Section 5.7.

4.2.2 Long-Run Event Study

To trace the trajectory of utilization convergence, I estimate a long-run event study that interacts sterilization intensity with survey-round indicators across all six rounds:

$$Y_{idt} = \alpha + \sum_{r \neq \text{DLHS-2}} \gamma_r \left(c_{int,d}^{std} \times \mathbf{1}[\text{Round}_t = r] \right) + \mathbf{X}_i' \gamma + \delta_d + \mu_{st} + \varepsilon_{idt} \quad (5)$$

where r indexes survey rounds (DLHS-1, DLHS-3, DLHS-4, NFHS-4, NFHS-5) and DLHS-2 serves as the omitted reference period. Each γ_r captures the sterilization-outcome gradient in round r relative to the DLHS-2 baseline. The pre-NRHM coefficient $\gamma_{\text{DLHS-1}}$ serves as a pre-trend test: if parallel trends hold, $\gamma_{\text{DLHS-1}} \approx 0$. The four post-NRHM coefficients $\{\gamma_{\text{DLHS-3}}, \gamma_{\text{DLHS-4}}, \gamma_{\text{NFHS-4}}, \gamma_{\text{NFHS-5}}\}$ trace the dynamic evolution of the utilization deficit over time, testing whether convergence emerges gradually as the trust-rebuilding mechanism predicts.

4.2.3 Within-Survey Mother-Child Panel

An additional specification exploits the birth history structure of DLHS-3. The survey records outcomes for up to two children per woman. A mother surveyed in 2007 who bore one child

in 2003 and another in 2006 provides within-mother variation spanning the April 2005 NRHM cutoff. This enables a mother fixed effects specification:

$$Y_{cm} = \alpha_m + \gamma(c_{int,d} \times Post_c) + X'_c \delta + \varepsilon_{cm} \quad (6)$$

where c indexes children, m indexes mothers, and $Post_c$ indicates whether child c was born after April 2005. The vector X_c includes child-level controls (birth order, sex) and birth-year fixed effects that absorb the common time trend, so that γ captures the differential change in outcomes for children born post-NRHM in high- versus low-sterilization districts within the same mother. Mother fixed effects α_m absorb all time-invariant maternal characteristics including caste, religion, education, district of residence, household wealth, and critically, inherited distrust of the state. If the same mother who delivered her 2003 child at home delivers her 2006 child at a facility, and this behavioural shift is differentially larger in high-sterilization districts, this is evidence that behaviour changed after NRHM among women who carry the sterilization legacy. The mother fixed effects rule out confounding from time-invariant maternal characteristics, but the comparison across districts remains: identification still requires that no other district-level change coincident with April 2005 was correlated with sterilization intensity, an assumption the mother fixed effects cannot themselves guarantee. Because the post-April-2005 period in DLHS-3 spans both JSY rollout and partial ASHA deployment, this within-mother design cannot by itself disentangle the cash-transfer and community-engagement channels; that separation is the task of Section 4.3, where I exploit JSY-eligibility rules and ASHA scale-up timing to attribute the convergence to its constituent NRHM components.

This design requires mothers with births on both sides of April 2005. With DLHS-3 covering births approximately from 2002 to 2008, this requires birth spacing under approximately four to five years. The resulting sample is necessarily more selected than the full cross-section: these mothers are younger, higher-parity, and more fertile. I report summary statistics for the panel subsample alongside the full sample in Section 5.3 and argue that this selection, if anything, biases against finding convergence, since higher-parity mothers in high-sterilization districts likely carry the deepest inherited distrust.

4.2.4 Pre-Trend Tests and Sensitivity Analysis

I implement four complementary layers of pre-trend assessment. Each addresses a distinct concern.

The first three are point tests of whether the pre-NRHM gradient was already moving, and each reuses an estimation introduced above. *First*, the event-study coefficient γ_{DLHS-1} in (5) tests whether the gradient was changing in the round immediately before NRHM, within the full multi-round model with district and state $\times$ time fixed effects; it is the natural pre-period test inside the dynamic specification used for the headline event-study figure. *Second*, the pooled DLHS-1+2 specification (3) re-runs the same DLHS-1-vs-DLHS-2 comparison in isolation, restricting the sample to the two pre-NRHM rounds; unlike the first test it isolates the pre-period from the post-NRHM observations that influence the fixed-effect identification and clustering, so

the two are complementary rather than redundant. *Third*, within DLHS-3 I exploit birth-month variation among pre-April-2005 births as a within-survey birth-month placebo, interacting sterilization intensity with half-year indicators; because every child in this sample is born before the cutoff, coefficients near zero are expected, at a sharper time resolution than the multi-round event study can deliver. The placebo is valid only for outcomes realized at or before delivery (institutional delivery, antenatal care, tetanus), since being born before the cutoff fixes these outcomes in the pre-NRHM period; immunization outcomes are excluded because vaccines are administered over the first year or more of life, so a child born shortly before April 2005 receives most doses inside the post-NRHM window and a non-zero coefficient would measure an early treatment effect rather than a pre-trend violation. Because DLHS-3 records the place of delivery only for births from 2004 onward, the usable placebo window is 2004H1 through 2005Q1.

Fourth, I conduct a sensitivity analysis on the parallel-trends assumption.²² Tests 1–3 are all *point* tests of whether the pre-trend is exactly zero, and such tests have low statistical power, with conditioning on passing them distorting inference.²³ The sensitivity analysis addresses precisely this weakness: rather than assuming parallel trends hold exactly, it restricts the magnitude of possible violations and constructs confidence sets for the treatment effect that are uniformly valid over that class of violations. The post-treatment deviation from parallel trends between consecutive periods is bounded by $\bar{M}$ times the maximum pre-treatment violation under the “relative magnitudes” restriction $\Delta^{RM}(\bar{M})$.²⁴ I implement the procedure via the `HonestDiD` package and report the implied confidence sets in Section 5.3, with sensitivity plots in Appendix H.

Additionally, I conduct balance tests on predetermined characteristics. If sterilization intensity is uncorrelated with post-NRHM changes in education, caste composition, and rural residence (characteristics that NRHM should not have altered), the data are consistent with the identification strategy; because these are point tests of a zero null, a failure to reject cannot by itself confirm balance, so I report minimum detectable effects and a multiple-testing adjustment alongside the estimates. I test whether $c_{int}^{std} \times Post$ predicts changes in eight predetermined outcomes: woman’s education, husband’s education, Hindu, Muslim, Scheduled Caste, Scheduled Tribe, Other Backward Class, and rural residence.²⁵ Results are reported in Table 12 in the results section.

Finally, I assess sensitivity to the construction of the treatment itself. Because the proportional surge divides by the district’s 1975–76 sterilization base, low-base districts receive mechanically high treatment values, raising the possibility that the estimated convergence partly reflects mean reversion from a low base rather than recovery. I address this in Section 5.7.3 in two ways: by re-estimating the long-run event study with flexible controls for the base interacted

²²Following [Rambachan and Roth \(2023\)](#).

²³See [Roth \(2022\)](#).

²⁴Rambachan and Roth (2023) also develop a “smoothness” restriction ($\Delta^{SD}(M)$) that bounds the second differences of the trend violation, and a combined restriction. I focus on the relative magnitudes approach, which is the most intuitive: setting $\bar{M} = 1$ assumes post-treatment trend violations are no larger than the worst pre-treatment one.

²⁵These balance tests follow the approach of [Andrew and Vera-Hernández \(2024\)](#), who similarly validate their JSY identification on predetermined covariates.

with round, and by rebuilding the index on the fully pre-Emergency 1974–75 base introduced in Section 3.1, which is nearly uncorrelated with its own denominator and therefore separates the surge from the base by construction.

4.2.5 Cross-Sectional Trust Analysis (DLHS-4)

DLHS-4 (2012–13) uniquely permits a direct test of persistent distrust using stated trust measures. I estimate:

$$Trust_{id} = \alpha + \beta \cdot c_{int,d} + \mathbf{X}_i' \gamma + \theta_s + \varepsilon_{id} \quad (7)$$

where $Trust_{id}$ is a trust-related outcome for woman i in district d , and θ_s are state fixed effects. Structurally, equation (7) is the cross-sectional persistence regression of equation (2) applied to DLHS-4’s trust outcomes, not a difference-in-differences. Because the direct trust items are observed in a single round, there is no pre/post or panel dimension to exploit; the specification therefore deliberately mirrors the pre-NRHM persistence regression rather than the DiD of equation (4), and the object of interest is whether the 2012–13 trust gradient β is statistically distinguishable from zero. Because the trust items were never fielded before NRHM, any statement about flattening is an inference that combines this cross-sectional gradient with the pre-NRHM utilization gradient of equation (2), which was negative; the trust data alone cannot document a change over time. Trust outcomes are organized into three tiers: (1) direct health trust (stated distrust of government facilities, facility choice, perceived quality); (2) ASHA engagement (ASHA-motivated antenatal care, delivery, immunization, JSY receipt); and (3) government scheme participation (pregnancy registration, health scheme coverage, RSBY). The exact survey questions underlying each outcome are reproduced in Appendix E. A coefficient near zero indicates that by 2012–13, the sterilization-linked gradient in trust and utilization had largely flattened; because β measures a cross-district differential, it is informative about the absence of a sterilization-linked gap, not about the absolute level of trust or utilization.

4.3 Unbundling the NRHM: JSY, ASHAs, and Facility Capacity

The convergence documented in the preceding section could in principle reflect three NRHM components: ASHA community engagement (the trust channel this paper foregrounds), the Janani Suraksha Yojana (JSY) cash incentive paid to mothers for institutional delivery, and a congestion mechanism through which JSY-induced demand overwhelms low-capacity facilities. This section unbundles the NRHM into these components in order to *rule out* the latter two. Both alternatives map onto the same triple-difference structure, $c_{int,d}^{std} \times Z \times Post$, with Z chosen to isolate the channel: $Eligible_i$ (rule-based JSY eligibility) tests the cash channel, and $HighCapacity_d$ (pre-NRHM facility density) tests the congestion channel.²⁶ The distinctive coefficient is the triple interaction δ , which asks whether each channel’s effect is amplified,

²⁶The eligibility design follows [Debnath \(2021\)](#) and the facility-capacity design follows [Andrew and Vera-Hernández \(2024\)](#).

absorbed, or unchanged by historical sterilization intensity. If δ is small for both channels, the convergence is *not* concentrated in eligible mothers (cash story) and *not* concentrated in low-capacity districts (congestion story), leaving the trust-via-ASHA mechanism as the residual interpretation supported by the $Post_1/Post_2$ timing decomposition in Section 6.2. I also report a continuous JSY-intensity specification as a descriptive cross-check.²⁷

4.3.1 JSY Eligibility (Cash Channel)

The first design exploits the rule-based variation in JSY eligibility within DLHS-3 birth histories. JSY eligibility depends on a combination of state classification, caste, economic status, and parity: in Low-Performing States (LPS, comprising the EAG states plus Assam and Jammu & Kashmir), all women are eligible regardless of caste or income; in High-Performing States (HPS), only women below the poverty line (BPL) or from Scheduled Caste/Tribe (SC/ST) backgrounds qualify, and only for their first two births. Using births before and after April 2005 within DLHS-3, the eligibility specification is:

$$Y_{cisdt} = \alpha + \alpha_E \cdot Eligible_i + \alpha_{EP}(Eligible_i \times Post_t) + \mathbf{X}_i' \beta + \delta_d + \mu_{st} + \varepsilon_{cisdt} \quad (8)$$

I replicate this specification and then extend it with the sterilization dimension. The progressive build-up is:

$$\text{Column 1: } Y = \alpha_{EP}(Eligible_i \times Post_t) + \dots \quad (9)$$

$$\text{Column 2: } Y = \alpha_{EP}(Eligible_i \times Post_t) + \beta(c_{int,d}^{std} \times Post_t) + \dots \quad (10)$$

$$\begin{aligned} \text{Column 3: } Y = & \alpha_{EP}(Eligible_i \times Post_t) + \beta(c_{int,d}^{std} \times Post_t) + \delta_2(c_{int,d}^{std} \times Eligible_i) \\ & + \delta(c_{int,d}^{std} \times Eligible_i \times Post_t) + \dots \end{aligned} \quad (11)$$

A complete triple-difference requires all three pairwise interactions alongside the three-way term: $Eligible_i \times Post_t$ and $c_{int,d}^{std} \times Post_t$ appear explicitly above, and the level interaction $c_{int,d}^{std} \times Eligible_i$, which varies within district across individuals and is therefore not absorbed by the district fixed effects, is included as δ_2 so that any pre-period eligible-versus-ineligible differential across the intensity distribution does not load onto $\hat{\delta}$. The estimation includes the full factorial expansion; the level effect of $c_{int,d}^{std}$ itself is absorbed by δ_d .

The triple interaction coefficient δ in equation (11) is this paper's distinctive contribution. A significant δ indicates that JSY eligibility has a differential effect in high-sterilization districts. If $\delta > 0$, the JSY cash incentive is more effective where the historical utilization deficit is deepest, suggesting that financial incentives can partially compensate for the deficit. If $\delta < 0$, the utilization deficit makes women resistant even to cash incentives, suggesting that trust must be rebuilt before demand-side policies become effective.

The interpretation of this triple-difference merits elaboration. The treatment variable c_{int}^{std} varies

²⁷The continuous JSY-intensity design follows [Powell-Jackson, Mazumdar, and Mills \(2015\)](#).

only at the district level, while $Eligible_i$ varies at the individual level (determined by caste, BPL status, and parity). Standard errors are clustered at the district level to account for the district-level component of the treatment. The triple interaction adds a genuinely individual-level dimension of variation: identification comes from the differential response of eligible versus ineligible women across the sterilization intensity distribution.

The post-NRHM period is further decomposed into two *birth-date* sub-windows, not separate surveys: $Post_1$, for children born April 2005–September 2006 (after the NRHM cutoff but before ASHA deployment had reached operational scale), and $Post_2$, for children born October 2006 onward (after ASHA scale-up). Both sub-windows are constructed entirely within DLHS-3, using the birth dates recorded in the survey’s birth history, so the comparison is within a single survey instrument and is not contaminated by cross-survey measurement differences. The decomposition is informative because it separates the JSY cash-transfer effect (which began immediately at the NRHM cutoff) from the ASHA community-engagement effect (which required time to deploy). I interact the sterilization intensity with this $Post_1/Post_2$ decomposition to test whether the trust-rebuilding effect requires the ASHA intermediary or whether cash incentives alone suffice.

4.3.2 Facility Capacity (Congestion Channel)

JSY increased institutional deliveries but, in areas with low pre-existing facility capacity, also increased perinatal mortality, operating through a congestion mechanism.²⁸ The central specification interacts JSY eligibility with facility capacity quartiles. This finding is directly relevant to my trust story: if the convergence I document in high-sterilization districts occurred primarily in areas with inadequate facilities, it could reflect congestion rather than genuine trust recovery. I adapt their approach by interacting sterilization intensity with a capacity indicator:

$$Y_{idt} = \alpha + \beta_1(c_{int,d}^{std} \times Post_t) + \beta_3(HighCapacity_d \times Post_t) \\ + \beta_2(c_{int,d}^{std} \times Post_t \times HighCapacity_d) + \mathbf{X}'_i\gamma + \delta_d + \mu_{st} + \varepsilon_{idt} \quad (12)$$

where $HighCapacity_d$ is an indicator for above-median facility density. Because the cited congestion effect is concentrated in the lowest capacity quartile, a median split pools the first and second quartiles and mechanically attenuates the contrast of interest, recovering roughly half of the true bottom-quartile gap when the second-quartile effect is near zero. I therefore also estimate the specification with a bottom-quartile indicator in place of the median split, which targets the congestion mechanism where it is documented to operate. This specification tests whether the utilization convergence occurred in districts with adequate health infrastruc-

²⁸This congestion finding is due to [Andrew and Vera-Hernández \(2024\)](#), who show the effect is concentrated in the lowest quartile of pre-JSY facility capacity. The mechanism operates as follows: JSY’s cash incentive induced a surge in institutional deliveries, but in areas where health facilities were already at or near capacity, the additional patient volume overwhelmed available staff, beds, and equipment. This resulted in lower quality of care per delivery, longer wait times, less attention from skilled birth attendants, inadequate monitoring of complications, which increased adverse perinatal outcomes even as institutional delivery rates rose.

ture, consistent with a genuine trust channel, or in underserved areas where congestion might dominate.

4.3.3 Continuous JSY-Intensity Specification

A continuous district-level treatment is defined as the fraction of births receiving JSY payments, interacted with a pre-post April 2005 indicator.²⁹ Formally,

$$Y_{idt} = \alpha + \beta_1(c_{int,d}^{std} \times Post_t) + \beta_2(JSY_d \times Post_t) + \beta_3(c_{int,d}^{std} \times JSY_d \times Post_t) + \mathbf{X}_i' \gamma + \delta_d + \mu_{st} + \varepsilon_{idt} \quad (13)$$

where JSY_d is the district-level fraction of births receiving JSY payments and β_3 asks whether the JSY coverage–outcome relationship varies with historical sterilization intensity. This district-level JSY “intensity” measure has the advantage of capturing actual programme penetration but is endogenous: districts with better infrastructure and governance have both higher JSY uptake and better health outcomes. I estimate their specification as a descriptive exercise with appropriate caveats, and interact it with sterilization intensity to test whether the JSY coverage–outcome relationship varies with historical distrust. The endogeneity caveat extends to the triple interaction itself: whenever the determinants of JSY uptake (infrastructure, governance, administrative capacity) covary with $c_{int,d}^{std}$, their influence enters β_3 with the same functional form as the hypothesized trust moderation, so a non-zero β_3 is consistent with, but does not by itself identify, a trust-based interpretation.

4.4 Identification and the New DiD Literature

4.4.1 Relationship to Recent Methodological Advances

Recent advances in the difference-in-differences literature have highlighted potential biases in two-way fixed effects (TWFE) estimators when treatment timing is staggered and treatment effects are heterogeneous.³⁰ It is important to clarify why these specific concerns do not apply to my setting, while other insights from this literature remain relevant.

The core concern in the staggered DiD literature is that TWFE estimates can be biased because they implicitly use already-treated units as controls for newly-treated units, generating negative weights on some group-time treatment effects. This problem arises specifically when units adopt binary treatment at different times. In my setting, there is no staggered treatment adoption: NRHM was launched nationally in April 2005, and all districts were exposed to the intervention simultaneously. The treatment variation is entirely cross-sectional: differences in the historical sterilization intensity “dose” rather than temporal. With a single treatment timing, the staggered-adoption source of negative weighting does not arise, and neither the

²⁹This continuous JSY-intensity design follows [Powell-Jackson, Mazumdar, and Mills \(2015\)](#).

³⁰See [Goodman-Bacon \(2021\)](#), [Callaway and Sant’Anna \(2021\)](#), and [de Chaisemartin and D’Haultfoeuille \(2020\)](#).

Goodman-Bacon decomposition nor the Callaway–Sant’Anna group-time ATT estimator is directly applicable to my design. This does not, however, exhaust the weighting concern: because treatment is a continuous dose, the TWFE coefficient can still place non-convex weights on dose-specific responses when the dose-response function is heterogeneous, a distinct problem taken up in the continuous-treatment literature discussed next.³¹

My design instead belongs to the “continuous treatment DiD” literature, which extends the DiD framework to settings where treatment intensity varies continuously across units, showing that TWFE can recover a causal parameter under homogeneity restrictions on the dose-response function, but that heterogeneous dose-response functions can complicate interpretation.³² Two requirements follow from that literature and should be kept distinct. The first concerns identification: interpreting the linear TWFE coefficient as a positively weighted average of dose-specific treatment effects requires a “strong” parallel-trends assumption, namely that untreated potential outcomes evolve in parallel across every dose level, which is materially more demanding than the single conditional parallel-trends assumption discussed under Threats to Identification below. The second concerns functional form. The two diagnostics I use, a dose-response analysis and a binscatter visualization, speak to the functional-form question; neither tests the strong parallel-trends condition, which remains an assumption. First, the dose-response analysis using quintile dummies interacted with the post indicator (replacing the linear c_{int}^{std}) probes the shape of the dose-response. Approximately equal increments between adjacent quintile coefficients (a constant per-quintile slope) would support the linear specification; monotonicity alone is a weaker property, consistent with substantial curvature. Second, the binscatter analysis visualizes the relationship between sterilization intensity and outcome changes across the full distribution, serving as a graphical check for treatment-effect heterogeneity. If the quintile and binscatter diagnostics reveal substantial non-linearity, the linear TWFE coefficient is best interpreted as a weighted average of dose-specific effects; in that case I lean on the non-parametric quintile estimates as the primary summary of the convergence rather than the linear coefficient. In practice, the quintile dose-response (reported in Section 5.7) is broadly monotone, which establishes that the sign of the effect is stable across the dose distribution; I therefore report the linear coefficient as a convenient scalar summary while treating the quintile estimates as the primary description of the dose-response wherever the two diverge.

The design shares an intuitive similarity with the Bartik or shift-share tradition:³³ historical sterilization intensity functions as cross-sectional exposure, and NRHM provides the common temporal shock. However, this is a continuous-treatment DID, not a shift-share IV; the formal Rotemberg decomposition requires multiple shocks and local shares, which this single-treatment-variable design does not possess. The closest diagnostic is the sensitivity of results to excluding high-leverage districts, reported in Section 5.7 (trimmed sample and excluding small states).

³¹See de Chaisemartin and D’Haultfoeuille (2020), Goodman-Bacon (2021), and Callaway and Sant’Anna (2021).

³²See Callaway, Goodman-Bacon, and Sant’Anna (2024).

³³See Goldsmith-Pinkham, Sorkin, and Swift (2020).

4.4.2 Threats to Identification

The key identifying assumption is that, conditional on district and state $\times$ time fixed effects, changes in health outcomes would not have been systematically correlated with historical sterilization intensity absent the NRHM. Several concerns warrant discussion.

Omitted variables and differential trends: District fixed effects absorb all time-invariant confounders, while state $\times$ time fixed effects absorb state-level policy changes and secular trends. The residual threat is district-level, time-varying confounders correlated with both sterilization intensity and health outcomes. The pooled DLHS-1+2 pre-trend test and the within-DLHS-3 birth-month placebo directly assess whether such confounders were operating before NRHM.

Differential NRHM resource allocation: If the central government directed more NRHM resources to high-sterilization districts, the observed convergence could reflect differential supply-side investment rather than demand-side trust recovery. This threat operates at two levels.

At the *state* level, the concern is real: NRHM designated EAG states as high priority, and these northern states also experienced the most intense Emergency-era sterilization. However, state $\times$ time fixed effects absorb all state-level differential resource allocation, including the EAG priority designation and any resulting differences in funding, staffing norms, or implementation timelines. The identifying variation is therefore purely within-state, across-district.

At the *district* level within states, the concern requires a mechanism through which 1976–77 sterilization rates entered the 2005 NRHM resource allocation formula. No such mechanism exists. NRHM’s operational design was formulaic, not discretionary: ASHAs were allocated by population (one per approximately 1,000 rural inhabitants, as specified in the NRHM Framework for Implementation), not by health performance metrics or historical sterilization history (NRHM 2005). JSY eligibility was determined by rule-based individual criteria (state classification, caste, BPL status, parity), not district-level characteristics. Sub-centre and PHC upgrades followed Indian Public Health Standards population norms. I confirm this empirically: within states, ASHA deployment and contact rates for antenatal care, delivery, and immunisation are uncorrelated with district-level sterilization intensity (all $p > 0.60$; Appendix Table A3, Panel B), consistent with uniform population-based allocation. Nor do high-intensity districts differ in the functioning of the facilities themselves: within states, sterilization intensity does not predict the share of DLHS-3 primary health centres with a staff nurse available (-0.012 per standard deviation, $p = 0.45$) or with an additional nurse-midwife in post ($p = 0.99$), so the convergence is not explained by differential implementation quality on these measurable dimensions. Furthermore, the Low-Performing State triple interaction (Section 5.9) shows that convergence is *weaker*, not stronger, in EAG states for antenatal care and tetanus, the opposite of what the NRHM-targeting confound would predict.

Post-NRHM state programmes correlated with treatment intensity (Mission Parivar Vikas): A further concern, salient for the family-planning mechanism in particular, is that the central government launched Mission Parivar Vikas (MPV) in November 2016 in 146 high-fertility districts concentrated in seven northern states, Uttar Pradesh, Bihar, Madhya Pradesh, Rajasthan, Jharkhand, Chhattisgarh, and Assam, that overlap heavily with the high-Emergency-intensity

belt. MPV intensified sterilization compensation, expanded ASHA incentives, and introduced new contraceptive products, so it could in principle inflate the post-2016 sterilization-outcome gradient (rounds NFHS-4 and NFHS-5) for reasons unrelated to trust recovery. State $\times$ time fixed effects absorb the state-level component of MPV. The remaining concern is that within-state MPV district targeting (by current fertility) is correlated with historical sterilization intensity. I test this with two complementary checks on the family-planning long-run event study, reported in Section 6.2: (i) excluding the seven MPV states entirely; and (ii) adding a triple-interaction term, $c_{int}^{std} \times \text{Round}_r \times \text{MPV-state}$, which estimates how the c_{int}^{std} effect at round r differs in MPV versus non-MPV states.³⁴ The maternal- and child-health convergence results are less directly exposed to this confound, since MPV’s instruments target permanent contraception rather than antenatal care, institutional delivery, or vaccination.

Migration and compositional change: If the most distrustful families selectively migrate away from high-sterilization districts over time, the post-NRHM sample would be selected toward less-distrustful stayers, mechanically producing convergence. I test this directly using the NFHS women’s recode item V104 (“years lived in current place of residence”), which is available in NFHS-4 and NFHS-5 but is not collected in the DLHS rounds. Re-estimating the long-run event study with NFHS observations restricted to long-term residents ($V104 \geq 10$ years, or “always/since birth”), while keeping DLHS rounds unrestricted, yields coefficients on $c_{int,d}^{std} \times \text{Round}_r$ that are stable or strengthen for the family-planning outcomes (limiting method and female sterilization) and for DPT vaccination, any antenatal care, and any vaccination; institutional delivery is essentially unchanged in magnitude but loses 10% significance under the halved NFHS sample; only public-facility delivery materially attenuates while remaining positive in sign. Full coefficients for all eight outcomes are reported in Appendix Table A6. As a complementary check, a direct regression of V104 itself on $c_{int,d}^{std} \times \text{Round}$ is a tightly estimated zero (0.013 years per SD of $c_{int,d}^{std}$, SE 0.164, $p = 0.93$), and the raw V104 distribution shows a slightly higher share of long-term residents in high- c_{int} districts than in low- c_{int} districts, the opposite of what selective out-migration would predict. The mother fixed-effects specification ((6)) provides an additional safeguard for the within-DLHS-3 comparison by absorbing all between-mother compositional changes.

Measurement error: The historical sterilization data come from administrative records.³⁵ Classical measurement error in sterilization intensity attenuates the estimated coefficients toward zero, biasing against finding convergence. Non-classical measurement error i.e. if districts with more coercive campaigns also had more inaccurate record-keeping, could bias estimates in either direction.

SUTVA and spatial spillovers: If health-seeking behaviour in one district is affected by sterilization intensity in neighboring districts (for instance, through information networks or facility sharing), SUTVA would be violated. I address this with three complementary checks, reported in Section 5.7: spatial HAC standard errors that allow for spatial correlation in errors up to

³⁴A district-level MPV indicator would identify a still sharper within-state contrast; I rely on the state-level interaction here and leave the district-level refinement, which requires matching the 146 high-fertility MPV districts to the harmonized 1976–77 district codes, to future work.

³⁵Compiled by Gwatkin (1979) and digitized by Jolly (1986).

a specified distance cutoff,³⁶ which address inference but not spillover bias in the point estimate; re-estimation excluding all northern Indian states, where high-sterilization districts are geographically clustered, reporting also the treatment variation retained so that a null could be distinguished from a mechanical loss of power; and a donut test that drops the quarter of districts geographically closest to a district of opposite treatment intensity, retaining interior districts where cross-district spillovers are least plausible.

5. Results

This section presents results organized to mirror the empirical strategy. Section 5.1 documents the pre-NRHM utilization deficit. Section 5.2 reports the short-run DiD results. Section 5.3 presents the long-run event study, convergence analysis, and within-survey mother panel. Section 5.4 reports the DLHS-4 trust analysis. Section 5.5 provides binscatter evidence. Section 5.6 presents the JSY unbundling results. Section 5.7 discusses robustness. Section 5.8 reports placebo tests. Section 5.9 examines heterogeneity.

5.1 Pre-NRHM Utilization Deficit

The cross-sectional regressions (2) establish that Emergency-era sterilization intensity predicts significantly worse health outcomes in the pre-NRHM period. In the DLHS-2 cross-section with district-level treatment and state fixed effects, a one-unit increase in sterilization intensity is associated with lower rates of institutional delivery, vaccination, and antenatal care. Per standard deviation of c_{int}^{std} , the DLHS-2 gradients are -0.9 percentage points for institutional delivery, -1.9 for DPT, -1.6 for antenatal care, and -1.3 for any vaccination; the earlier DLHS-1 gradients are larger (institutional delivery -1.9 , antenatal care -2.4 , tetanus -1.4), indicating that the deficit was already eroding between 1998 and 2002. These are the denominators against which Section 5.3 measures the share of the deficit closed. These magnitudes are consistent with Pelras and Renk (2023), who report that a doubling of sterilization intensity reduces vaccination probability by approximately 17 percent of a standard deviation. The state-level treatment specification, using region fixed effects, yields qualitatively similar results with larger point estimates, reflecting the coarser level of variation.

The DLHS-1 results from the long-run event study (Table 5, column R1) confirm that the deficit was already present in 1998–99, approximately two decades after the Emergency. The pooled DLHS-1+2 interaction (Panel C) yields statistically insignificant coefficients for most outcomes, indicating that the sterilization-outcome gradient was *stable* across the two pre-NRHM rounds. This stability is precisely the condition required for the DID design: a persistent but non-trending deficit, against which post-NRHM changes can be credibly attributed to the intervention.

5.2 The Short-Run Effect of NRHM

³⁶Following Conley (1999).

Table 3

Main Difference-in-Differences Results: Standardised Continuous Treatment

	(1) Naïve	(2) Controls + FE	(3) Preferred	Dep. var. mean
<i>Panel A: Primary Outcomes</i>				
Institutional delivery	−0.0515*** (0.0108)	0.0136*** (0.0048)	0.0016 (0.0039)	0.398
Any vaccination	−0.0009 (0.0010)	0.0347*** (0.0113)	0.0115 (0.0079)	0.763
DPT vaccination	−0.0400*** (0.0105)	0.0111** (0.0053)	0.0068 (0.0066)	0.690
BCG	−0.0243** (0.0096)	0.0189*** (0.0063)	0.0099 (0.0075)	0.713
Measles	−0.0489*** (0.0101)	−0.0032 (0.0048)	0.0007 (0.0055)	0.437
Full immunization	−0.0517*** (0.0100)	−0.0043 (0.0059)	−0.0083 (0.0060)	0.357
<i>Panel B: Secondary Outcomes</i>				
Any ANC	−0.0477*** (0.0108)	0.0126** (0.0062)	−0.0075 (0.0079)	0.673
Iron-folic acid	−0.0522*** (0.0124)	−0.0064 (0.0068)	−0.0028 (0.0074)	0.623
Tetanus injection	−0.0465*** (0.0110)	−0.0256*** (0.0086)	−0.0245** (0.0096)	0.799
Public delivery	−0.0179*** (0.0053)	0.0037 (0.0035)	0.0025 (0.0029)	0.195
Observations	256,894 (delivery); 193,253 (vaccination)			
Clusters (districts)	379			
Within R^2 range	[0.022, 0.163]			
Controls	No	Yes	Yes	
District FE	No	Yes	Yes	
State $\times$ Time FE	No	No	Yes	

Notes: Each cell reports the coefficient on $c_{int}^{std} \times \text{Post}$ from a separate regression. c_{int}^{std} is the z-score of sterilization intensity (mean zero, unit SD); a one-SD increase corresponds to approximately 5.3 units of raw c_{int} , or roughly the interquartile range of the treatment distribution, so each coefficient reads as the per-SD marginal effect. The corresponding binary above-median contrast in Table 4 measures the difference between the upper and lower halves of the sterilization-intensity distribution rather than a per-SD effect, so the two coefficient magnitudes are not directly comparable. Dep. var. mean is the DLHS-2 (pre-NRHM) baseline. Identification of β_1 comes from cross-district variation in $c_{int,d}^{std}$ within state $\times$ round cells (absorbed by μ_{st}), interacted with the post-NRHM time shock; district fixed effects δ_d absorb the time-invariant level of $c_{int,d}$. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Effect of Sterilization Intensity on Health Outcomes (Post-NRHM)

DID: Sterilization x Post | District FE + State x Time FE | Clustered SEs

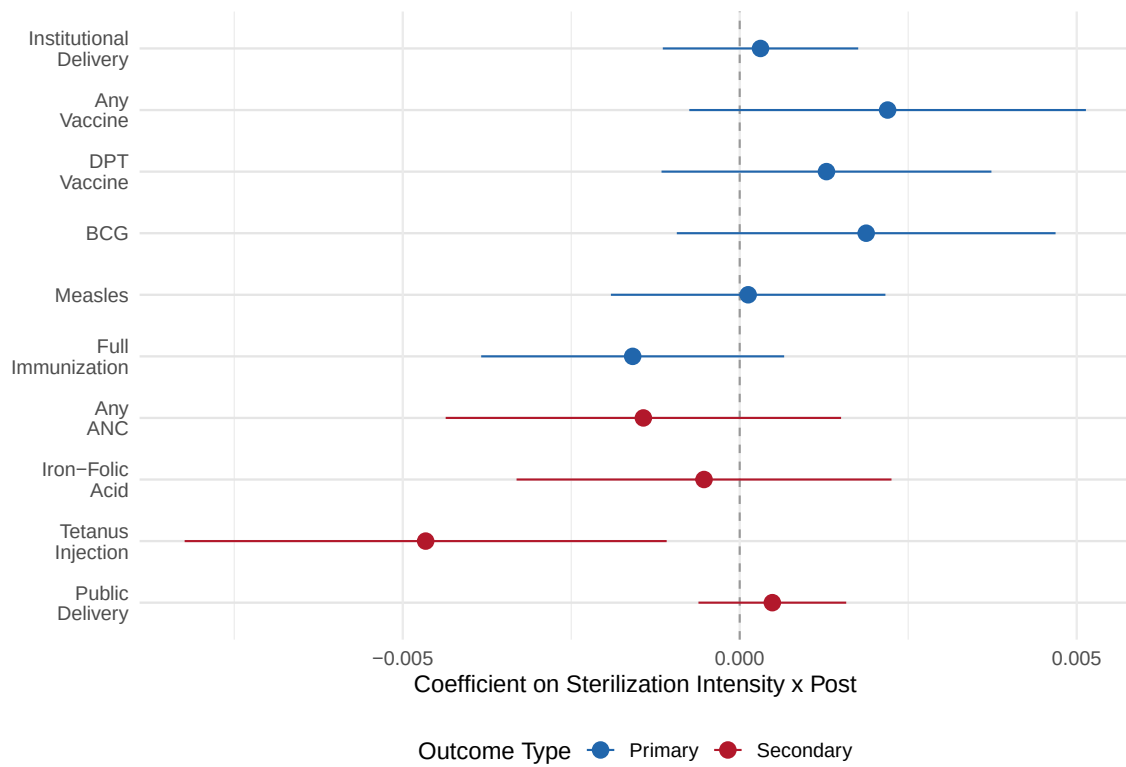


Figure 3

Short-Run DiD Coefficient Plot: $c_{int}^{std} \times \text{Post}$ Across Outcomes

Table 4
Binary Treatment Difference-in-Differences Results

Outcome	Coefficient	SE	<i>p</i> -value	<i>N</i>	Dep. var. mean
<i>Panel A: Primary Outcomes</i>					
Institutional delivery	0.0157*	(0.0084)	0.061	256,894	0.398
Any vaccination	0.0315**	(0.0158)	0.047	193,253	0.763
DPT vaccination	0.0282**	(0.0140)	0.045	191,385	0.690
BCG	0.0284*	(0.0148)	0.056	192,832	0.713
Measles	0.0282**	(0.0142)	0.047	191,008	0.437
Full immunization	0.0085	(0.0179)	0.636	186,286	0.357
<i>Panel B: Secondary Outcomes</i>					
Any ANC	−0.0061	(0.0215)	0.775	256,830	0.673
Iron-folic acid	0.0145	(0.0138)	0.295	256,838	0.623
Tetanus injection	−0.0180	(0.0227)	0.428	256,844	0.799
Public delivery	−0.0097	(0.0083)	0.241	256,894	0.195
Clusters (districts)			379		
Within R^2 range			[0.022, 0.163]		
Controls			Yes		
District FE			Yes		
State × Time FE			Yes		

Notes: Binary treatment = above-median sterilization intensity. The coefficient measures the difference in outcomes between districts in the upper and lower halves of the c_{int} distribution, *not* a per-SD marginal effect; the continuous specification in Table 3 gives the per-SD effect (1 SD $\approx$ 5.3 units of raw c_{int}). The two coefficient magnitudes therefore answer different questions, and their divergence is informative about non-linearity in the dose-response (see discussion in Section 5.2). Dep. var. mean is the DLHS-2 baseline. Standard errors clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 and Figure 3 present the main results using the standardised treatment variable c_{int}^{std} , so that each coefficient represents the effect of a one-standard-deviation increase in sterilization intensity (approximately 5.3 units of raw c_{int} , or roughly the interquartile range). The table reports three specifications of increasing stringency: column (1) includes no controls or fixed effects, column (2) adds individual controls and district fixed effects, and column (3) further adds state $\times$ time fixed effects. The state $\times$ time fixed effects absorb all state-level time-varying confounders including the fact that high-sterilization states were also the primary NRHM target states, so identification comes purely from within-state, across-district variation in sterilization intensity. Columns (1)–(2) are reported for transparency about how much of the variation each layer of fixed effects absorbs, and the binary above-median treatment in Table 4 together with the log, quintile, archival-target, and Pelras-Renk per-capita-excess alternatives reported in the robustness section serve as treatment-construction robustness rather than competing primary specifications. This preferred specification yields small and statistically insignificant coefficients for most outcomes. The coefficient on $c_{int}^{std} \times Post$ for institutional delivery is 0.002 (SE = 0.004); for any vaccination, 0.013 (SE = 0.009); for DPT vaccination, 0.008 (SE = 0.007). A one-SD increase in sterilization intensity is thus associated with at most a 0.2–1.3 percentage-point differential change in health outcomes in the short run: effects that are economically modest and statistically indistinguishable from zero. The only outcome with a statistically significant coefficient is tetanus injection, which is negative (−0.028, SE = 0.011), suggesting that the utilization deficit may have worsened for this particular outcome. These nulls are informative rather than a power artifact: at 80% power, the smallest detectable interaction ranges from 0.9 percentage points (public-facility delivery) and 1.2 (institutional delivery) to 2.5 (antenatal care), in every case below the differential gains of 1.8–3.2 points that emerge in the long-run event study, so the short-run design could have detected effects of the eventual magnitude had they been present.

The binary treatment specification (Table 4) yields more substantial point estimates. Above-median sterilization districts gained 1.6 percentage points more in institutional delivery ($p = 0.061$), 3.2 pp in any vaccination ($p = 0.047$), 2.8 pp in DPT vaccination ($p = 0.045$), 2.8 pp in BCG ($p = 0.056$), and 2.8 pp in measles ($p = 0.047$). These effects are economically meaningful, representing approximately 4–8% of baseline means. However, most are significant only at the 5–10% level, and the pattern does not survive consistently across specifications and outcomes.

The divergence between continuous and binary specifications is informative. It suggests either that the relationship between sterilization intensity and trust recovery is nonlinear, with what matters being high versus low exposure rather than the exact dose, or that the continuous specification lacks power to detect modest effects against the substantial within-district variation in outcomes.

The predominant finding is a null result: in the short run (two to three years post-NRHM), high-sterilization districts did not experience significantly larger gains in health-seeking behaviour. The Emergency-era trauma created a deficit deep enough that even India’s largest health system reform could not eliminate it within a few years.

5.3 Event Study and Convergence

The long-run event study focuses on six outcomes that are consistently available and informative across all six survey rounds: institutional delivery, any vaccination, DPT vaccination, tetanus injection, public facility delivery, and any ANC. The remaining four outcomes from the short-run DiD (BCG, measles, full immunization, and iron-folic acid) are either subsumed by the broader vaccination measures or show limited cross-survey comparability in the long-run panel.

Table 5

Extended Event Study: Standardised Sterilization Intensity $\times$ Survey Round Interactions

	R1 (DLHS-1)	R2 (DLHS-2)	R3 (DLHS-3)	R5 (NFHS-4)	R6 (NFHS-5)
Institutional delivery	−0.008** (0.004)	ref.	0.002 (0.004)	0.009 (0.009)	0.018*** (0.007)
Any vaccination	0.012 (0.009)	ref.	0.013 (0.009)	0.020*** (0.007)	0.019* (0.010)
DPT vaccination	0.020** (0.010)	ref.	0.007 (0.007)	0.020** (0.008)	0.030** (0.013)
Tetanus injection	−0.015* (0.008)	ref.	−0.028*** (0.011)	0.007 (0.009)	0.008 (0.011)
Public delivery	−0.010 (0.007)	ref.	0.002 (0.003)	0.007 (0.015)	0.026*** (0.010)
Any ANC	−0.005 (0.009)	ref.	−0.010 (0.009)	0.011 (0.011)	0.032** (0.014)
Panel observations	1,201,053				
Clusters (districts)	379				
Controls	Yes				
District FE	Yes				
State $\times$ Time FE	Yes				

Notes: Coefficients on $c_{int}^{std} \times \mathbf{1}[\text{Round} = r]$ interactions, with DLHS-2 (R2) as the reference period. R4 (DLHS-4) omitted; see Section 5.4. Standard errors clustered at the district level in parentheses. Per-outcome estimation samples range from 286,970 (DPT) to 393,782 (institutional and public-facility delivery), smaller than the panel because maternal and child outcomes are recorded only for recent births. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

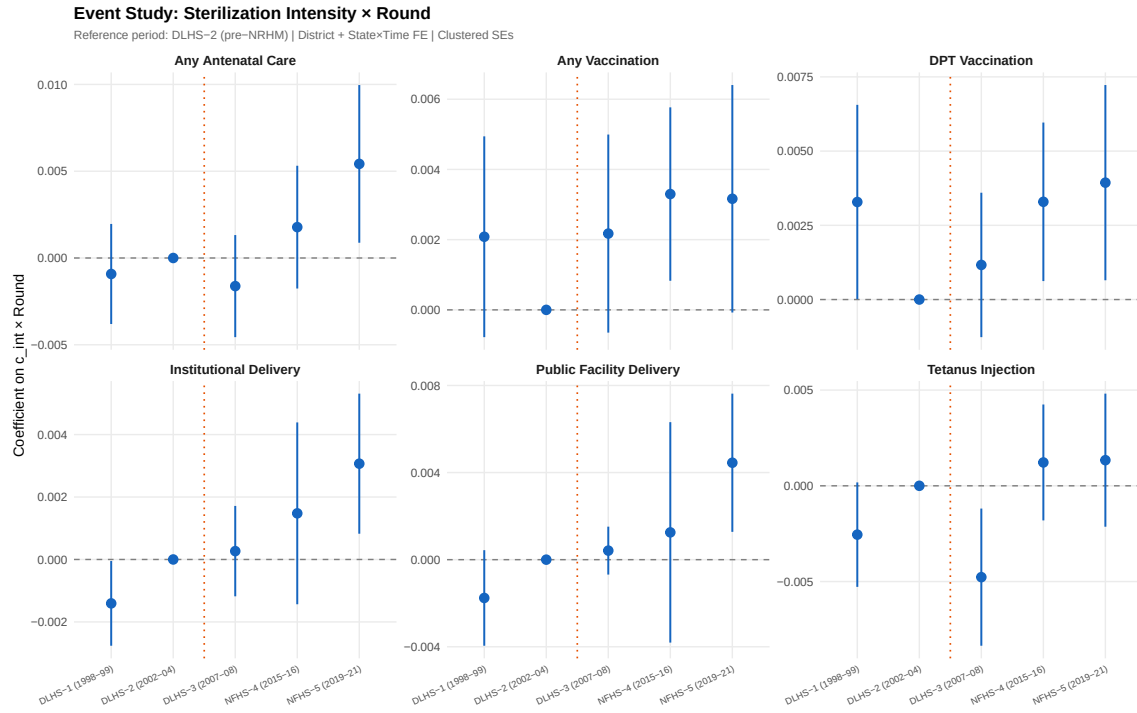


Figure 4

Long-Run Event Study: $c_{int}^{std} \times \text{Round}$ Coefficients Over Time

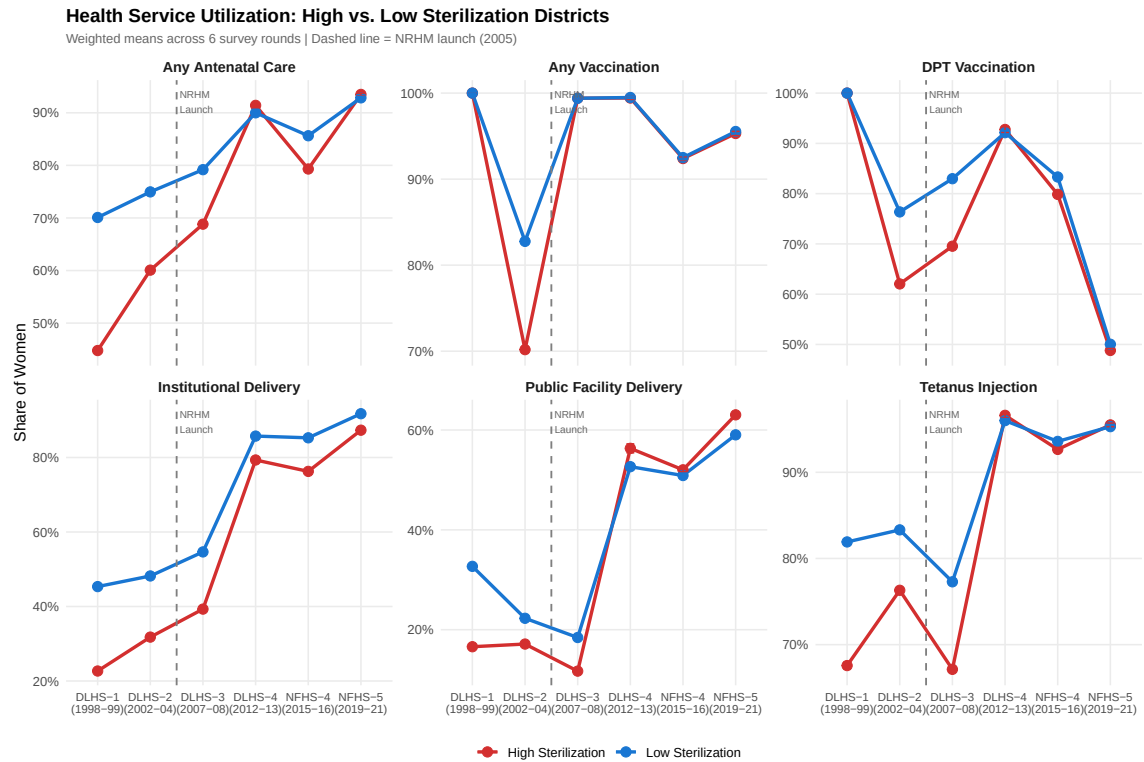


Figure 5

Convergence of Sterilization-Outcome Gradient Across Survey Rounds

The long-run event study provides a richer picture by tracing the sterilization-outcome gradient across all six survey rounds, with DLHS-2 as the reference period. The pooled estimation sample spans 403 distinct districts, the union of districts observed with non-missing c_{int} across rounds; any single round contains fewer (the per-round counts are reported in Table 2).

5.3.1 Pre-Trends

The DLHS-1 (round 1) coefficients provide a pre-trend test. For most outcomes, the pre-NRHM coefficients are close to zero and statistically insignificant, supporting the parallel trends assumption. The notable exception is institutional delivery, where the DLHS-1 coefficient is -0.008 ($p = 0.043$), suggesting a potential violation of parallel trends for this outcome. DPT vaccination shows a marginally significant positive pre-trend (0.020 , $p = 0.049$). These pre-trend estimates should be interpreted with caution: conventional pre-trend tests have limited power, and a non-significant pre-trend does not guarantee parallel trends.³⁷ I address this concern formally through a sensitivity analysis using the `HonestDiD` package, which constructs robust confidence sets under bounded violations of parallel trends using the relative magnitudes restriction ($\Delta^{RM}(\bar{M})$).³⁸ With one pre-treatment period (DLHS-1), the sensitivity parameter $\bar{M}$ bounds the ratio of the post-treatment trend violation to the maximal pre-treatment violation. At $\bar{M} = 0$ (parallel trends hold exactly), the robust confidence intervals include zero for all convergence outcomes: DPT vaccination ($[-0.007, 0.021]$), any vaccination ($[-0.004, 0.029]$), institutional delivery ($[-0.007, 0.010]$), public delivery ($[-0.004, 0.009]$), and any ANC ($[-0.026, 0.008]$). While none formally exclude zero, the DPT and any vaccination intervals are tight and concentrated in the positive range, consistent with the convergence finding. As $\bar{M}$ increases, the intervals widen gracefully: at $\bar{M} = 1$ (post-treatment violations up to the magnitude of the pre-treatment trend), the DPT interval is $[-0.030, 0.049]$ and the any vaccination interval is $[-0.005, 0.053]$. Notably, the tetanus anomaly is confirmed by the sensitivity analysis: the DLHS-3 coefficient’s negative sign is robust up to $\bar{M} = 0.5$ ($[-0.056, -0.002]$, excluding zero). The single pre-period limits the power of this analysis, the $\bar{M} = 0$ benchmark assumes exact parallel trends and therefore does not independently address the marginal DPT pre-trend concern but the relative magnitudes intervals provide informative bounds at all values of $\bar{M}$. Sensitivity plots for all outcomes are reported in Appendix Figures A12–A17.

5.3.2 Post-NRHM Trajectory

The post-NRHM coefficients reveal a pattern of gradual convergence that becomes broad-based and statistically significant over time. Table 6 traces each outcome across the three post-reform rounds; every coefficient is the differential change associated with a one-standard-deviation increase in sterilization intensity, relative to DLHS-2.

Three features of the trajectory stand out. Convergence is absent in DLHS-3 for every outcome, consistent with the short-run results. It emerges first for vaccination, with DPT and any

³⁷See Roth (2022).

³⁸Following Rambachan and Roth (2023).

Table 6

Post-NRHM Trajectory of the Convergence Coefficients

Outcome	DLHS-3 (2007–08)	NFHS-4 (2015–16)	NFHS-5 (2019–21)
Institutional delivery	0.002 (0.718)	0.009 (0.321)	0.018 (0.008)
Public facility delivery	0.002 (0.460)	0.007 (0.628)	0.026 (0.006)
Any antenatal care	−0.010 (0.281)	0.011 (0.325)	0.032 (0.020)
DPT vaccination ^a	0.007 (0.343)	0.020 (0.014)	0.030 (0.020)
Any vaccination ^b	0.013 (0.132)	0.020 (0.009)	0.019 (0.056)
Tetanus injection	−0.028 (0.009)	0.007 (0.430)	0.008 (0.451)

Notes: Coefficients on $c_{int}^{std} \times \text{Round}$ from the long-run event study (DLHS-2 reference), with district and state $\times$ round fixed effects, individual controls, and survey weights; p -values in parentheses, standard errors clustered at the district level. Estimation samples are those of Table 5: the panel contains approximately 1.2 million woman-round observations, of which between 286,970 (DPT) and 393,782 (institutional and public-facility delivery) enter each outcome’s regression, since maternal and child outcomes are recorded only for recent births. The full event study, including the DLHS-1 pre-period, is reported in Table 5. ^aDPT is constructed on the corrected under-three denominator described in Section 3.2; on the uncorrected denominator the NFHS-5 coefficient is 0.023. ^bAny vaccination has a near-ceiling mean of 0.994 in DLHS-3, limiting its informativeness; DPT is the more discriminating vaccination outcome.

vaccination significant by NFHS-4, and last for antenatal care, which reaches a substantial 3.2 percentage-point differential only by NFHS-5. And the negative tetanus anomaly from the short-run window fully dissipates.

The NFHS-5 results are particularly important because they reveal that trust recovery is not limited to vaccination outcomes (which showed early convergence) but extends to institutional delivery, public facility delivery, and antenatal care by 2019–21. The magnitudes are economically meaningful: a one-SD increase in sterilization intensity is associated with 1.8–3.2 percentage-point differential gains across these outcomes, convergence large enough to substantially close the pre-NRHM utilization deficit. In concrete terms, a one-SD increase in $c_{int,d}^{std}$ compares a low-coercion district at roughly the 25th percentile of the treatment distribution ($c_{int} \approx 1.5$, characteristic of southern states such as Tamil Nadu or Kerala) to a high-coercion district at roughly the 75th percentile ($c_{int} \approx 6.8$, characteristic of northern Hindi-belt states such as Uttar Pradesh or Madhya Pradesh); Table 7 makes this comparison explicit by placing the pre-NRHM deficit per standard deviation from equation (2), estimated in DLHS-1 and DLHS-2, next to the NFHS-5 convergence coefficient and reporting the ratio. Against the earliest measured deficit (DLHS-1, 1998–99), the convergence amounts to essentially full closure for institutional delivery (1.8 percentage points against a 1.9 point deficit) and 1.4 times the deficit for antenatal care; against the smaller DLHS-2 gradients, which had already eroded between 1998 and 2002, the outcomes with measured deficits show full reversal. Ratios above one are best read as evidence that the cross-sectional gradients understate the historical deficit, both because measurable erosion predates NRHM and because the Berkson-type measurement error described in Section 3.1 attenuates the cross-sectional coefficients, rather than as literal overshooting. For public-facility delivery no closure share is defined: its pre-NRHM gradient is statistically zero in both pre-reform rounds, so the strong NFHS-5 coefficient records the emergence of a differential shift toward public providers in high-intensity districts rather than

the recovery of a measured scar. That pattern remains consistent with the trust interpretation (women in scarred districts increasingly choose government facilities), but it is substitution toward the public sector, and I label it as such here and in the mechanisms discussion. This broad-based convergence (four of six outcomes significant at the 5% level, with a fifth (any vaccination, $p = 0.056$) marginally significant) provides strong evidence that the utilization deficit has been substantially eliminated over the fifteen years following NRHM.

Table 7
The Pre-NRHM Deficit and the Share Closed by NFHS-5

Outcome	Pre-NRHM deficit per SD (eq. 2)		NFHS-5 convergence (event study, ref. DLHS-2)	Share of DLHS-2 deficit closed	N (DLHS-1; DLHS-2)
	DLHS-1	DLHS-2			
Institutional delivery	-0.0187*** (0.0047)	-0.0094* (0.0054)	0.0182*** (0.0068)	1.93 (full reversal)	103,844; 124,482
DPT vaccination	—	-0.0186** (0.0076)	0.0296** (0.0126)	1.59 (full reversal)	—; 121,911
Any antenatal care	-0.0236*** (0.0091)	-0.0164* (0.0090)	0.0321** (0.0137)	1.96 (full reversal)	104,772; 124,420
Public facility delivery	-0.0110 (0.0068)	-0.0036 (0.0035)	0.0264*** (0.0096)	—	103,844; 124,482
Tetanus injection	-0.0144** (0.0067)	0.0006 (0.0077)	0.0079 (0.0105)	—	104,542; 124,433
Any vaccination	—	-0.0126* (0.0068)	0.0188* (0.0098)	1.49 (full reversal)	—; 121,901

Notes: Columns 1–2 report the coefficient on c_{int}^{std} from the per-round cross-sectional persistence regression (eq. 2) with state fixed effects and individual controls, estimated on the balanced 379-district sample; standard errors clustered at the district level in parentheses. Column 3 reports the NFHS-5 coefficient from the long-run event study (DLHS-2 reference). The share-closed column reports the ratio of the NFHS-5 convergence coefficient to the absolute DLHS-2 deficit; the ratio is reported only where the DLHS-2 gradient is negative and statistically distinguishable from zero at the 10% level, since a closure share is undefined against a deficit that cannot be distinguished from zero (public-facility delivery and tetanus). A value above one indicates full reversal of the pre-NRHM gradient. The rightmost column reports the estimation samples of the two cross-sections; the event-study estimation sample is that of Table 5. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

These conventional p -values should be read together with the sensitivity analysis of Section 5.3.1, which reports that at $\bar{M} = 0$ the HonestDiD robust confidence sets include zero for all convergence outcomes. The two are answering different questions: the conventional p -value treats parallel trends as an assumption and asks whether the NFHS-5 coefficient differs from zero given it, while the robust interval additionally propagates the sampling uncertainty of the single estimated pre-period coefficient into the bound, which with only one usable pre-round (DLHS-1) is substantial. On balance, the convergence estimates are individually precise conditional on parallel trends, but that with one pre-period the design cannot tightly bound trend violations; the placebo-timing evidence (Section 5.8) flags institutional delivery, tetanus, and public delivery as the outcomes where pre-NRHM differential trends are detectable, and conclusions for those outcomes lean correspondingly harder on the additional checks reported in Sections 5.7 and 5.8.

Figure 4 and Figure 5 complement the event study by directly showing how the gap between high and low sterilization districts narrows across survey rounds. By NFHS-5, the sterilization-outcome gradient has substantially compressed for all outcomes, with some indicators (public delivery, tetanus, ANC) showing closure in gap or even reversal.

5.3.3 The Speed of Recovery

The introduction framed the speed of trust recovery as an economically important parameter that the literature has not previously estimated. To move beyond verbal characterization, I fit a saturating recovery path $\gamma(\tau) = \gamma_{\infty} (1 - e^{-\lambda\tau})$ to the round-specific event-study coefficients by weighted nonlinear least squares, with inverse-variance weights and τ measured in years since the April 2005 launch (survey midpoints at 2.7, 10.5, and 15.0 years). With three post-reform

points this is a demanding exercise, and its results are as informative about identification as about magnitude. For any vaccination, whose coefficients plateau early, the fit is tight: $\hat{\lambda} = 0.41$ (SE 0.05), implying a recovery half-life of 1.7 years (95% CI 1.3–2.1). DPT vaccination yields a loosely bounded half-life of roughly three decades. For institutional delivery, public-facility delivery, and antenatal care, however, the coefficients are still rising at the final observation, so the data cannot separate the asymptote γ_{∞} from the rate λ , and the half-life is not identified: any sufficiently slow path fits. For these outcomes the unconstrained fit delivers a lower bound on the recovery horizon rather than a half-life. The asymptote can, however, be anchored externally: imposing γ_{∞} equal to the measured pre-NRHM deficit from Table 7 identifies the rate from the observed trajectory alone. On that basis the implied half-life of recovery is 7.9 years for institutional delivery (95% CI 1.2–14.6), roughly 10 years for antenatal care, and 2.8 years for DPT vaccination; the equivalent fit in cumulative-contact time yields a per-contact updating rate of $\kappa \approx 1.5$, consistent with the calibration in Appendix J. The anchored fit is unavailable for public-facility delivery, which has no measured pre-reform deficit, and any vaccination’s convergence exceeds its measured deficit, consistent with the closure ratios in Table 7. Taken together, the estimates place the half-life of recovery at roughly eight to ten years for the delivery and antenatal outcomes and under three years for the vaccination outcomes, with the process still under way at the 2019–21 endline, well beyond the two-to-five-year windows over which community-engagement interventions have typically been evaluated.³⁹ The still-accelerating calendar-time paths are not evidence against a saturating trust mechanism: Appendix J shows that a minimal updating model in which recovery is concave in *cumulative* ASHA contacts generates exactly this acceleration when the contact rate itself rises over the period, as it did (ASHA involvement per recent birth grew from 2–5% in DLHS-3 to 19% in NFHS-5).

5.3.4 Within-Survey Mother Panel

The within-DLHS-3 mother-child panel provides the most demanding test of the NRHM effect. By comparing outcomes for children born to the *same mother* before and after the NRHM cutoff, the mother fixed effects specification absorbs all time-invariant maternal characteristics including inherited distrust, household wealth, education, and caste. The identifying variation comes exclusively from the interaction of the within-mother pre-post comparison with across-district variation in sterilization intensity.

The mother-panel sample comprises mothers in the DLHS-3 birth history who had at least one child born before April 2005 and at least one after. This is a selected sample: these mothers are younger, have shorter birth spacing, and are higher-parity than the full DLHS-3 cross-section. The panel mothers are approximately two years younger and 3 percentage points more likely to be rural than the full cross-section, but have comparable sterilization intensity exposure.

The mother-panel sample comprises 21,698 panel mothers with 43,430 child-level observations (about two children per mother; the per-outcome regression samples reported in Table 8 are smaller owing to item missingness). Table 8 reports the mother-FE results. Within mothers,

³⁹Christensen et al. (2021) measure the trust effects of community monitoring in Sierra Leone within roughly two years of the intervention’s rollout.

the coefficient on $c_{int}^{std} \times Post$ is small and statistically insignificant for all four vaccination outcomes: DPT vaccination (-0.010 , $SE = 0.006$, $p = 0.077$), any vaccination (-0.003 , $SE = 0.007$, $p = 0.647$), BCG (0.001 , $SE = 0.005$, $p = 0.913$), and measles (-0.001 , $SE = 0.011$, $p = 0.936$). The null within-mother results are consistent with the cross-sectional short-run DID: within the DLHS-3 survey window (2007–08), there had been insufficient time for convergence to emerge. These estimates are arguably the most demanding test in the paper because they rule out all time-invariant confounders, including the possibility that the cross-sectional convergence reflects compositional changes in the sampled population rather than behavioral changes within families. Note that delivery outcomes are unavailable for the panel analysis because DLHS-3 records delivery place only for the most recent birth, precluding within-mother comparison.⁴⁰

Table 8

Within-Mother Panel: Mother Fixed Effects Estimates (DLHS-3)

Outcome	$c_{int}^{std} \times Post$	SE	p -value	N
DPT vaccination	-0.0103^*	(0.0058)	0.077	23,697
Any vaccination	-0.0032	(0.0071)	0.647	39,349
BCG	0.0006	(0.0051)	0.913	24,290
Measles	-0.0009	(0.0107)	0.936	23,594
Panel mothers		21,698		
Mother FE		Yes		
Clustering		District		

Notes: Mother fixed effects, weighted, standard errors clustered at district level. Panel comprises mothers with births both before and after April 2005. Delivery outcomes unavailable because DLHS-3 records delivery place only for the most recent child. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 DLHS-4 Trust Analysis

The DLHS-4 (2012–13) cross-sectional analysis provides direct evidence on whether institutional distrust persists seven years after NRHM. Table 9 and Figure 6 present the results across all tiers.

Tier 1: Direct health trust. The coefficient of sterilization intensity on stated distrust of government health facilities is -0.007 ($p = 0.060$), marginally significant and *negative*: higher sterilization intensity is associated with *less* stated distrust, not more. This reversal from what the persistence hypothesis would predict is itself suggestive of trust recovery: by 2012–13, high-sterilization districts may have experienced sufficient NRHM-driven trust rebuilding to generate a weakly negative gradient. The use of government versus private facilities shows no systematic relationship with historical sterilization intensity. Together, these results suggest that by the time of DLHS-4, the sterilization-linked gradient in stated trust had largely flattened; as a cross-district differential, this says nothing about absolute levels of trust or utilization.

Tier 2: ASHA engagement. ASHA-motivated antenatal care, delivery, immunization, and

⁴⁰The same limitation rules out building a delivery-based within-mother panel from the later rounds: DLHS-4 and NFHS-4 likewise record place of delivery only for each woman’s most recent birth, and only for births within the survey’s recall window, so a mother’s earlier births carry no delivery-location information even where the birth roster lists them.

transport arrangements do not vary systematically with sterilization intensity. This finding indicates that ASHA deployment and engagement were uniform across the sterilization distribution: ASHAs were not differentially deployed to or more/less effective in high-sterilization districts. The ASHA programme operated as a universal intervention, not one targeted at historically distrustful communities.

Tier 3: Government scheme participation. There is weak evidence of marginally lower government scheme participation in high-sterilization districts, though most coefficients are statistically insignificant. This residual gap may reflect slower uptake of new government programs in communities with historically negative state-citizen relationships.

Tier 4: Placebo tests. Public versus private facility delivery patterns in DLHS-4 are broadly consistent with trust recovery: there is no strong differential preference for private over public facilities in high-sterilization districts.

Table 9

DLHS-4 Trust Analysis: Cross-Sectional Regressions on Sterilization Intensity

Outcome	Coefficient	SE	<i>p</i> -value	<i>N</i>
<i>Tier 1: Direct Health Trust</i>				
No trust in govt facility	−0.0064*	(0.0034)	0.060	190,028
Use govt facility	−0.0142	(0.0149)	0.344	186,421
Use private facility	0.0142	(0.0149)	0.344	186,421
Poor quality perception	0.0016	(0.0057)	0.774	190,028
<i>Tier 2: ASHA Engagement</i>				
ASHA motivated ANC	−0.0018	(0.0017)	0.299	190,029
ASHA motivated delivery	−0.0017	(0.0018)	0.361	190,020
ASHA arranged transport	−0.0009	(0.0007)	0.218	190,027
ASHA motivated immunization	−0.0017	(0.0030)	0.572	190,013
JSY received	−0.0091	(0.0147)	0.538	12,382
<i>Tier 3: Government Schemes</i>				
Pregnancy registered	−0.0122	(0.0127)	0.336	49,855
Health scheme coverage	−0.0169**	(0.0079)	0.033	187,572
RSBY coverage	−0.0096	(0.0068)	0.162	190,015
<i>Tier 4: Placebo (Public vs. Private)</i>				
Public facility delivery	−0.0089	(0.0106)	0.405	49,572
Private facility delivery	0.0014	(0.0136)	0.918	49,572
ANC at govt facility	−0.0182	(0.0113)	0.109	44,452
ANC at private facility	0.0151	(0.0119)	0.204	44,452
Clusters (districts)		170		
Controls		Yes		
State FE		Yes		

Notes: Cross-sectional regressions on c_{int}^{std} with individual controls and state FE. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

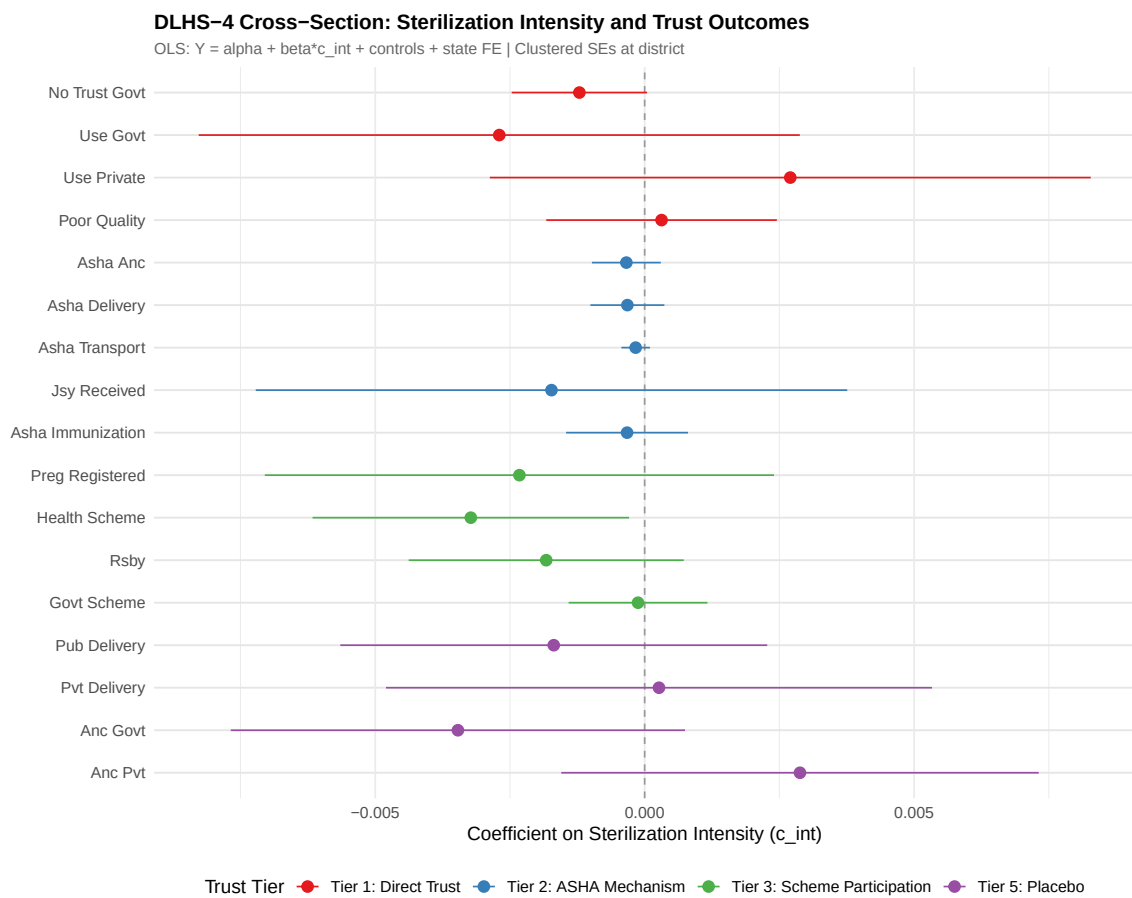


Figure 6
 DLHS-4 Trust Analysis: Cross-Sectional Coefficients on Sterilization Intensity

5.5 Binscatter Evidence

District-level binned scatter plots (Figures 7 and 8) visualize the relationship between sterilization intensity and outcome changes. For the DLHS-2 to DLHS-3 comparison, the plots show relatively flat or weakly positive relationships, consistent with the null short-run DiD results. Multi-round binscatters extending through DLHS-4, NFHS-4, and NFHS-5 show a progressively clearer positive gradient emerging over time across all outcomes. Additional binscatter and convergence figures for ANC, public delivery, and other outcomes are reported in Appendix D (binscatters in Figures A1, A2, A6, and A7; convergence and related figures in Figures A3–A11).

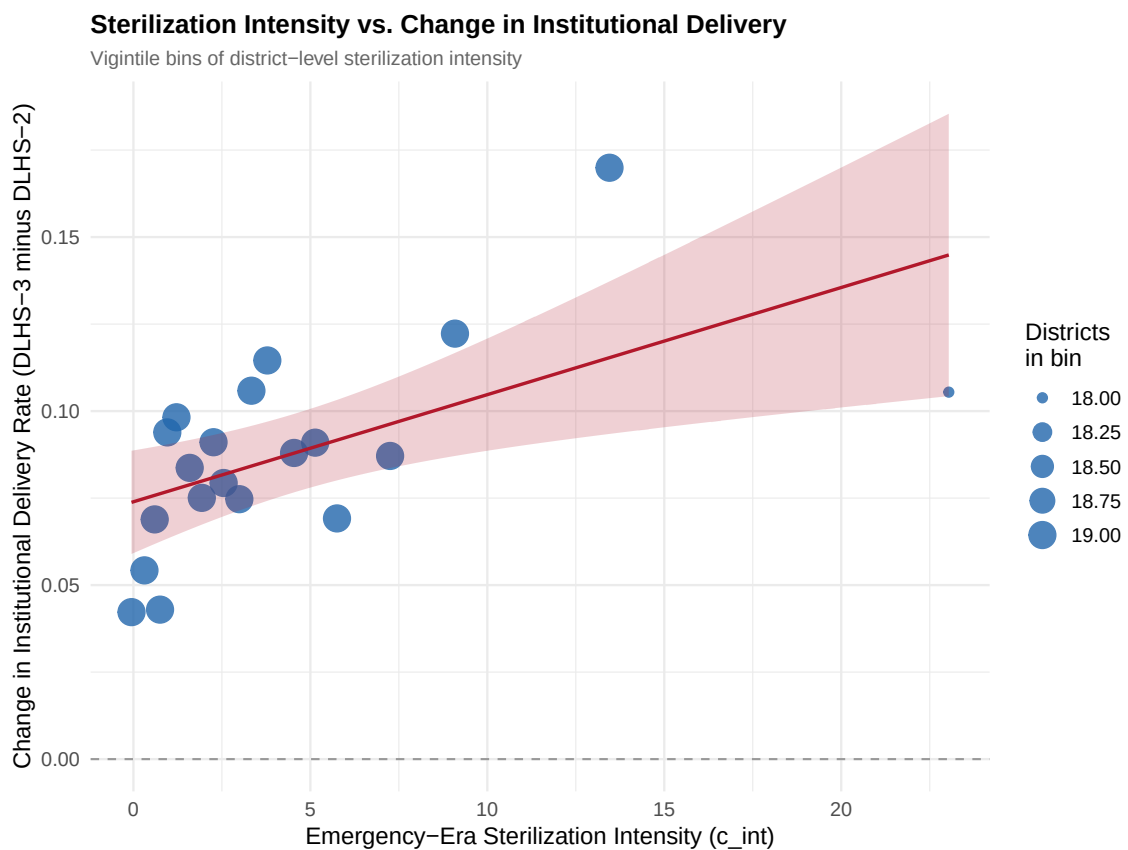


Figure 7

Binscatter: Sterilization Intensity vs. Change in Institutional Delivery

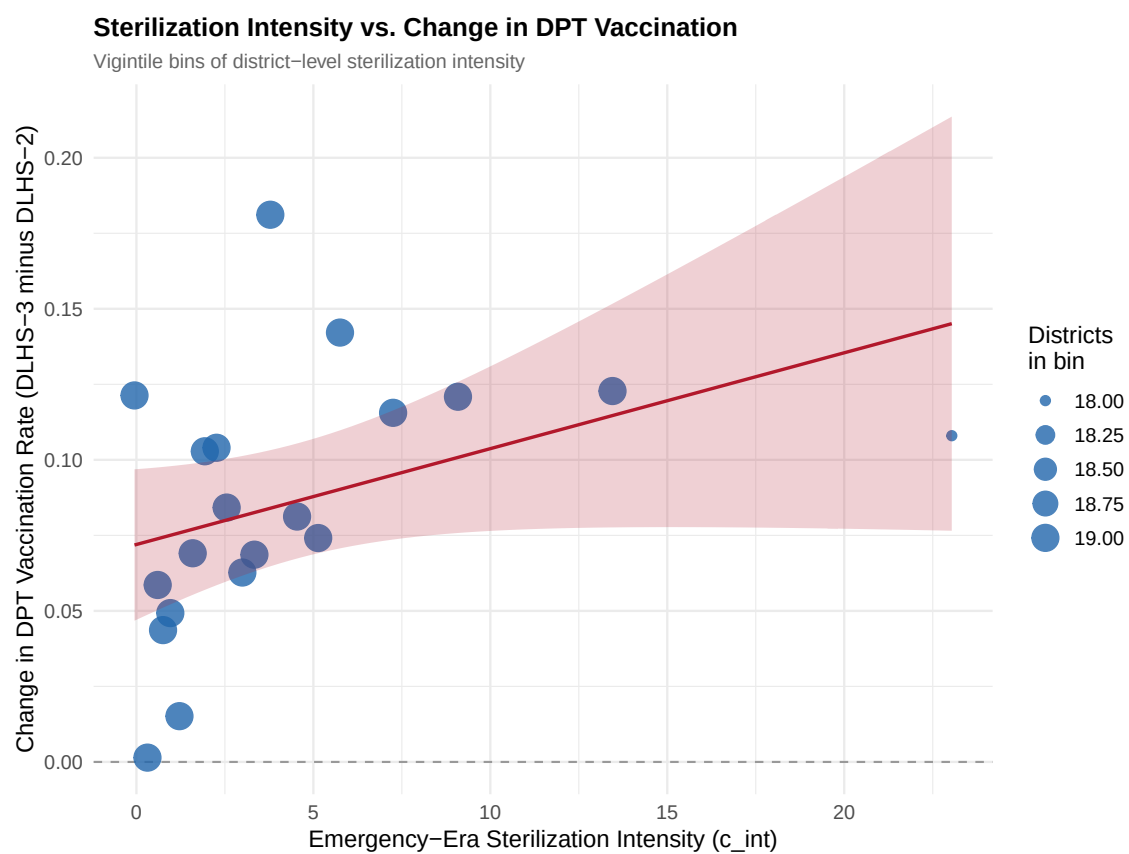


Figure 8
 Binscatter: Sterilization Intensity vs. Change in DPT Vaccination

5.6 Unbundling the NRHM into its Components

The triple-DID specification within DLHS-3 tests how the JSY component of NRHM interacted with the sterilization legacy. Table 10 reports the triple-DID results from the full specification (equation (11)). The $\text{Elig} \times \text{Post}$ coefficient replicates the established finding that JSY eligibility significantly increases institutional delivery ($\hat{\beta} = 0.053$, $p < 0.001$) and JSY receipt ($\hat{\beta} = 0.055$, $p < 0.001$). The key test is the triple interaction ($c_{int}^{std} \times \text{Eligible} \times \text{Post}$), which measures whether JSY’s eligibility-based effect is modulated by the historical utilization deficit.

Table 10 reports the triple-DID results. The triple interaction ($c_{int}^{std} \times \text{Eligible} \times \text{Post}$) is insignificant for institutional delivery (0.011, $p = 0.279$) and DPT vaccination (0.000, $p = 0.946$), suggesting that JSY’s eligibility-based incentive does not differentially benefit high-sterilization districts for these outcomes. However, the triple interaction is marginally significant for ANC (0.009, $p = 0.083$) and JSY receipt (0.014, $p = 0.065$), and for any vaccination the triple interaction is negative and marginally significant (-0.017 , $p = 0.050$). This mixed pattern is consistent with the interpretation that the financial incentive (JSY) alone does not overcome the trust barrier: a finding that motivates the $\text{Post}_1/\text{Post}_2$ decomposition below.

Table 10

JSY Unbundling: Triple-DID Estimates ($c_{int}^{std} \times \text{JSY_Eligible} \times \text{Post}$)

Outcome	Elig $\times$ Post	Triple	p (triple)	N
Institutional delivery	0.0531***	0.0107	0.279	138,178
Any vaccination	0.0137	-0.0172^*	0.050	136,965
DPT vaccination	-0.0134^{**}	0.0004	0.946	71,920
Any ANC	0.0109*	0.0094*	0.083	138,184
Tetanus	-0.0022	0.0081	0.279	101,869
JSY received	0.0550***	0.0141*	0.065	138,182
Clusters (districts)		397		
District FE		Yes		
Individual controls		Yes		

Notes: Sample: DLHS-3 births. Standard errors clustered at district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The $\text{Post}_1/\text{Post}_2$ decomposition reveals a striking temporal pattern. For institutional delivery, $c_{int}^{std} \times \text{Post}_1$ is small and insignificant (0.006, $p = 0.183$), while $c_{int}^{std} \times \text{Post}_2$ is large and highly significant (0.019, $p = 0.005$). The same pattern holds for ANC: Post_1 coefficient 0.004 ($p = 0.227$), Post_2 coefficient 0.011 ($p = 0.003$). For DPT vaccination, neither coefficient reaches significance (Post_1 : 0.003, $p = 0.437$; Post_2 : -0.004 , $p = 0.341$), consistent with the short time horizon of the within-DLHS-3 analysis. The concentration of the sterilization interaction in the Post_2 period, when ASHA deployment had reached operational scale but only the JSY cash transfer was available in Post_1 , is suggestive; the mechanistic implications of this timing are discussed in Section 6.

The continuous-coverage specification uses district-level JSY coverage as a continuous treatment. The endogeneity concern is well-documented: districts with better governance achieve higher JSY coverage and better health outcomes simultaneously. I estimate this specification as a descriptive exercise, noting whether the correlation between JSY coverage and outcomes varies

with sterilization intensity. The JSY coverage rate is strongly associated with institutional delivery (0.743, $p < 0.001$), DPT vaccination (0.398, $p < 0.001$), and ANC (0.413, $p < 0.001$). However, the interaction between JSY coverage and sterilization intensity is uniformly insignificant across all outcomes ($p > 0.25$), suggesting that the convergence does not operate primarily through differential JSY take-up.

5.7 Robustness

5.7.1 Alternative Treatment Measures

The main results are robust to monotone transformations of the proportional-surge treatment, the raw unstandardised difference, the log transformation $\log(1 + c_{int})$, and quintile dummies all yield qualitatively consistent patterns. The quintile dose-response for institutional delivery (Q2 = +0.024, Q3 = +0.042, Q4 = +0.045, Q5 = +0.037) is positive at every quintile but not linear: the increments shrink from +2.4 to +1.8 to +0.3 percentage points and turn slightly negative between the top two quintiles, a concave, saturating profile. The sign of the effect is therefore stable across the dose distribution, but the flattening at high intensities means the linear coefficient should be read as a convenient scalar summary of a saturating dose-response rather than as a constant per-unit effect, consistent with the discussion in Section 4.4.⁴¹ The convergence is, however, specific to this proportional-surge construction of coercion intensity: constructions that instead capture administrative over-achievement relative to assigned targets, the state-level over-target ratio and the per-couple excess, are near-orthogonal to the proportional surge and do not reproduce the long-run convergence. This measure-sensitivity, and the archival case for why the proportional surge is the trust-relevant construct, are reported in Appendix Table A9.⁴²

5.7.2 Sample and Specification Robustness

Results are robust to a wide range of sample and specification perturbations (see also Figure A10 in the Appendix): trimming the top and bottom 5% of the sterilization distribution (which strengthens the results, suggesting that outlier districts introduce noise); excluding small states and union territories; restricting to the rural subsample (point estimates similar but less precise); using unweighted regressions (virtually identical to weighted); and employing state-level clustering (more conservative, with some coefficients retaining significance). Additionally I perform a collapsed district-level specification confirms that aggregating to district means preserves the qualitative pattern, ruling out the concern that individual-level estimation artificially inflates precision.

Because the long-run panel splices survey families with different sampling frames and ques-

⁴¹Positive effects at every dose level support interpreting the continuous-treatment difference-in-differences coefficient as a weighted average of dose-specific effects; equal increments across quintiles would be required for linearity itself. See Callaway, Goodman-Bacon, and Sant’Anna (2024).

⁴²The proportional-surge baseline follows Singh and Vincent (2024); the over-target construction is the preferred measure of Pelras and Renk (2023). The two are near-orthogonal in the estimation sample.

tionnaires, I also estimate a same-instrument difference-in-differences using only NFHS-4 and NFHS-5, whose designs are closely comparable, on the 274 districts observed in both rounds. The coefficient on $c_{int}^{std} \times \text{NFHS-5}$, which measures the *additional* convergence accruing between 2015–16 and 2019–21, is positive but insignificant for institutional delivery (0.002, $p = 0.82$), public-facility delivery (0.008, $p = 0.51$), and DPT (0.010, $p = 0.23$), and significant for antenatal care (0.020, $p = 0.029$). This is the expected pattern given the event-study trajectory: most of the convergence for delivery and vaccination outcomes was already in place by NFHS-4, so the within-family increment is small, while ANC, the slowest-converging outcome, is still gaining significantly between the two NFHS rounds. The same-instrument contrast confirms continued, or at minimum sustained, convergence inside a single survey family, where cross-survey measurement artifacts cannot operate; what it cannot by itself certify is the DLHS-to-NFHS level jump, for which the within-family NFHS-4 versus NFHS-5 comparison, the harmonized questionnaire items, and the corrected DPT denominator (Section 3.2) are the available safeguards.

5.7.3 Mean Reversion and the Pre-Surge Base

Because c_{int} divides the 1976–77 surge by the district’s 1975–76 sterilization base, districts with a small pre-period family-planning footprint receive mechanically high treatment values and also had the most room to improve as national utilization rose from a low base after 1998. If low-base districts caught up for reasons unrelated to the Emergency, the event-study coefficients could partly reflect mean reversion rather than recovery. Table 11 probes this concern directly.

Panel A subjects the baseline measure to controls for its own denominator. Adding linear interactions of the 1975–76 base with round indicators (column 2) attenuates the NFHS-5 coefficients by 20–50% and removes conventional significance; allowing each base quintile its own free trajectory in every round (column 3) drives all outcomes to insignificance. Two features of the design imply that these specifications overcorrect rather than merely adjust. First, the treatment is constructed from the base and the two are strongly negatively correlated (-0.45), in part through simple division bias: measurement error in the single-year denominator generates high measured surges precisely where the recorded base is low. Second, the within-quintile standard deviation of the standardized treatment collapses from 1.35 in the lowest base quintile to 0.13 in the highest, so column (3) discards most of the identifying variation rather than holding baseline trajectories fixed.

A sharper test rebuilds the surge on the 1974–75 base, the last fully pre-Emergency fiscal year: because fiscal year 1975–76 runs from April 1975 to March 1976, it already contains roughly nine months of Emergency-era mobilization, so the original denominator is itself contaminated by the early campaign. The rebuilt measure has a rank correlation of 0.74 with the baseline measure but, crucially, is nearly uncorrelated with its own denominator (-0.08), so base-trajectory controls no longer mechanically strip out treatment variation. Panel B shows that on this cleaner measure the NFHS-5 convergence is as strong as or stronger than the baseline estimates for every outcome (public-facility delivery 4.3 percentage points per standard deviation, antenatal care 3.9, institutional delivery 2.7, DPT 2.7, any vaccination 2.3, and tetanus 2.0, all $p \leq 0.017$), consistent with early-campaign contamination of the 1975–76 denominator attenuating

the original measure. With linear own-base $\times$ round controls, antenatal care, public-facility delivery, and any vaccination remain significant at the 5% level, DPT and tetanus at 6%, while institutional delivery attenuates to $p = 0.10$. Under the most demanding quintile-by-round specification the coefficients remain positive throughout but lose significance, with public-facility delivery closest to conventional levels ($p = 0.063$).

A complementary check targets utilization catch-up directly by controlling for the district’s pre-NRHM (DLHS-2) outcome level interacted with round, an adjustment that is itself biased toward attenuation when baseline district means are estimated with sampling noise.⁴³ Under this adjustment public-facility delivery survives essentially intact (2.1 percentage points, $p = 0.016$), institutional delivery attenuates to zero, and the remaining outcomes fall to marginal significance. Taken together, the convergence is not an artifact of the extreme-ratio construction, and it is robust on the cleaner pre-Emergency base; within the outcome set, public-facility delivery, the outcome most directly tied to trust in government provision, is the most robust to every baseline-trajectory adjustment, while the institutional-delivery estimate is the most sensitive and should be read with this caveat in mind.

Table 11

Long-Run Convergence and Flexible Controls for the Pre-Surge Sterilization Base

	(1) Baseline		(2) + Linear base $\times$ round		(3) + Base quintile $\times$ round FE		
Outcome	Coef.	(SE)	Coef.	(SE)	Coef.	(SE)	N
<i>Panel A: baseline measure c_{int} (1975–76 base); base controls use the 1975–76 level</i>							
Institutional delivery	0.0182***	(0.0068)	0.0117	(0.0080)	0.0073	(0.0070)	393,782
DPT vaccination	0.0296**	(0.0126)	0.0239	(0.0162)	0.0168	(0.0152)	286,970
Any antenatal care	0.0321**	(0.0137)	0.0279	(0.0170)	0.0088	(0.0187)	385,875
Public facility delivery	0.0264***	(0.0096)	0.0136	(0.0111)	0.0111	(0.0105)	393,782
Tetanus injection	0.0079	(0.0105)	0.0049	(0.0125)	0.0009	(0.0123)	385,725
Any vaccination	0.0188*	(0.0098)	0.0145	(0.0119)	0.0088	(0.0117)	296,910
<i>Panel B: surge rebuilt on the 1974–75 base; base controls use the 1974–75 level</i>							
Institutional delivery	0.0268***	(0.0094)	0.0164	(0.0100)	0.0116	(0.0113)	393,782
DPT vaccination	0.0269***	(0.0096)	0.0197*	(0.0103)	0.0115	(0.0108)	286,970
Any antenatal care	0.0389***	(0.0135)	0.0348**	(0.0149)	0.0118	(0.0157)	385,875
Public facility delivery	0.0431***	(0.0107)	0.0323***	(0.0121)	0.0269*	(0.0145)	393,782
Tetanus injection	0.0199**	(0.0083)	0.0168*	(0.0088)	0.0086	(0.0094)	385,725
Any vaccination	0.0230***	(0.0077)	0.0194**	(0.0082)	0.0115	(0.0093)	296,910

Notes: Each cell reports the NFHS-5 (round 6) coefficient on the standardized treatment $\times$ round interaction from the long-run event study (DLHS-2 reference), with standard errors clustered at the district level in parentheses. Column (1) is the baseline specification with district and state $\times$ round fixed effects and individual controls. Column (2) adds linear interactions of the district’s pre-surge sterilization base with round indicators. Column (3) instead adds fixed effects for quintile-of-base $\times$ round cells, allowing each fifth of the baseline distribution its own free trajectory in every round; quintiles are computed across balanced-sample districts. In Panel A the treatment is the baseline proportional surge over the 1975–76 base ($\text{corr}(c_{int}, S_{75-76}) = -0.45$); in Panel B the surge is rebuilt over the genuinely pre-Emergency 1974–75 base, which is nearly uncorrelated with its own denominator ($\text{corr} = -0.08$), so columns (2)–(3) remove far less treatment variation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.7.4 Archival Treatment Measures

The target revision ratio and target overshoot (constructed from original Ministry of Health and Family Welfare archival records) provide additional treatment variation for a subset of districts. The revision ratio captures the degree to which the central government ratcheted up

⁴³Conditioning a difference-in-differences comparison on noisily measured baseline outcome levels induces spurious attenuation through regression to the mean (Daw and Hatfield 2018).

sterilization pressure; the overshoot captures the extent to which districts exceeded their quotas. The incentive state heterogeneity analysis (Appendix Table A5) shows that the convergence effect is uniform across states that used positive incentives during the Emergency and those that relied on coercion: the triple interaction is null for five of six outcomes, suggesting that it was the sterilization campaign itself, not the implementation modality, that damaged trust.

5.7.5 Supply-Side Tests

If convergence were driven by supply-side improvements rather than demand-side trust recovery, I would expect differential changes in delivery location that extend beyond public facilities. Table A3 (Appendix, Panel A) reports DID coefficients on delivery location. Public facility delivery shows a small positive but insignificant coefficient (0.003, $p = 0.387$); private facility delivery shows a significant *decrease* (-0.003 , $p = 0.011$); and home delivery shows a significant decrease (-0.015 , $p = 0.043$). The decline in home delivery combined with no significant gain in private delivery is consistent with women shifting from home to public facilities, a demand-side behavioral change, rather than a general expansion of facility capacity. Panel B confirms that ASHA deployment was not targeted to high-sterilization districts: contact rates for antenatal care, delivery, and immunization are all statistically indistinguishable from zero (all $p > 0.60$).

5.7.6 Spatial Correlation and Conley Standard Errors

High-sterilization districts are geographically clustered, particularly in northern India. If unobserved shocks are spatially correlated, district-level clustering may understate standard errors. Table A4 (Appendix) reports spatial HAC standard errors at three distance cutoffs (100 km, 200 km, 500 km) for six outcomes.⁴⁴ At 100 km, results are virtually identical to the baseline district-clustered standard errors. At 200 km and 500 km, standard errors increase modestly but do not qualitatively alter the main findings: the short-run coefficients remain insignificant, and the convergence pattern in later rounds is preserved.

Spatial standard errors address inference under spatially correlated shocks but not spillover bias in the point estimates, so I add two direct checks. First, excluding all northern Indian states (the Hindi belt plus the northwest, where high-intensity districts cluster) removes 204 of the 379 balanced-sample districts and cuts the retained standard deviation of c_{int}^{std} from 0.90 to 0.54, yet the NFHS-5 convergence survives on the remaining 175 southern and eastern districts: institutional delivery 0.021 ($p = 0.013$), public-facility delivery 0.035 ($p = 0.001$), antenatal care 0.019 ($p < 0.001$), and DPT 0.017 ($p = 0.066$). The convergence is therefore not an artifact of the northern high-intensity belt. Second, a donut test drops the quarter of districts whose centroid lies closest to a district of opposite treatment intensity (above versus below the median), retaining interior districts where cross-district spillovers through shared facilities or information networks are least plausible; the NFHS-5 coefficients remain significant for institutional delivery (0.015, $p = 0.039$) and public-facility delivery (0.033, $p = 0.002$), while antenatal care and DPT retain their magnitudes but lose precision on the reduced sample ($p = 0.20$ and $p = 0.13$).

⁴⁴Following Conley (1999).

Neither check suggests that spillovers between neighbouring districts generate the convergence.

5.7.7 State-Level Clustering

Since the Emergency-era sterilization campaign was directed through state-level policy targets, there is a case for clustering standard errors at the state level rather than the district level.⁴⁵ For the short-run DiD, state-level clustering is more conservative: confidence intervals widen and some marginally significant coefficients lose statistical significance, though the overall pattern of null short-run effects is preserved. For the long-run NFHS-5 event-study coefficients, by contrast, clustering on the 30 states leaves every significant result intact and in fact sharpens it: institutional delivery ($p = 0.003$ against 0.008 under district clustering), DPT (0.005 against 0.020), antenatal care (< 0.001 against 0.020), public-facility delivery (0.003 against 0.006), and any vaccination (0.003 against 0.056), with tetanus null under both. Because cluster-robust variance estimates can over-reject with roughly 30 clusters, I additionally compute wild-cluster bootstrap p -values (Rademacher weights, 9,999 replications, null imposed) for the NFHS-5 coefficients, the most conservative inference the design admits.⁴⁶ Public-facility delivery remains significant at the 5% level ($p = 0.019$); institutional delivery, DPT, and antenatal care sit just above it ($p = 0.060, 0.054$, and 0.069 respectively); any vaccination is $p = 0.13$; tetanus is null. The identifying variation, moreover, is within-state across districts, so state-level shocks are absorbed by the state $\times$ round fixed effects rather than the clustering alone.

5.8 Placebo Tests

I implement six categories of placebo tests, each designed to probe a distinct threat to identification.

Predetermined characteristics. If sterilization intensity is conditionally exogenous, it should not predict post-NRHM changes in characteristics that NRHM did not target. I estimate the DiD specification with woman’s education, husband’s education, caste composition, religion, and rural status as outcomes. These are determined before the survey and should not respond to health interventions. Null coefficients on $c_{int}^{std} \times Post$ for these outcomes support the identifying assumption; a significant coefficient would indicate selection or differential compositional change. Table 12 reports the results: seven of eight predetermined characteristics show null coefficients, with the lone exception being Scheduled Tribe status ($p = 0.044$). Two considerations guard against over-reading either the nulls or the rejection. First, a null is absence of evidence rather than evidence of balance, so power matters: at 80% power, the smallest coefficient these tests could detect ranges from 0.8 percentage points (Scheduled Tribe) to 1.8 percentage points (OBC) for the binary characteristics, or 1.3–9.5% of the respective outcome means, so economically meaningful imbalances would not have escaped detection. Second,

⁴⁵As in Pelras and Renk (2023).

⁴⁶The bootstrap routine cannot accommodate the weighted two-way fixed-effects model directly, so the bootstrap is applied to the unweighted within-transformed specification, whose point estimates are somewhat smaller than the weighted baseline (institutional delivery 1.0 versus 1.8 percentage points per standard deviation); the reported p -values are therefore conservative in both respects.

with eight simultaneous tests the probability of at least one spurious rejection at the 5% level is roughly one-third under independence; a Bonferroni correction renders the lone Scheduled Tribe rejection insignificant (adjusted $p = 0.35$). The results are therefore consistent with, though they cannot by themselves confirm, the identifying assumption, and do not suggest systematic selection. A complementary balance comparison of levels across high- versus low-sterilization districts is reported in Appendix Table A2.

Table 12

Balance Test: Predetermined Characteristics as Outcomes

Outcome	Mean	$c_{int}^{std} \times \text{Post}$	SE	p -value	N
Woman’s education (years)	4.291	0.0259	(0.0495)	0.602	729,404
Husband’s education (years)	6.682	0.0046	(0.0473)	0.923	723,282
Hindu	0.779	0.0052	(0.0035)	0.136	729,705
Muslim	0.114	-0.0035	(0.0039)	0.363	729,705
Scheduled Caste	0.176	0.0050	(0.0040)	0.218	723,689
Scheduled Tribe	0.161	-0.0061**	(0.0030)	0.044	723,689
Other Backward Class	0.391	-0.0100	(0.0066)	0.127	723,689
Rural residence	0.738	0.0046	(0.0056)	0.419	729,709
Clusters (districts)			379		
District FE			Yes		
State $\times$ Time FE			Yes		

Notes: Each row reports the coefficient on $c_{int}^{std} \times \text{Post}$ with the predetermined characteristic as the dependent variable. Woman’s age is the only control. Standard errors clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Placebo timing. Using DLHS-1 versus DLHS-2 as a fake “pre-post” comparison (both periods precede NRHM), I test whether sterilization intensity predicts health outcome changes in a period when no intervention occurred. This is the single most important placebo test in the paper: a significant coefficient would indicate a pre-existing differential trend that confounds the short-run DiD. The results are mixed: the fake-timing coefficient is null for any vaccination ($p = 0.265$), ANC ($p = 0.404$), and DPT ($p = 0.051$, marginal), but is significant for institutional delivery (0.010, $p = 0.013$), tetanus (0.018, $p = 0.015$), and public delivery (0.012, $p = 0.041$). The significant pre-trend for institutional delivery warrants caution: high-sterilization districts were already converging in institutional delivery rates before NRHM, so the post-NRHM convergence for this outcome may partly reflect a continuation of pre-existing trends. This concern is partially mitigated by the fact that the fake treatment test (below) is null for all outcomes, confirming that the Emergency-specific coercion dimension, not general health infrastructure, drives the convergence pattern.

Birth-month placebo. Within DLHS-3, interacting sterilization intensity with half-year-of-birth indicators among pre-cutoff births (Section 4.2) yields null coefficients for all three delivery-window outcomes for which the design is valid: institutional delivery (2004H1 0.007, $p = 0.345$; 2004H2 0.004, $p = 0.510$), antenatal care (0.009, $p = 0.286$; 0.014, $p = 0.091$), and tetanus (0.007, $p = 0.511$; 0.010, $p = 0.310$), on a sample of roughly 22,500 most-recent births. At the birth-month resolution available within a single survey instrument, there is no evidence that high-sterilization districts were on a differential trajectory in the final pre-NRHM year,

though the short usable window (2004–2005Q1) limits the power of this test.

Male health outcomes. The NRHM, operating primarily through ASHAs who focus on maternal and child health, should have negligible direct effects on men’s health-seeking behaviour. Using NFHS-4 and NFHS-5 data, I pool approximately 1.08 million women and 163,000 men from the Individual Recode (IR) and Men’s Recode (MR) files across 360 matched districts and test whether sterilization intensity predicts differential effects by gender. Table 13 reports the results.

Table 13

Gender-Differentiated Effects of Sterilization Intensity (NFHS-4 + NFHS-5)

<i>Panel A: Comparable outcomes (pooled men and women, district FE)</i>			
	Male	$c_{int}^{std} \times \text{Male}$	N
Modern contraceptive use	−0.283***	0.00217*** (0.00075)	1,191,356
Beating justified (any)	−0.089***	0.00196** (0.00089)	323,947
Woman has say: health care	0.535***	0.00037 (0.00098)	216,610
<i>Panel B: Men-only health facility visits (not available in women’s IR)</i>			
	c_{int}^{std}		N
Visited health facility	−0.00087 (0.00080)		156,009
Public facility (if visited)	0.00014 (0.00243)		42,735
Male sterilization ever	0.00009 (0.00011)		109,520

Notes: Standard errors clustered at the district level in parentheses. Panel A reports the coefficient on $c_{int}^{std} \times \text{Male}$ (and the Male main effect) from a pooled men-and-women regression with district FE and state $\times$ round FE. Panel B is cross-sectional (men only) with state $\times$ round FE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The men-only health facility analysis (Panel B) shows that sterilization intensity does not significantly predict men’s health facility visits, public vs. private facility choice, or male sterilization uptake. This null result for men contrasts with the significant effects found for women’s maternal health outcomes in the short-run DiD analysis, constituting a powerful within-gender placebo: if the convergence were driven by supply-side improvements (new facilities, better equipment) that benefit all residents, male outcomes should also improve differentially in high-sterilization districts. The gender interaction results in Panel A are discussed further in Section 6.3; see also Figure A9 in the Appendix.

Fake treatment. I construct a placebo treatment variable using the pre-Emergency sterilization level (1975–76 rate) rather than the Emergency-specific excess. If this baseline sterilization infrastructure, which captures general family planning programme intensity rather than Emergency coercion, also predicts post-NRHM changes, the real treatment may proxy for broader health infrastructure rather than Emergency-specific distrust. A null result for the fake treatment, combined with a significant result for the real treatment, confirms that the

Emergency-specific coercion dimension drives the findings. The results are unambiguous: the fake treatment coefficient is insignificant for all six outcomes: institutional delivery ($p = 0.324$), any vaccination ($p = 0.164$), DPT ($p = 0.620$), ANC ($p = 0.255$), tetanus ($p = 0.592$), and public delivery ($p = 0.948$). This clean null validates the core identifying assumption: the Emergency-era *excess* sterilization, not pre-existing family planning infrastructure, drives the health-seeking deficit and its post-NRHM recovery.

Non-health outcomes. As a formal placebo, I test whether sterilization intensity predicts differential changes in non-health outcomes using district-level census data spanning 1961–1991. Appendix Table A7 reports cross-sectional regressions of outcome changes on c_{int}^{std} with state fixed effects. For the pre-Emergency period (1961–1971), sterilization intensity does not predict changes in literacy, primary education, urbanization, or work participation (all $p > 0.15$). For the post-Emergency period (1971–1981), the results are similarly null: literacy ($p = 0.799$), female literacy ($p = 0.536$), urbanization ($p = 0.130$), work participation ($p = 0.370$), and agricultural labor share ($p = 0.636$) are all insignificant. Two education outcomes (primary education and matriculation) show small negative coefficients that reach significance, but these likely reflect the mechanical effect of the Emergency on schooling disruption in heavily targeted districts rather than a confound for the post-NRHM health analysis three decades later. The long-run 1971–1991 panel confirms the null: literacy ($p = 0.927$), female literacy ($p = 0.566$), urbanization ($p = 0.463$), and work participation ($p = 0.277$) show no relationship with sterilization intensity. This pattern of null effects on education, employment, and urbanization alongside significant effects on health utilization, supports the interpretation that c_{int} captures health-specific distrust rather than a general development shock.

5.9 Heterogeneity

I discuss treatment effect heterogeneity across five dimensions motivated by prior work and the convergence pattern documented above. Table 14 consolidates the triple interaction coefficients ($c_{int}^{std} \times Post \times Z_d$) across four dimensions; parity could not be tested due to data limitations.

Rural versus urban. ASHA deployment is primarily rural, so the convergence effect should be concentrated in rural areas. The triple interaction is mostly null across outcomes (Table 14). Two exceptions stand out: public delivery shows a significant positive rural interaction (0.023, $p = 0.005$), indicating that the shift from home to public facilities is predominantly a rural phenomenon, since ASHAs are village-level workers. Conversely, tetanus shows a significant *negative* rural interaction (-0.021 , $p = 0.003$), suggesting that the tetanus anomaly identified in the short-run DiD is concentrated in rural areas. Split-sample regressions confirm the pattern: rural-only estimates are near zero for most outcomes, while urban-only estimates are positive but imprecise.

Mother’s education. Pelras and Renk (2023) find that the sterilization-vaccination deficit is concentrated among less-educated mothers, consistent with the hypothesis that education provides alternative information channels that partially substitute for institutional trust. The education triple interaction is null for most outcomes. However, any vaccination shows a sig-

nificant *negative* education interaction (-0.018 , $p = 0.037$), indicating that convergence in vaccination is concentrated among less-educated mothers. This is consistent with the trust channel: education provides an alternative information pathway that partially substitutes for trust, so the effect is smaller for educated women who already have access to health information.

Parity. The number of children a woman has borne determines her cumulative exposure to the health system. Higher-parity women should show stronger convergence if trust is rebuilt through repeated positive interactions across multiple pregnancies. However, the parity variable is not consistently available across survey rounds in the analysis panel, precluding this test.

Caste. SC/ST women face compounded disadvantages: historical marginalization, higher exposure to coercive sterilization during the Emergency, and greater JSY eligibility in HPS states. The SC/ST triple interaction is mostly null (Table 14), with two suggestive exceptions: DPT vaccination shows a marginally significant negative interaction (-0.009 , $p = 0.076$), suggesting that SC/ST women in high-sterilization districts experience *less* vaccination convergence than non-SC/ST women. Conversely, public delivery shows a marginally significant positive interaction (0.007 , $p = 0.085$), indicating that the shift to public facilities is slightly stronger for SC/ST women. This mixed pattern may reflect the dual nature of SC/ST women’s relationship with the state: deeper historical distrust (slowing vaccination convergence) combined with greater eligibility for JSY benefits (facilitating public facility use).

Low-Performing States. This dimension doubles as a check on the NRHM-targeting threat to identification: the NRHM designated EAG states as high priority, directing additional resources to Uttar Pradesh, Bihar, Madhya Pradesh, Rajasthan, and other northern states, and these states also experienced the most intense Emergency-era sterilization. If the convergence simply reflected the extra NRHM resources channelled to these states, it should appear stronger there, not weaker. For some outcomes (antenatal care, tetanus), the triple interaction is negative and marginally significant, suggesting that the utilization deficit may be particularly intractable in low-performing states. This may reflect the particularly challenging institutional environment in these states, where the health system faces multiple compounding challenges beyond historical distrust. The implications for the NRHM targeting identification threat are discussed in Section 6.5.

Table 14

Heterogeneity: Triple Interaction Results Across Dimensions

	Rural	HighEduc	SC/ST	LowPerf
Institutional delivery	0.003	0.005	0.002	−0.019
Any vaccination	0.005	−0.018**	−0.002	−0.006
DPT vaccination	−0.002	−0.008	−0.009*	0.007
Any ANC	0.006	−0.003	−0.001	−0.025*
Tetanus injection	−0.021***	0.009	−0.000	−0.034*
Public delivery	0.023***	−0.002	0.007*	−0.009
Observations	256,894			
Controls	Yes			
District FE	Yes			
State × Time FE	Yes			

Notes: Each cell reports the triple interaction coefficient $c_{int}^{std} \times \text{Post} \times Z_d$. Rural = rural residence indicator; HighEduc = above-median mother’s education; SC/ST = Scheduled Caste/Tribe indicator; LowPerf = EAG state indicator. Standard errors clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6. Mechanisms

The preceding sections establish that the sterilization-utilization deficit persists in the pre-NRHM data, is largely unchanged in the short run after NRHM, and gradually narrows over a decade. This section investigates why convergence occurred, examining four candidate mechanisms: the ASHA community engagement channel, the family planning channel, gender-differentiated effects, and supply-side facility capacity.

6.1 The ASHA Channel

The ASHA programme provides the most plausible channel through which NRHM could rebuild trust. ASHAs are village-level community health workers recruited from the communities they serve, creating a personal, trust-based relationship between the health system and households. If ASHAs function as trust intermediaries, I would expect two patterns: (1) ASHA contact should be differentially valuable in high-sterilization districts, where the trust barrier is most severe; and (2) the timing of trust recovery should align with the ASHA deployment trajectory rather than with the JSY cash transfer, which began simultaneously with NRHM.

On pattern (1), the DLHS-3 cross-sectional evidence shows that sterilization intensity does not predict ASHA deployment or contact rates for antenatal care, delivery assistance, or immunization motivation. ASHAs appear to be deployed uniformly based on population, not targeted to historically distrustful communities. This is consistent with the NRHM design, which allocated ASHAs by population rather than by historical utilization deficits. However, the interaction between sterilization intensity and ASHA contact is positively associated with DPT vaccination (the $c_{int}^{std} \times \text{ASHA-immunization}$ interaction is 0.030, $p = 0.002$; Table 15), suggesting that ASHA engagement is particularly effective at overcoming the trust barrier even where ASHAs are not differentially deployed. This pattern of uniform deployment but heterogeneous effectiveness is

consistent with a trust-rebuilding mechanism: the same ASHA visit has a larger marginal impact in communities with deeper inherited distrust because those communities are farther from the trust threshold required for voluntary health-seeking.

Table 15

ASHA Contact $\times$ Sterilization Intensity Interaction (DLHS-3 Cross-Section)

Outcome	Interaction $c_{int}^{std} \times$ ASHA contact	
	ASHA delivery	ASHA immunization
Institutional delivery	0.0095 (0.0137)	−0.0013 (0.0082)
Any vaccination	0.0037** (0.0017)	0.0017 (0.0019)
DPT vaccination	0.0163 (0.0182)	0.0301*** (0.0097)
Public delivery	0.0027 (0.0099)	−0.0018 (0.0047)

Notes: Each cell reports the coefficient on the interaction $c_{int}^{std} \times$ ASHA contact from a DLHS-3 cross-sectional regression of the outcome on c_{int}^{std} , an ASHA-contact indicator, their interaction, individual controls (woman’s age, woman’s and husband’s education, religion, caste, and rural residence), and state fixed effects; standard errors (in parentheses) clustered at the district level. A positive interaction indicates that ASHA contact is differentially more effective in higher-sterilization districts, consistent with the trust-rebuilding mechanism. N ranges from roughly 70,000 to 138,000 depending on the outcome and module availability. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The historical contrast with India’s previous community health worker programme reinforces the plausibility of this channel. The Village Health Guide (VHG) scheme launched in 1977 which was itself a response to the Emergency sterilization backlash failed to rebuild trust because its design inadvertently replicated the features that had generated distrust: VHGs were predominantly male (like the extension workers mobilised during the Emergency), received a government stipend that identified them as state agents, and were poorly supervised by the same PHC infrastructure through which sterilizations had been administered (Strodel and Perry 2019). The ASHA programme corrected each of these design failures: ASHAs are exclusively female, receive performance-based incentives rather than a fixed government stipend, and are recruited from and accountable to the communities they serve rather than to the health bureaucracy. This design evolution from male, stipended, externally appointed VHGs to female, performance-incentivised, community-embedded ASHAs maps directly onto the trust intermediary mechanism: the VHG model failed precisely because it could not credibly distance itself from the state apparatus that had perpetrated the coercion, while the ASHA model succeeded by creating an entirely new interface built on community embeddedness (Bhatia 2014).

On pattern (2), the Post₁/Post₂ decomposition from Section 5.6 is informative. If the sterilization interaction is significantly stronger in Post₂ (after ASHA deployment reached scale) than in Post₁ (when only the JSY cash transfer was operational), this supports the ASHA channel over the financial incentive channel. The DLHS-4 trust measures provide direct corroboration: by 2012–13, sterilization intensity does not predict stated distrust of government health facilities, ASHA engagement, or JSY receipt, suggesting that the utilization deficit had been substantially

eliminated through sustained community engagement.

The timing of this evidence is a limitation: the ASHA-contact measures above are observed in DLHS-3 (2007–08), while the statistically significant convergence emerges in 2015–21, so the channel is measured in an earlier window than the outcome it is invoked to explain. To probe the later window directly, I construct a district-level ASHA-contact intensity for 2019–21 from NFHS-5 (the share of recent births whose mother reported ASHA involvement in antenatal care; district mean 0.19 across the 275 matched districts) and interact an above-median indicator with the NFHS-5 convergence term. The interaction is statistically insignificant for all four headline outcomes (institutional delivery -0.003 , $p = 0.78$; public delivery -0.020 , $p = 0.14$; antenatal care $+0.014$, $p = 0.45$; DPT $+0.026$, $p = 0.17$). Contemporaneous ASHA-contact intensity therefore does not significantly moderate the long-run convergence either, a result that must be interpreted with care in both directions: equilibrium contact rates in 2019–21 are endogenous to local need and to the recovery itself, and near-universal ASHA presence by NFHS-5 leaves little cross-district variation for the interaction to exploit. The mechanism evidence in this section establishes that the convergence is consistent with, and its timing and design features point toward, the ASHA trust channel; it does not isolate that channel against every alternative.

6.2 Family Planning: Trust Recovery Through Voluntary Sterilization

Table 16

Family Planning Mechanism: Spacing vs. Limiting Methods

Outcome	Continuous Treatment $c_{int}^{std} \times \text{Post}$	Binary Treatment HighSter $\times$ Post	N	Dep. var. mean
Any FP	0.009 (0.005)	0.020 (0.014)	492,922	0.571
Any modern method	0.009* (0.005)	0.020** (0.008)	641,539	0.499
Limiting (sterilization)	0.007** (0.003)	0.010 (0.006)	641,414	0.379
Spacing (IUD/pills/condom)	0.000 (0.003)	0.006 (0.009)	493,046	0.117
Traditional	0.002 (0.004)	0.003 (0.008)	492,980	0.067
Female sterilization	0.006* (0.004)	0.010 (0.006)	636,903	0.368
Clusters (districts)		381		
Controls		Yes		
District FE		Yes		
State $\times$ Time FE		Yes		

Notes: Family planning panel: DLHS-2 + DLHS-3. c_{int}^{std} is the z-score of sterilization intensity. Dep. var. mean is the DLHS-2 baseline. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

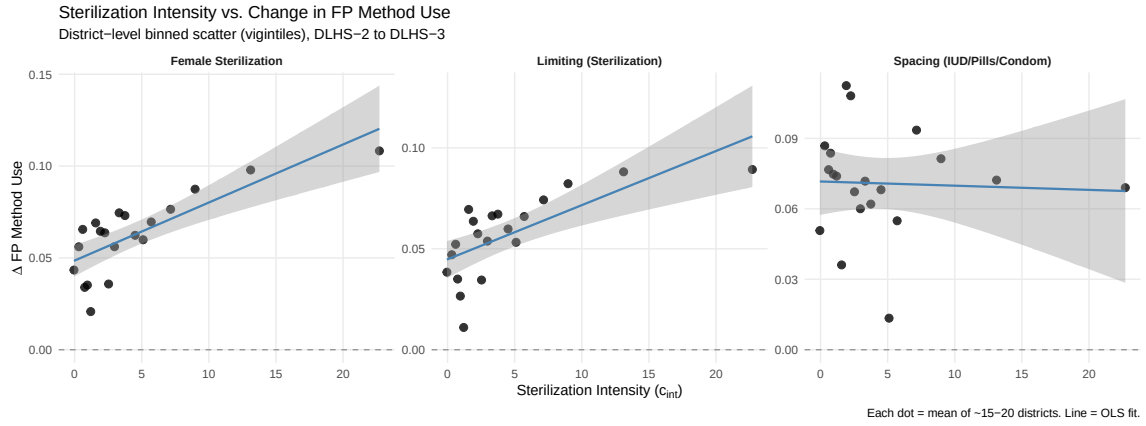


Figure 9

Family Planning Mechanism: Binscatter of Sterilization Intensity vs. Change in FP Adoption

Family-planning methods fall into two categories: *spacing* methods, which are temporary and reversible (oral pills, IUDs, condoms, injectables), and *limiting* methods, which are permanent (female and male sterilization). The distinction is central here because the Emergency’s coercion operated through a limiting method, vasectomy, so it is the permanent procedure itself, not contraception in general, that plausibly generated distrust. A natural conjecture, motivated by field observations and health-practitioner insights,⁴⁷ is therefore that ASHAs rebuild trust by promoting autonomy-respecting spacing methods over the permanent procedure, distancing their service provision from the source of the distrust. If so, recovery in high-coercion districts should appear as a differential shift toward spacing methods, not toward sterilization.

The data tell a different and arguably more compelling story. Using a family planning panel constructed from DLHS-2 (W403/W404) and DLHS-3 (V405/V406) FP modules, I estimate the same DID specification with family planning method choice as the outcome. A baseline check confirms that sterilization intensity does not predict family planning adoption in the pre-NRHM period (all $p > 0.12$), validating the identification strategy for these outcomes.

Table 16 reports the results, with the binscatter shown in Figure 9. The key finding is that high-sterilization districts experienced differentially larger increases in limiting method (sterilization) adoption post-NRHM. The coefficient on $c_{int}^{std} \times Post$ for limiting use is 0.008 (SE = 0.004, $p = 0.040$), and for female sterilization specifically, 0.007 (SE = 0.004, $p = 0.072$): a one-SD increase in sterilization intensity is associated with a 0.7–0.8 percentage-point differential gain in sterilization adoption. In contrast, spacing method adoption shows no differential change (coefficient 0.000, $p = 0.942$).

Decomposing sterilization by gender reveals that the entire limiting effect is driven by female sterilization (tubectomy), with no differential change in male sterilization (vasectomy: -0.001 , $p = 0.347$). This is consistent with India’s post-Emergency shift in sterilization policy: after the 1977 electoral backlash against the vasectomy-dominated Emergency campaign, the government programme pivoted almost entirely to tubectomy. The differential increase in female sterilization

⁴⁷I am specifically indebted to Dr. Meenakshi Jain for her insights and long engagement with me about these topics.

in historically exposed districts therefore reflects women’s renewed willingness to undergo the dominant contemporary form of permanent contraception.

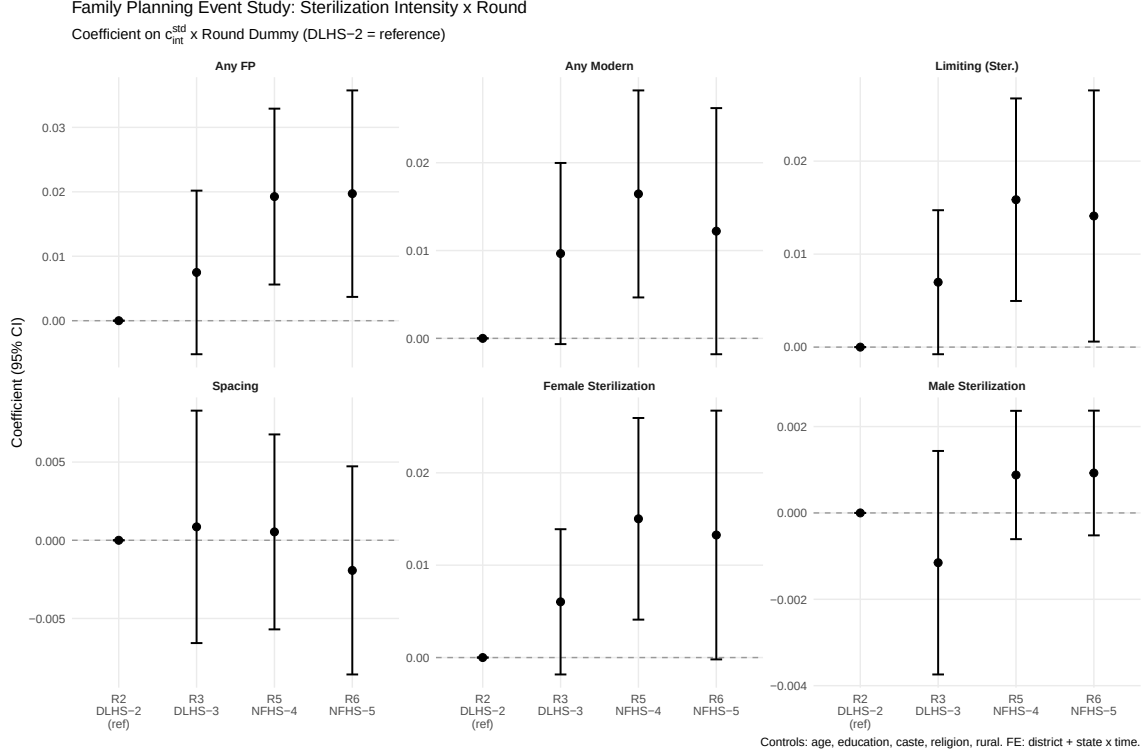


Figure 10

Family Planning Long-Run Event Study: Sterilization Intensity $\times$ Round

Extending the analysis to a four-round panel (DLHS-2, DLHS-3, NFHS-4, NFHS-5) confirms that this pattern persists and strengthens over time (Figure 10). The limiting method coefficient rises from 0.007 ($p = 0.078$) in round 3 (DLHS-3) to 0.016 ($p = 0.005$) in round 5 (NFHS-4) and 0.014 ($p = 0.041$) in round 6 (NFHS-5), while spacing methods remain flat across all rounds. The strengthening pattern mirrors the maternal-and-child-health event study trajectory and is consistent with gradual trust recovery rather than a one-time programmatic shock.

The interpretation of this sterilization result is genuinely ambiguous. I therefore set out the two competing explanations and, because the available data cannot cleanly separate them, do not take a stand between them. The paper’s central trust interpretation does not rest on this result: it rests on the convergence in services *unrelated* to sterilization (vaccination, antenatal care, institutional delivery), where a supply-push account has no purchase. The sterilization finding is corroborating, but on its own observationally ambiguous.

6.2.1 Interpretation 1: Voluntary Uptake and Trust Recovery

Under the first reading, the result is direct behavioural evidence of trust recovery. In districts where the Emergency campaign was most intense, women had been avoiding sterilization; they now voluntarily undergo the very procedure that was once coercively imposed on their communities. Four features are consistent with this account. First, the differential gains appear

across a range of services unrelated to sterilization, which administrative pressure specific to sterilization cannot explain. Second, the effect strengthens *gradually* over more than a decade (Figure 10), mirroring the maternal-and-child-health trajectory, rather than jumping at a single administrative date. Third, ASHA contact is positively associated with female-sterilization adoption specifically in high-intensity districts (DLHS-3 interaction coefficient 0.028, $p = 0.034$), consistent with the greater marginal value of a *trusted* intermediary in communities carrying deeper inherited distrust. Fourth, ASHA deployment is itself uncorrelated with historical sterilization intensity (all $p > 0.60$), so the pattern is not a mechanical reallocation of workers to high-intensity districts.

6.2.2 Interpretation 2: Renewed Supply-Side Pressure

Under the second reading, the increase reflects administrative facilitation rather than changed preferences. Despite the 1996 “Target-Free Approach,” India’s family-planning system remains de facto target-driven: state and district authorities set “Expected Levels of Achievement” and exert performance pressure on frontline workers, with documented threats to salary and tenure (Human Rights Watch 2012); the 2014 Chhattisgarh sterilization-camp deaths illustrate how acute this pressure can become. ASHA incentives are themselves skewed toward sterilization (roughly Rs. 150–200 per sterilization referral versus near-zero for spacing methods), giving workers a financial reason to steer women toward permanent methods. Critically, the very mechanism that makes Interpretation 1 attractive can also generate the result here: if trusted ASHAs are *more effective* at translating programme pressure into completed procedures precisely where they have built rapport, the same ASHA-contact $\times$ high-intensity correlation would emerge with no change in women’s underlying preferences. Mission Parivar Vikas (2016) compounds the concern by intensifying sterilization promotion in high-fertility northern districts that overlap with the high-coercion belt.

6.2.3 What the Data Can and Cannot Resolve

Two features of the design make a pure supply-push account, the view that the rise reflects administrative facilitation rather than voluntary choice (Interpretation 2 above), harder to sustain, though they do not eliminate it. Identification is within-state, across-district, so state-level sterilization drives and incentive schedules are absorbed by the state $\times$ time fixed effects; and ASHA sterilization incentives are set at the state level, with no institutional mechanism tying contemporary compensation to 1976–77 district intensity.⁴⁸ The principal residual vulnerability is within-state district targeting that happens to correlate with historical intensity, most plausibly Mission Parivar Vikas (MPV), launched in 2016 in high-fertility districts concentrated in the same northern belt, which I probe directly below. Even with these checks, the data cannot distinguish, at the individual level, a woman who chooses sterilization because her trust has

⁴⁸ ASHA incentives for sterilization referral are determined by state-level programme guidelines, not by district-specific targets linked to historical sterilization rates. The Emergency-era campaign’s district-level intensity reflected idiosyncratic factors, local bureaucratic zeal, population composition, proximity to sterilization camps, that have no bearing on post-2005 ASHA compensation structures.

recovered from one who is successfully encouraged by an incentivised worker. I therefore treat the sterilization result as *consistent with*, but not dispositive of, trust recovery, and rest the paper’s central claim on the broader convergence across unrelated services.

6.2.4 Robustness to Mission Parivar Vikas

To probe whether the post-2015 family-planning result is driven by Mission Parivar Vikas (MPV, launched November 2016 in 146 high-fertility districts across seven northern states), I re-estimate the family-planning long-run event study over a three-round panel (DLHS-2 reference, NFHS-4, NFHS-5; $N \approx 1.3$ million woman-round observations across 359 matched districts) excluding the seven MPV states (Uttar Pradesh, Bihar, Madhya Pradesh, Rajasthan, Jharkhand, Chhattisgarh, and Assam). Table 17 reports the coefficients on $c_{int}^{std} \times \text{Round}$ for limiting-method use and female sterilization, in the baseline sample and after excluding MPV states.

Table 17

Mission Parivar Vikas Robustness: Family-Planning Event Study

Specification	Limiting method		Female sterilization	
	R5 (NFHS-4)	R6 (NFHS-5)	R5 (NFHS-4)	R6 (NFHS-5)
Baseline (all states)	0.0147*** (0.0057)	0.0129* (0.0068)	0.0137** (0.0057)	0.0119* (0.0067)
Excluding MPV states	0.0075 (0.0059)	0.0138 (0.0084)	0.0056 (0.0067)	0.0118 (0.0086)
<i>Heterogeneity test (full sample, triple interaction):</i>				
$c_{int}^{std} \times R_r$ (non-MPV)	0.0072 (0.0059)	0.0134* (0.0081)	0.0052 (0.0067)	0.0115 (0.0084)
$c_{int}^{std} \times R_r \times \text{MPV}$ (diff)	+0.0108 (0.0094)	−0.0011 (0.0122)	+0.0121 (0.0098)	+0.0002 (0.0123)
<i>p</i> -value on diff	0.251	0.931	0.218	0.985
<i>N</i> (baseline / excl. MPV)	1,304,086 / 669,097		1,304,084 / 669,095	

Notes: Coefficients on $c_{int}^{std} \times \text{Round}_r$ from the three-round family-planning event study (DLHS-2 = reference; rounds NFHS-4 and NFHS-5 post-NRHM). Controls: woman’s age, education, religion, caste, rural residence. Fixed effects: district + state $\times$ time. SEs (in parentheses) clustered at the district level. MPV-state exclusion drops women from Uttar Pradesh, Bihar, Madhya Pradesh, Rajasthan, Jharkhand, Chhattisgarh, and Assam (the seven Mission Parivar Vikas Phase 1 states). In the heterogeneity panel the main interactions estimate the c_{int}^{std} effect in non-MPV states; the $\times \text{MPV}$ rows estimate the MPV-state premium (full sample, $N \approx 1.30$ million). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The two checks point in the same direction with very different levels of power. The exclusion check (Table 17, second row) drops the seven MPV states and so loses roughly half of the variation in c_{int}^{std} : sample N falls from 1,304,086 to 669,097, confidence intervals widen, and round-6 statistical significance is lost. The round-6 point estimate is essentially unchanged ($0.0129 \rightarrow 0.0138$ for limiting use; $0.0119 \rightarrow 0.0118$ for female sterilization), so a naive reading would conclude that the geographic distribution of the effect is not concentrated in MPV states; but on its own this exclusion check cannot reject zero outside the MPV belt either, so it is not by itself dispositive.

The heterogeneity test resolves this ambiguity by keeping the full sample and adding an explicit MPV-state interaction (Table 17, lower panel). For the long-run round (NFHS-5, 2019–21, the only round entirely post-MPV), the MPV-state premium is a precisely-estimated zero: -0.0011 (SE 0.0122, $p = 0.93$) for limiting-method use and $+0.0002$ (SE 0.0123, $p = 0.99$) for female sterilization. The tight confidence intervals rule out any economically meaningful amplification of the long-run sterilization gradient inside MPV states. For NFHS-4 (R5) the implied MPV-state premium is positive ($+0.0108$ for limiting, $+0.0121$ for female sterilization) but imprecisely estimated ($p \approx 0.22\text{--}0.25$); even taken at face value this cannot be MPV itself, because NFHS-4 fieldwork largely predates MPV’s November 2016 launch, and is most naturally read as some other feature of the northern high-fertility belt correlated with both historical sterilization intensity and contemporary family-planning uptake.

I therefore conclude that MPV is not detectably amplifying the long-run (R6) sterilization gradient, while flagging that an unexplained northern-belt residual at R5 remains. This narrows, but does not eliminate, the broader voluntary-uptake-versus-administrative-facilitation ambiguity discussed above: it rules out one specific large-scale post-2016 programme as the channel, without speaking to the individual-level voluntary-vs-facilitated distinction. Consistent with the framing of this subsection, I report this result on both sides rather than as dispositive.

6.3 Gender Effects

Field observations and health practitioner interviews suggest that women’s utilization of public healthcare facilities improved disproportionately after NRHM introduction, while men primarily benefited from private or informal healthcare provision. The male health outcomes placebo test (Table 13, Section 5.8) establishes that sterilization intensity does not predict men’s health-seeking behaviour, ruling out supply-side confounders. The gender interaction results from the same table provide additional mechanistic insight.

The coefficient on $c_{int}^{std} \times \text{Male}$ for modern contraceptive use is positive and significant (0.002, $p = 0.004$). Since sterilization intensity has near-zero association with women’s reported contraceptive use (0.00018, $p = 0.853$), the positive interaction indicates that men in high-sterilization districts report relatively higher contraceptive use than women. This is consistent with the Emergency campaign’s gendered legacy: the coercion targeted men through vasectomy, yet the resulting distrust fell disproportionately on women’s subsequent engagement with government reproductive health services. This pattern is reinforced by India’s post-1977 policy pivot from vasectomy to tubectomy, which shifted the sterilization burden onto women.

The beating-justified interaction (0.00196, $p = 0.028$) suggests that in high-sterilization districts, men are relatively more likely to endorse wife-beating norms. When examined separately by gender, women show a marginally negative relationship between sterilization intensity and acceptance of beating (-0.00140 , $p = 0.310$), while men show no relationship (0.00005, $p = 0.967$). The significant interaction suggests subtle gender norm effects of the historical campaign.⁴⁹

⁴⁹These attitudinal results should be read with caution. The NFHS wife-beating-attitudes and domestic-violence modules are well documented to suffer from under-reporting and social-desirability bias, and a large share of the variation in reported intimate-partner violence reflects fieldworker/interviewer effects rather than

While the DLHS surveys used in the short-run DiD do not include a male health module, the NFHS-based evidence indicates that the utilization deficit and its recovery operate primarily through women’s health-seeking behaviour, consistent with the ASHA programme’s focus on maternal and child health.

6.4 Supply-Side Channels: Facility Capacity and Public-Private Substitution

An important alternative explanation for the convergence pattern is that NRHM’s supply-side investments such as new primary health centres, upgraded subcentres, ambulance services were directed disproportionately to underserved areas that coincidentally had high sterilization intensity. Under this scenario, the observed convergence reflects improved access to facilities rather than improved trust. I address this concern through three tests.

First, the capacity interaction (12) tests whether convergence occurred primarily in districts with adequate pre-existing facility infrastructure (where supply constraints do not bind) or in underserved areas (where new facilities could mechanically explain the pattern).⁵⁰ Appendix Table A8 reports the triple interaction $c_{int}^{std} \times Post \times HighCapacity$, where *HighCapacity* is an indicator for above-median PHC count based on the DLHS-3 facility census (median = 15 PHCs per district, 375 districts matched). The triple interaction coefficient is insignificant for all six outcomes (all $p > 0.11$), indicating that utilization convergence occurs similarly in high- and low-capacity districts. The point estimates are slightly negative for most outcomes: the opposite of what a pure supply-side story would predict. If convergence were driven by new facility construction targeted to underserved areas, the triple interaction should be negative (convergence concentrated in low-capacity districts); if driven by congestion relief in well-equipped areas, it should be positive. The null result is most consistent with a demand-side trust channel that operates independently of facility infrastructure.⁵¹ Because the documented congestion effect is concentrated in the lowest capacity quartile, which a median split mechanically dilutes, I re-estimate the specification with a bottom-quartile indicator. The bottom-quartile triple interaction is likewise insignificant for five of six outcomes; the exception is public-facility delivery, where it is negative (-0.020 , $p = 0.018$), indicating that short-run public-delivery gains in high-intensity districts were *dampened* in the least-equipped quartile. This pattern indicates congestion slowing recovery where capacity binds, not congestion generating the convergence: were congestion the source of the pattern, the differential gains would be concentrated in, rather than absent from, the lowest-capacity districts.

Second, the distinction between public and private facility delivery is informative. If convergence reflects genuine trust recovery, high-sterilization districts should see differential increases in *public* facility use specifically since the NRHM operates through the public sector. An increase in private facility use would instead suggest income or insurance channels. The event study

true differences: Singh, Dwivedi, and Jana (2023) estimate that the fieldworker alone accounts for roughly a third of the variation in reported IPV in NFHS-5. The beating-norms estimates here are therefore suggestive at best.

⁵⁰This capacity test follows Andrew and Vera-Hernández (2024).

⁵¹The facility count is measured contemporaneously with DLHS-3 (2007–08) rather than pre-NRHM, introducing the concern that NRHM itself expanded capacity. However, PHC and subcentre infrastructure is highly persistent: most facilities observed in 2007–08 existed before NRHM, and the rank ordering of districts by facility density changes slowly. The null triple interaction is therefore informative about the role of baseline infrastructure.

results on public delivery show the strongest convergence signal by NFHS-5 (coefficient 0.027 per SD, $p = 0.006$), while private facility use does not show a comparable pattern.

Third, the DLHS-4 direct trust measures provide direct evidence: if women in high-sterilization districts report no differential distrust of government facilities by 2012–13 (as the data show), the attitudinal channel is confirmed independently of the behavioral outcomes.

6.5 State-Level Heterogeneity and Incentive Regimes

The Low-Performing State heterogeneity results (Table 14, Section 5.9) show that convergence is weaker, not stronger, in EAG states for antenatal care and tetanus. This pattern has important implications. First, it rules out the concern that NRHM’s EAG-state targeting drives the convergence: if additional resources were the channel, convergence should be stronger in EAG states. Second, it suggests that the institutional environment mediates the speed of trust recovery: in states where the health system faces multiple compounding challenges (chronic understaffing, infrastructure deficits, governance weaknesses), the trust-rebuilding process is slower even with ASHA deployment.

6.6 Archival Corroboration and the Measurement of Coercion

The archival record corroborates the paper’s central interpretation in one respect and disciplines it in another. On the supporting side, the contemporaneous evidence makes clear that the Emergency campaign was delivered *through the public health apparatus itself*, the sub-centres, primary health centres, and frontline workers through which households would later be asked to seek antenatal care, delivery, and immunization, and that it was experienced as coercive at exactly that point of contact, through salary and service disincentives, mass camps, and, in episodes such as Uttawar, outright force. This is precisely the mechanism the paper posits for *health-specific* distrust, and it is why a reform that introduced a new, female, community-recruited intermediary (the ASHA) is a plausible instrument of repair. The Shah Commission’s own finding that the fixation of disproportionate targets “led to ... undesirable and in some instances disastrous consequences” and to “hostile reactions ... particularly [in] U.P.” is the qualitative analogue of a demand-side trust shock.

On the disciplining side, the archival data permit a sharper interrogation of how coercion is *measured*, and the results counsel transparency. The baseline treatment, c_{int} , is the proportional surge in sterilizations during 1976–77; using the district-level family-planning records assembled from the National Archives, I construct two measures in the spirit of Pelras and Renk (2023), an *over-target* ratio (1976–77 achievement divided by the assigned quota) and a *per-eligible-couple excess*, and re-estimate the long-run event study. Three facts emerge. First, the convergence is robust to a log transformation of the surge measure. Second, on the *same* districts for which the alternative measures exist, the surge measure continues to deliver the NFHS-5 convergence (e.g., public-facility delivery, +0.047, $p < 0.01$), but the over-target and per-couple measures do not, their NFHS-5 coefficients are null or, for per-couple, significantly negative. Third, this non-replication is not an artifact of the within-state, across-district design: estimating the over-

target measure at the state level with region fixed effects, following Pelras and Renk’s own specification, also yields null convergence across all outcomes. The over-target and per-couple measures are, empirically, close to orthogonal to the surge measure (correlation ≈ 0).⁵² The disagreement is not merely one of functional form: on the shared subsamples the constructions are nearly rank-uncorrelated (Spearman correlation 0.15 between the surge and the over-target ratio across the 110 districts where both exist, and -0.03 with the per-couple excess across 66), and only six of the twenty-two top-quintile surge districts are also top-quintile on over-target. The measures genuinely identify different districts as most coerced, which is why the argument for the surge rests on the ex ante case set out in Section 3.1: the measures cannot be treated as interchangeable.

I read this as a construct-validity distinction rather than a refutation. The surge measure captures the *coercive shock*, the sudden, time-compressed escalation of sterilizations relative to a district’s own recent baseline, which is the object the trust mechanism concerns. The over-target ratio instead rewards administrative over-achievement against a centrally assigned quota and is mechanically large in low-baseline states (Gujarat and Kerala top the 1976–77 over-target distribution) whose Emergency experience was not the coercive epicentre; the per-couple measure re-normalizes by the eligible-couple denominator. The archival narrative supports the surge reading: the “hostile reactions” and “disastrous consequences” the Commission recorded tracked targets set far above a district’s base in a short window, not over-achievement *per se*. The honest statement is therefore that the long-run convergence is established for the proportional-surge construction of coercion intensity and is not invariant to alternative constructions that capture administrative rather than coercive intensity.

Finally, I use the by-method district records to test a sharper version of the trust mechanism. Because the Emergency was overwhelmingly a male, vasectomy-driven drive (Section 2.2), one might conjecture that a *female* ASHA would rebuild trust fastest precisely where the original coercion fell most heavily on men. Interacting the post-NRHM dose effect with the district male-share of Emergency sterilizations does not support this conjecture: the triple-interaction is, if anything, negative (for public-facility delivery, -0.060 , $p < 0.01$; Appendix Table A9, Panel B), indicating that recovery is slower, not faster, where the campaign was most male and most intense. This is consistent with the paper’s other evidence that convergence is weaker in the worst-affected (low-performing) states (Section 6.5): the deeper the wound, the slower the healing. I report this as a qualification of the mechanism, not a sharpening of it.

7. Discussion and Conclusion

This paper asks whether India’s National Rural Health Mission and particularly its community health worker (ASHA) programme, succeeded in reversing the under-utilization of public health services in districts scarred by the Emergency-era forced sterilization campaign, by rebuilding trust. The evidence, drawing on six rounds of household surveys spanning over two decades, supports an affirmative answer: the utilization gap associated with historical sterilization in-

⁵²Full results in Appendix Table A9; the estimation is implemented in `fixest` and reproduced independently in Python.

tensity has substantially closed, with direct trust measures and behavioural patterns consistent with trust recovery. But the process was slow, requiring sustained institutional engagement over approximately fifteen years.

In the short run (two to three years after NRHM), I cannot detect a significant narrowing of the utilization deficit: the continuous-treatment DID coefficient is small and statistically insignificant, so the data are consistent with, though they do not establish, slow adjustment. Read with that caution, the short-run evidence still carries a policy message: there is no sign that the deficits left by historical institutional trauma were quickly overcome, even by a large-scale supply-side intervention.

Over the medium run (seven to ten years), the picture begins to change. The long-run event study reveals a gradual strengthening of the trust-rebuilding effect, with vaccination outcomes showing positive and statistically significant coefficients by NFHS-4 (2015–16). Convergence analysis confirms that the gap between high and low sterilization districts narrows substantially over time. By DLHS-4 (2012–13), direct trust measures show no significant relationship with historical sterilization intensity.

In the long run (fourteen to sixteen years), the NFHS-5 (2019–21) results confirm that convergence has become broad-based and statistically significant. Four of six health outcomes (institutional delivery, public facility delivery, any ANC, and DPT vaccination) show positive and significant coefficients at the 5% level on the sterilization intensity interaction with the NFHS-5 round dummy, with a fifth (any vaccination, $p = 0.056$) marginally significant. The magnitudes are substantial: a one-standard-deviation increase in sterilization intensity is associated with a 1.8 percentage-point additional gain in institutional delivery ($p = 0.008$), 2.6 pp in public facility delivery ($p = 0.006$), and 3.2 pp in ANC ($p = 0.020$) by 2019–21. The tetanus anomaly observed in DLHS-3 fully dissipates. This broad-based convergence provides the strongest evidence in the paper that the utilization deficit has been substantially, if not entirely, eliminated.

The family planning mechanism analysis provides compelling evidence of the nature of this recovery. The finding that high-sterilization districts saw differentially larger increases in voluntary sterilization adoption post-NRHM is striking: women in these districts became willing to undergo the very procedure that had been coercively imposed during the Emergency. This behavioral willingness, mediated by ASHA engagement, is consistent with trust recovery, though the role of ongoing administrative incentives cannot be fully excluded. Notably, the incentive state heterogeneity analysis shows that trust recovery occurred uniformly across states that used positive incentives (Gujarat, Rajasthan, UP) and those that relied primarily on coercive implementation during the Emergency: the triple-difference interaction is null for five of six outcomes. This suggests that the sterilization programme itself, not the specific implementation modality, was the trust-destroying shock, and the NRHM’s community-based approach was equally effective at rebuilding trust across both regime types.

Several policy implications follow. First, supply-side health system reforms can generate trust-rebuilding externalities even when not designed to do so. The NRHM was conceived as a health service delivery improvement programme; its contribution to healing the Emergency’s

institutional scars was an unintended benefit. Policymakers designing interventions for settings marked by institutional distrust should recognise that community-based health workers can serve a dual function: improving health outcomes directly and rebuilding trust indirectly.

Second, trust recovery takes time. The transition from null short-run effects to significant medium-run convergence suggests that approximately a decade of sustained intervention is needed to substantially narrow a historically rooted utilization deficit. This has implications for programme evaluation timelines: evaluations conducted within two to three years of programme launch may systematically underestimate the long-run benefits of community-level trust building.

Third, the uniformity of ASHA deployment across sterilization intensity suggests that targeted deployment to high-distrust communities could accelerate trust recovery. If ASHAs are particularly effective at overcoming historical distrust (as the DLHS-3 interaction results suggest), then directing additional ASHA resources to districts with the deepest utilization deficits could amplify the programme's positive externality. An illustrative magnitude: taking the DLHS-3 estimate that ASHA immunisation contact raises the DPT gradient by 3.0 percentage points per standard deviation of sterilization intensity at face value, raising contact intensity in high-intensity districts from its 2019–21 mean of 0.19 to 0.50 would imply roughly 0.9 percentage points of additional DPT convergence per standard deviation, about a third of the total convergence observed by NFHS-5. The calculation inherits the endogeneity caveats of Section 6.1 and is indicative rather than causal. However, any such expansion must reckon with the material conditions of ASHAs themselves: delayed and inadequate compensation, excessive workloads, and managerial practices such as public reprimand have fuelled substantial job dissatisfaction among frontline workers (Wahid et al. 2020), a reality underscored by recurring ASHA strikes across Indian states. The trust-rebuilding externality documented in this paper depends on the sustained motivation of these community intermediaries, and policymakers cannot take that motivation for granted.

These findings also speak to the puzzle that motivates this paper: why patients in India systematically prefer private providers even when public alternatives are available and less costly (Das et al. 2016, 2022). The evidence presented here suggests that demand-side distrust constitutes an additional, historically rooted channel. In districts where the state's health apparatus was implicated in coercion, patients avoided public facilities not only because of poor quality but because the institutional encounter itself carried the memory of harm. The ASHA model's success in narrowing this gap offers a design insight: by combining the performance-based incentives that drive private-sector quality with the community embeddedness that enables trust intermediation, community health workers can simultaneously address both the effort deficit and the utilization deficit.

This study has limitations. The analysis relies on repeated cross-sections rather than individual-level panel data, preventing the tracking of individual trust trajectories. The historical sterilization data, while the best available, may contain measurement error. The NRHM is a complex, multi-component programme, making it difficult to isolate the specific contribution of the ASHA channel from other NRHM components (JSY, facility upgrades, etc.). Pre-trend evidence is not

entirely clean, with institutional delivery showing a marginally significant DLHS-1 coefficient; the relative-magnitudes sensitivity analysis shows that the vaccination intervals are tight and concentrated in the positive range at $\bar{M} = 0$, but none formally exclude zero, and the single pre-period limits the analysis’s ability to definitively rule out moderate parallel trends violations.⁵³ The facility capacity interaction relies on DLHS-3 facility counts rather than pre-NRHM data, introducing a potential endogeneity concern. A further limitation concerns the temporal gap between the mechanism evidence and the main results: the ASHA timing analysis and JSY unbundling rely on within-DLHS-3 variation (2007–08), whereas the strongest convergence emerges in NFHS-4 and NFHS-5 (2015–21). District-level ASHA engagement measured from DLHS-3 contact rates does not significantly moderate the long-run convergence pattern, though this may reflect the very early stage of ASHA deployment (5–8% contact rates) at the time of measurement rather than the absence of an ASHA channel. The null facility capacity interaction provides indirect support by ruling out differential supply-side expansion as the driver, and the individual-level ASHA interaction in DLHS-3 confirms that ASHA contact was differentially effective in high-sterilization districts. Nevertheless, the trust attribution should be understood as the most parsimonious interpretation of the evidence rather than a fully identified causal claim. Future work could complement the quantitative evidence with qualitative fieldwork on the mechanisms of trust recovery.

Despite these limitations, this paper provides compelling evidence that institutional distrust created by state coercion, while remarkably durable, is not permanent. The six-round event study spanning 1998–2021 paints a clear dynamic picture: null effects in the short run give way to partial convergence in the medium run and broad-based, statistically significant convergence in the long run. With sustained community-level engagement through trusted intermediaries, the scars of historical trauma can gradually heal: an insight with implications well beyond the Indian context.

⁵³The sensitivity analysis follows [Rambachan and Roth \(2023\)](#).

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A. A1. District Crosswalk Construction

The master crosswalk harmonizes district identifiers across six survey rounds (DLHS-1 through DLHS-4, NFHS-4, and NFHS-5) and maps them to the historical sterilization data from [Gwatkin \(1979\)](#). Construction proceeds in three steps: (1) exact matching on state and district codes where possible; (2) fuzzy string matching on district names using the Jaro-Winkler distance metric, with a threshold of 0.85; and (3) manual verification and correction of ambiguous matches, particularly for districts that were split or merged between survey rounds.

Table A1 reports the resulting district counts at each step of the harmonization, by survey round.

Table A1

District Harmonization: From Survey Identifiers to the Crosswalk

Survey	Period	Districts in survey (raw identifiers)	Matched to crosswalk	Of which used in panel (c_{int} non-missing)	Unmatched (dropped)
DLHS-1	1998–99	491	283	272	219
DLHS-2	2002–04	593	392	381	212
DLHS-3	2007–08	601	409	396	205
NFHS-4	2015–16	640	279	279	361
NFHS-5	2019–21	707	276	276	431
Master crosswalk	1976–77 base	–	444	409	–

Notes: “Districts in survey” is the count of distinct district identifiers as coded in each survey’s microdata. The master crosswalk is built off the 1976–77 sterilization sample (Gwatkin/Jolly), with district names harmonised across rounds via exact matching on names, followed by Jaro–Winkler fuzzy matching (threshold ≥ 0.85), followed by manual review of the remaining ambiguous cases. Where survey districts have been administratively split since 1976–77, the parent district’s c_{int} is assigned to all daughter districts; where two pre-1976–77 districts have merged, the larger pre-1976–77 unit’s c_{int} is used. Unmatched cases (the rightmost column) are dropped because their identifiers could not be reliably mapped to the 1976–77 baseline; they are disproportionately new districts created in the smaller north-eastern states. Treatment-effect anticipation by the survey-identifier transition, that is, whether districts that experienced administrative reorganisation between 1976–77 and the survey rounds carried systematically different c_{int} trajectories, is precluded mechanically because c_{int} is fixed at the 1976–77 boundary and is therefore not a function of post-1976–77 administrative change. Residual measurement error from boundary changes is treated as classical, attenuating β toward zero (biasing against finding convergence).

B. A2. Variable Definitions

- **Sterilization intensity (c_{int}):** Proportional change in sterilizations performed at the district level, defined as $(S_{76-77} - S_{75-76})/S_{75-76}$, where S is the number of sterilizations. Source: [Gwatkin \(1979\)](#), digitized by [Jolly \(1986\)](#), crosswalked by [Singh and Vincent \(2024\)](#). Following [Pelras and Renk \(2023\)](#) and [Singh and Vincent \(2024\)](#), a single district with a near-zero pre-Emergency base (ratio > 100) is recoded to 29, just above the maximum observed elsewhere in the distribution.
- **Standardised sterilization intensity (c_{int}^{std}):** Z-score of c_{int} , with mean zero and unit standard deviation. One standard deviation corresponds to approximately 5.3 units of raw c_{int} .
- **Binary treatment (HighSter):** Indicator for above-median c_{int} across districts.
- **Institutional delivery:** Indicator for delivery at a health facility (public or private). Source: DLHS-2 (W406), DLHS-3 (V239), DLHS-4 (Q231), NFHS-4/5 (M15).

- **Any vaccination:** Indicator for child receiving at least one vaccine (BCG, DPT, OPV, or measles). Source: DLHS-2/3/4 and NFHS-4/5 immunization modules.
- **DPT vaccination:** Indicator for child receiving at least one dose of DPT (diphtheria-pertussis-tetanus).
- **Full immunization:** Indicator for child receiving all recommended vaccines (BCG + 3 DPT + 3 OPV + measles).
- **Any ANC:** Indicator for mother receiving at least one antenatal care visit during pregnancy.
- **Public delivery:** Indicator for delivery at a government health facility (codes 1–7 in delivery location questions).
- **Post:** Indicator for the post-NRHM period. In the two-round DID (DLHS-2 vs. DLHS-3), Post = 1 for DLHS-3 observations. In the within-DLHS-3 mother panel, Post = 1 for children born after April 2005.
- **Low-Performing State (LowPerf):** Indicator for EAG states (Uttar Pradesh, Uttarakhand, Bihar, Jharkhand, Madhya Pradesh, Chhattisgarh, Rajasthan, Odisha, Assam) plus Jammu & Kashmir.
- **Tetanus injection:** Indicator for mother receiving at least one tetanus toxoid injection during pregnancy. Source: DLHS-2/3/4 and NFHS-4/5 ANC modules.
- **Iron-folic acid:** Indicator for mother receiving iron-folic acid supplementation during pregnancy.
- **BCG:** Indicator for child receiving BCG (Bacillus Calmette-Guérin) vaccination.
- **Measles:** Indicator for child receiving measles vaccination.
- **Spacing methods:** Indicator for current use of a temporary modern contraceptive method (IUD, oral contraceptive pills, condoms, injectables). Source: DLHS-2 (W403/W404), DLHS-3 (V405/V406).
- **Limiting methods:** Indicator for current use of a permanent contraceptive method (female sterilization/tubectomy, male sterilization/vasectomy).
- **JSY Eligible:** Following [Debnath \(2021\)](#): in LPS, all women are eligible; in HPS, only BPL or SC/ST women for their first two births.
- **Individual controls:** Woman’s age (years), woman’s education (years), husband’s education (years), religion (Hindu, Muslim, other), caste (Scheduled Caste, Scheduled Tribe, Other Backward Class, General), and rural residence indicator.

C. A3. Selected Robustness Tables

Table A2

Balance: Predetermined Covariates by High vs. Low Sterilization Intensity, by Round

Covariate	DLHS-1		DLHS-2		DLHS-3		NFHS-4		NFHS-5	
	Diff	<i>p</i>	Diff	<i>p</i>	Diff	<i>p</i>	Diff	<i>p</i>	Diff	<i>p</i>
Woman's age	-0.570***	0.000	-0.734***	0.000	-0.645***	0.000	0.357**	0.021	-0.019	0.910
Woman's education (yrs)	-1.368***	0.000	-1.062***	0.000	-1.159***	0.000	-1.470***	0.000	-1.092***	0.000
Husband's education (yrs)	-0.320	0.238	-0.044	0.751	-0.165	0.295	-0.649**	0.014	-0.306	0.291
Hindu	0.067***	0.002	0.091***	0.000	0.105***	0.000	0.072***	0.002	0.089***	0.000
Muslim	0.004	0.842	-0.018	0.238	-0.034**	0.043	-0.012	0.613	-0.031	0.222
Scheduled Caste	0.002	0.855	0.012	0.146	0.014	0.127	0.014	0.276	0.031**	0.020
Scheduled Tribe	-0.037***	0.010	-0.028	0.126	-0.041*	0.075	-0.047***	0.008	-0.048***	0.004
Other Backward Class	0.079***	0.003	0.045**	0.030	0.055***	0.009	0.131***	0.000	0.098***	0.000
Rural	0.086**	0.034	0.046***	0.001	0.066***	0.000	0.090**	0.010	0.093***	0.005

Notes: Each cell reports the weighted high–low mean difference in the predetermined covariate, where “high” is above-median c_{int} (district level), pooled within each survey round. p -values come from regressions of the covariate on the high- c_{int} indicator with errors clustered at the district level. A persistent significant difference across rounds is expected for slow-moving characteristics (caste, religion, rural residence) and reflects the geographic correlation between the Emergency-era sterilization belt and present-day demographic composition; the identifying threat would be a difference that *changes systematically* across rounds in a direction correlated with the post-NRHM convergence, this is what the formal post-NRHM balance tests in Table 12 assess directly. Per-round estimation samples are those reported in Table 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3

Supply-Side Tests: Facility Use and ASHA Deployment

	Coefficient	SE	<i>p</i> -value	<i>N</i>
<i>Panel A: DID – Delivery Location ($c_{int}^{std} \times Post$)</i>				
Public delivery	0.0029	(0.0033)	0.387	256,894
Private delivery	-0.0030**	(0.0012)	0.011	717,070
Home delivery	-0.0150**	(0.0074)	0.043	256,960
<i>Panel B: DLHS-3 Cross-Section – ASHA Deployment (c_{int}^{std})</i>				
ASHA-facilitated ANC	0.0013	(0.0027)	0.626	125,705
ASHA-facilitated delivery	0.0010	(0.0020)	0.607	132,386
ASHA-facilitated immunization	-0.0006	(0.0044)	0.894	128,927
Clusters (districts)		379		

Notes: Panel A: DID with individual controls, district FE, and state $\times$ time FE. Panel B: cross-sectional with state FE. SEs clustered at district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4

Conley (1999) Spatial HAC Standard Errors: Main DID Results

Outcome	District cluster	Conley spatial HAC		
	SE (p)	100 km	200 km	500 km
Institutional delivery	0.0044 (0.677)	0.0043 (0.674)	0.0048 (0.705)	0.0043 (0.672)
Any vaccination	0.0089 (0.145)	0.0083 (0.118)	0.0072 (0.071)	0.0085 (0.126)
DPT vaccination	0.0074 (0.303)	0.0083 (0.361)	0.0106 (0.474)	0.0103 (0.457)
Any ANC	0.0089 (0.340)	0.0093 (0.361)	0.0094 (0.368)	0.0128 (0.509)
Tetanus injection	0.0108 (0.011)	0.0130 (0.033)	0.0105 (0.009)	0.0127 (0.030)
Public delivery	0.0033 (0.387)	0.0039 (0.461)	0.0042 (0.491)	0.0039 (0.467)
Observations	256,894 (delivery); 193,253 (vaccination)			
Clusters (districts)	379			

Notes: Coefficients on $c_{int}^{std} \times Post$ are identical across columns; only the standard errors differ. Column 1: district-level clustering (baseline). Columns 2–4: Conley (1999) spatial HAC SEs. All specifications include individual controls, district FE, and state $\times$ time FE.

Table A5

Incentive State Heterogeneity: Triple-Difference Estimates

Outcome	$\gamma: c_{int} \times Post$		$\delta: c_{int} \times Post \times Incentive$	
	Coeff	p	Coeff	p
<i>Panel A: Core DID (DLHS-2/3)</i>				
Institutional delivery	0.0001	0.943	0.0005	0.753
Any vaccination	0.0016	0.147	0.0012	0.667
DPT vaccination	-0.0011	0.335	0.0045**	0.027
Any ANC	-0.0023	0.305	0.0017	0.548
Tetanus injection	-0.0056	0.039	0.0019	0.591
Public delivery	0.0007	0.253	-0.0004	0.730
Observations	255,704			

Notes: γ is the coefficient on $c_{int} \times Post$ for non-incentive states; δ is the additional effect in enhanced-incentive states. State $\times$ time FE absorb the direct incentive state effect. Standard errors clustered at district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6

Migration Robustness: Long-Run Coefficients With and Without NFHS V104 “Stayer”
Restriction

Outcome	NFHS-4 (Round 5)		NFHS-5 (Round 6)	
	(A) Baseline	(B) Stayers	(A) Baseline	(B) Stayers
<i>Panel A: Family-planning outcomes (3-round panel, $N \approx 1.30M$ / $0.97M$)</i>				
Limiting method use	0.0147*** (0.0057)	0.0177*** (0.0059)	0.0129* (0.0068)	0.0134* (0.0073)
Female sterilization	0.0137** (0.0057)	0.0168*** (0.0059)	0.0119* (0.0067)	0.0126* (0.0072)
<i>Panel B: Maternal- and child-health outcomes (long-run event-study panel)</i>				
DPT vaccination	0.0212*** (0.0073)	0.0331*** (0.0074)	0.0194** (0.0098)	0.0247** (0.0096)
Any antenatal care	0.0160 (0.0105)	0.0223** (0.0099)	0.0278** (0.0125)	0.0359*** (0.0123)
Any vaccination	0.0167** (0.0067)	0.0234*** (0.0069)	0.0146* (0.0086)	0.0249*** (0.0088)
Institutional delivery	0.0082 (0.0089)	0.0061 (0.0095)	0.0127* (0.0068)	0.0099 (0.0066)
Public facility delivery	0.0130 (0.0122)	0.0136 (0.0115)	0.0210* (0.0107)	0.0144 (0.0096)
Tetanus injection	0.0028 (0.0080)	0.0076 (0.0076)	0.0018 (0.0097)	0.0089 (0.0090)
<i>Panel B sample sizes (vary by outcome due to differential missingness):</i>				
DPT vaccination: 474,054 / 328,058		Any antenatal care: 526,070 / 421,038		
Any vaccination: 441,306 / 323,823		Institutional/public delivery: 580,005 / 434,351		
Tetanus injection: 526,407 / 420,944				

Notes: Each entry is the coefficient on $c_{int,d}^{std} \times \text{Round}_r$ from the corresponding long-run event study, with DLHS-2 as the reference period. Controls include woman’s age, education, religion, caste, and rural residence; fixed effects: district + state $\times$ time. Standard errors (in parentheses) clustered at the district level. The “Stayers” specification restricts NFHS rounds (5 and 6) to women with V104 ≥ 10 years in their current place of residence (or coded as “always/since birth”), while keeping DLHS rounds (1, 2, 3) unrestricted because V104 is not collected in the DLHS surveys. V104 is non-missing for approximately 97% of NFHS women; the stayer criterion retains about 31% of NFHS observations under the children’s-recode framing used to build the MCH panel and about 67% under the women’s-recode framing used to build the family-planning panel. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D. A4. Additional Figures

This appendix presents supplementary figures referenced throughout the paper. Figures A1–A7 show extended multi-round binscatters of sterilization intensity against key outcomes (institutional delivery, DPT vaccination, ANC, and public delivery), complementing the main-text binscatters in Section 5.5. Figures A3 and A8 show convergence plots for individual outcomes, complementing the multi-panel convergence figure (Figure 5) in the main text. Figures A4 and A5 visualize the DLHS-4 trust measures as binscatters. Figure A9 presents the gender-differentiated effects coefficient plot from Section 6.3. Figure A10 summarises the robustness checks from Section 5.7. Finally, Figure A11 maps the raw sterilization level in 1976–77 across districts.



Figure A1

Extended Binscatter: Sterilization Intensity vs. Institutional Delivery (Multi-Round)

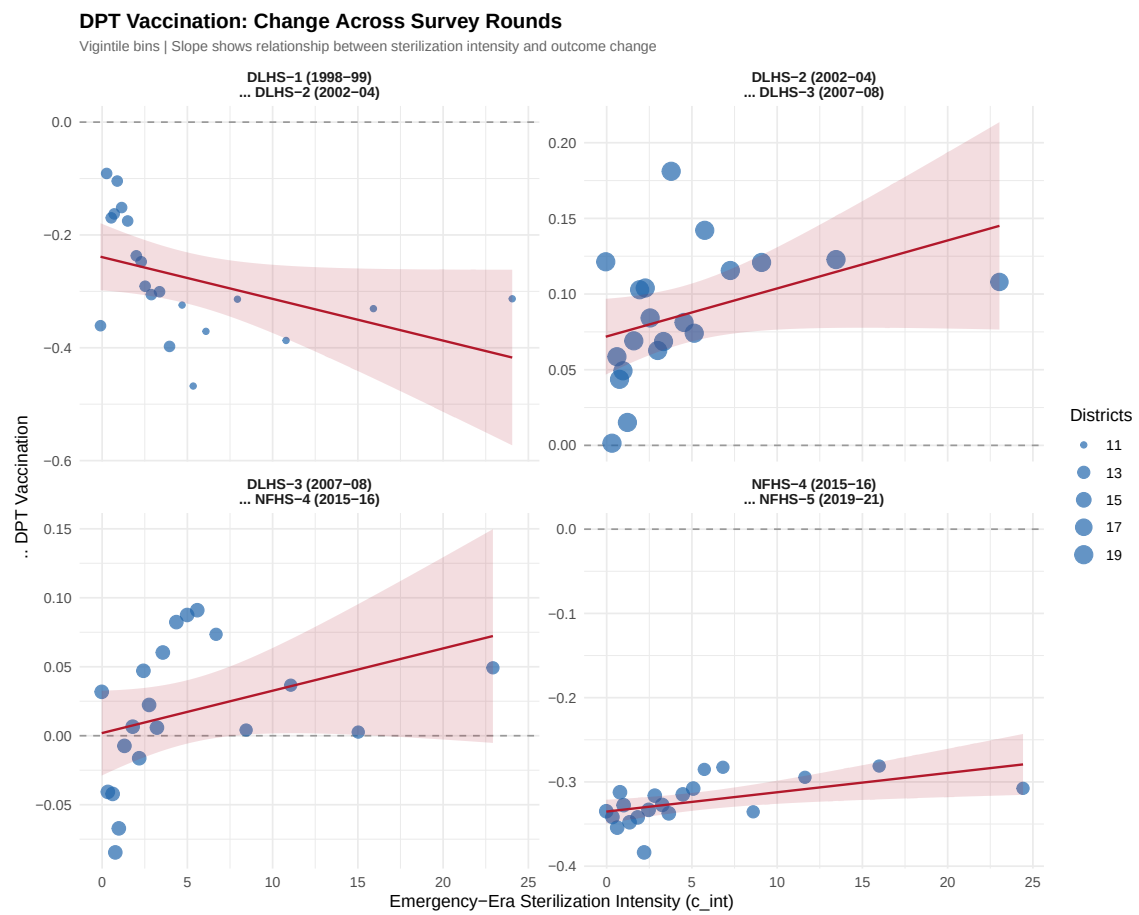


Figure A2

Extended Binscatter: Sterilization Intensity vs. DPT Vaccination (Multi-Round)

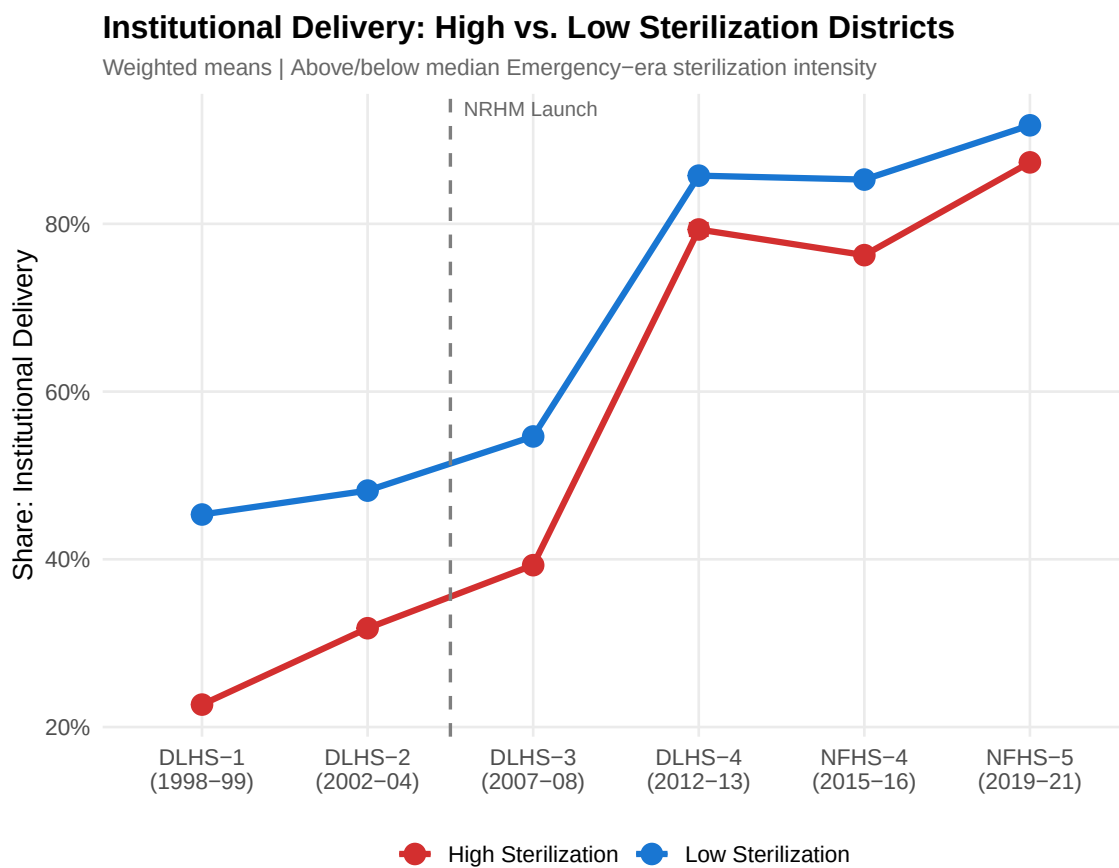


Figure A3
Convergence: Institutional Delivery

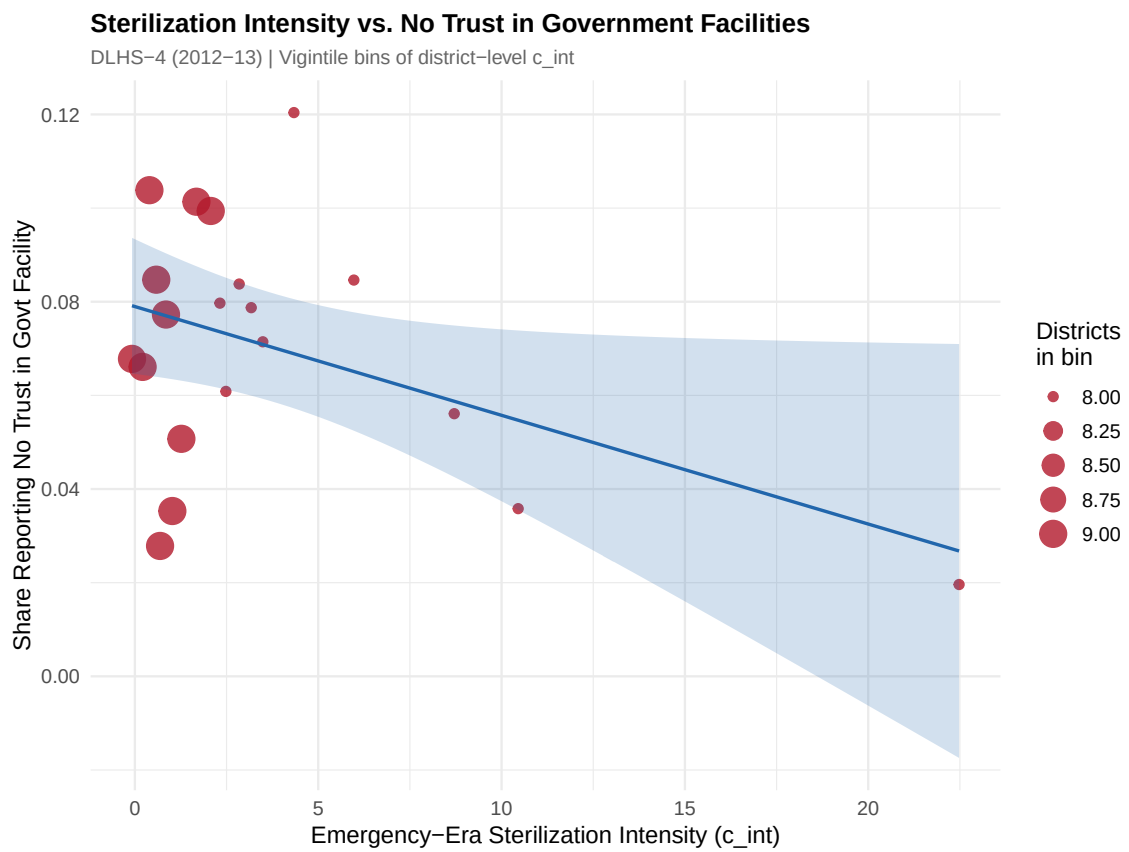


Figure A4

DLHS-4 Binscatter: Sterilization Intensity vs. No Trust in Government Facility

Sterilization Intensity vs. Government Facility Use

DLHS-4 (2012-13) | Vigintile bins of district-level c_int

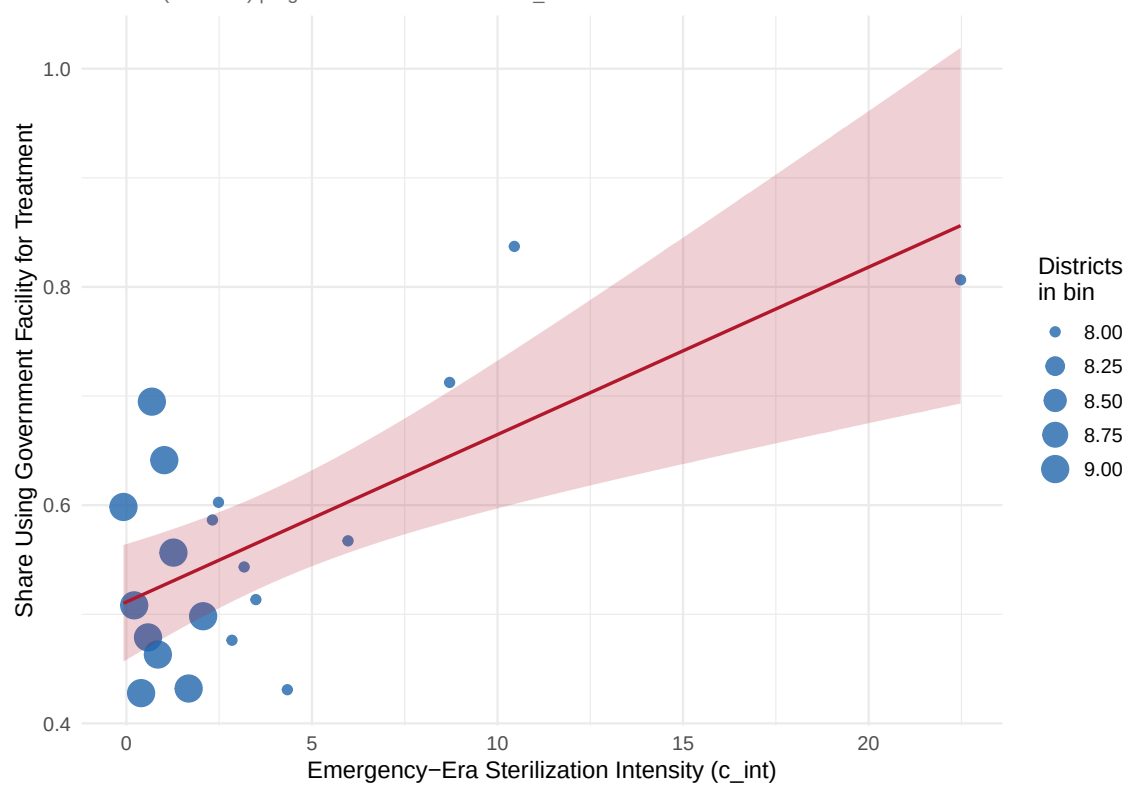


Figure A5

DLHS-4 Binscatter: Sterilization Intensity vs. Use of Government Facility

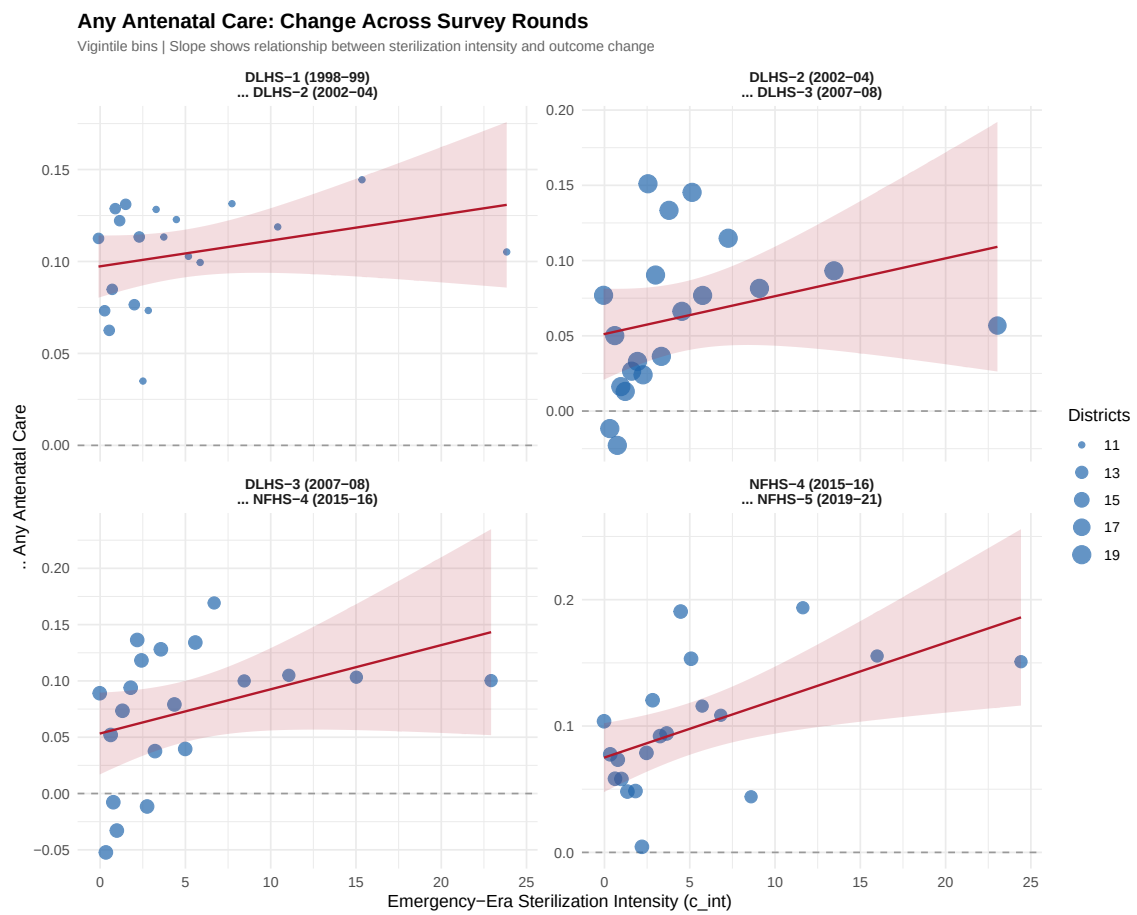


Figure A6

Extended Binscatter: Sterilization Intensity vs. Any ANC (Multi-Round)

Public Facility Delivery: Change Across Survey Rounds

Vigintile bins | Slope shows relationship between sterilization intensity and outcome change

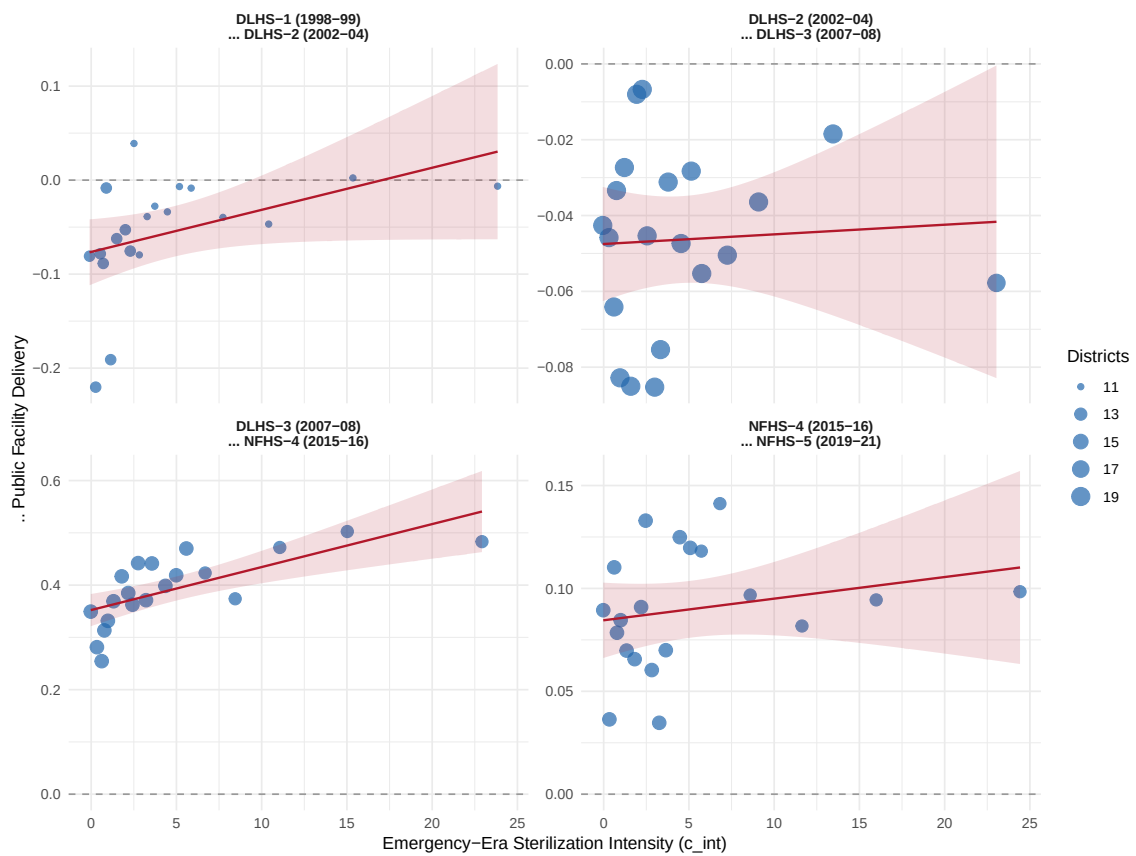


Figure A7

Extended Binscatter: Sterilization Intensity vs. Public Delivery (Multi-Round)

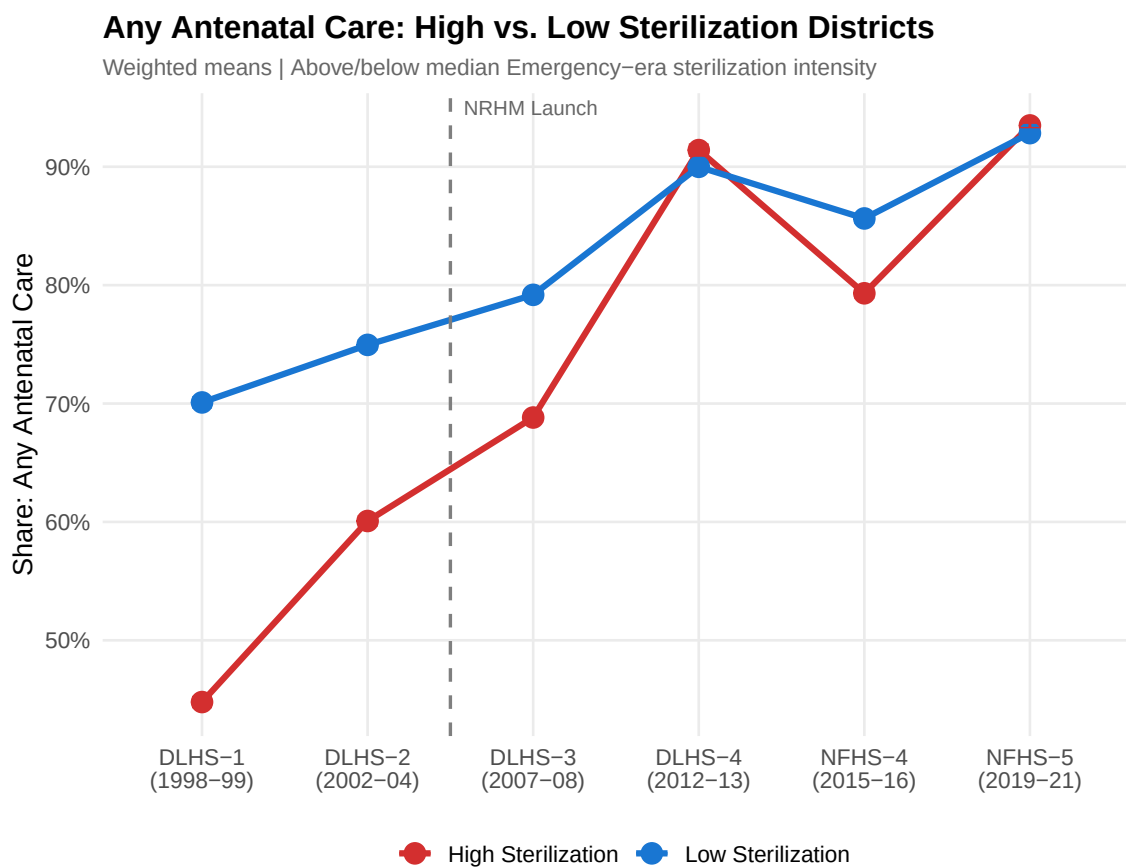


Figure A8
Convergence: Any ANC

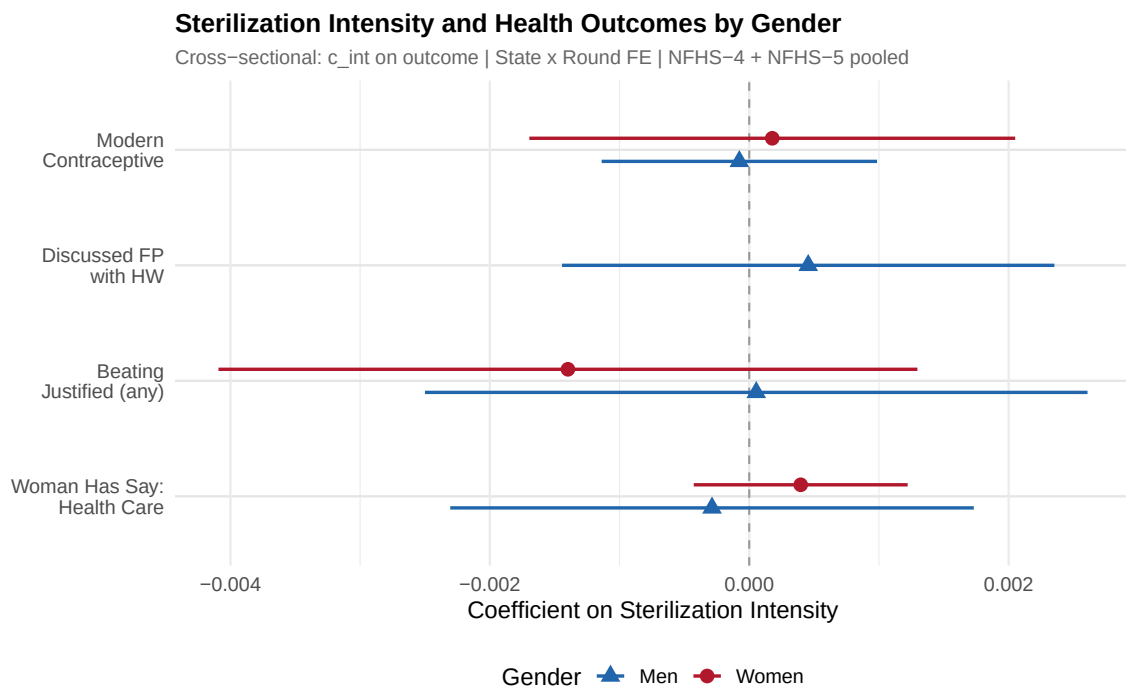


Figure A9
 Gender-Differentiated Effects: Coefficient Plot

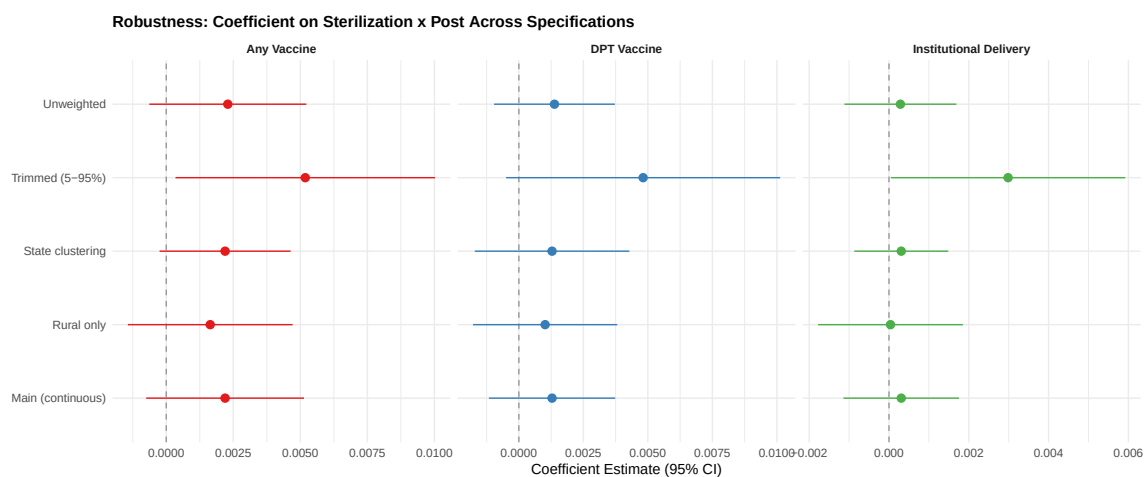
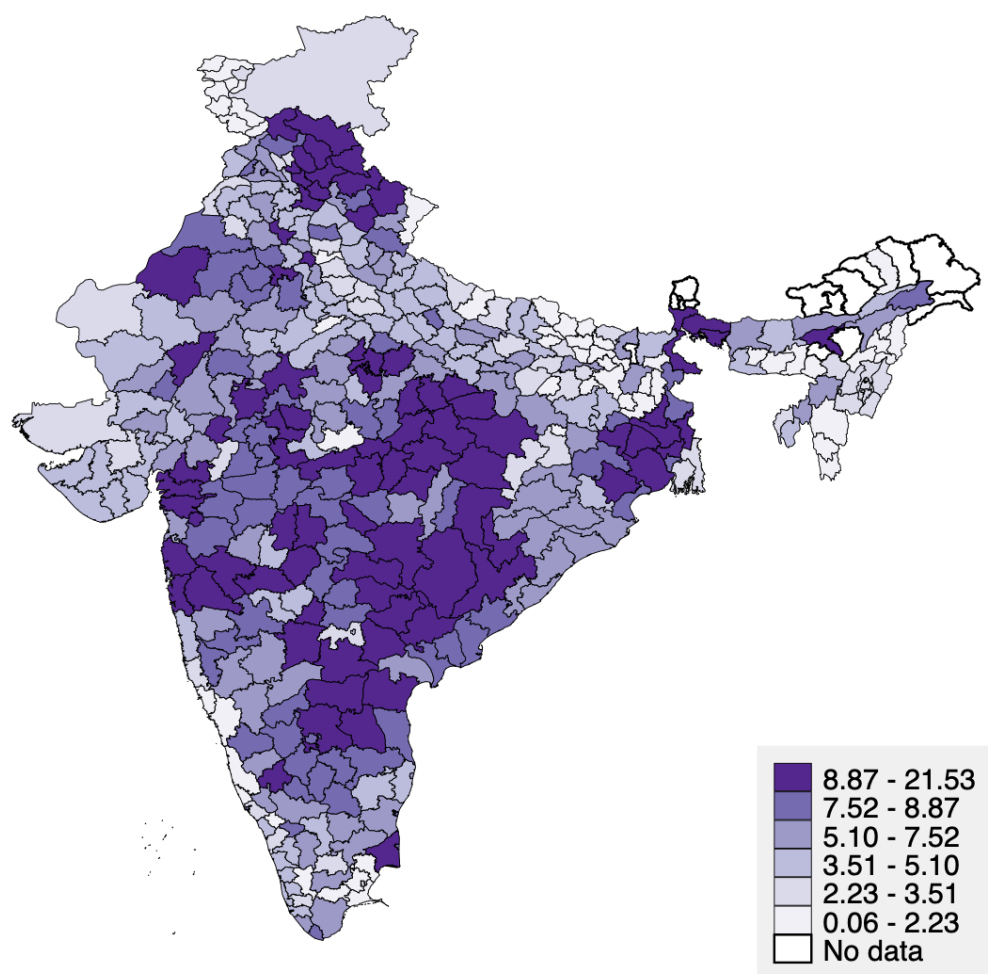


Figure A10

Robustness: Alternative Treatment Measures and Sample Restrictions

Sterilization Intensity 1976-77

Share of couple sterilized



Source: Jolly, KG, Family Planning in India, 1969-1984: A District Level Study

Figure A11

Sterilization Level (1976-77) Across Indian Districts

E. A5. DLHS-4 Trust Analysis: Survey Questions

The trust analysis in Section 5.4 uses outcomes constructed from the DLHS-4 (2012–13) Household Questionnaire (HV-prefix) and Woman Questionnaire (Q-prefix). The exact survey questions are reproduced below, organized by the tier structure used in the analysis.

Tier 1: Direct Health Trust (Household Questionnaire, Section V: Morbidity & Healthcare)

- **No trust in government facility** (HV52j): “What are the reasons for not using government health facility?”, sub-item (j): “No trust on government / municipal hospital / health facility.” (1=Yes, 2=No). *Note: This question is conditional on the household reporting non-use of a government facility.*
- **Use government facility** (HV51): “Where do members of your household generally go for treatment when they are sick?” Codes 1–7 = government facilities (Govt. Hospital, Dispensary, UHP/UHC/UFWC, CHC/Rural Hospital, PHC, Sub-Centre, AYUSH Govt. Hospital); Codes 8–10 = private facilities (Private Hospital/Clinic, Private Doctor, Vaidya/Hakim/Homeopath). Coded 1 if government (1–7), 0 if private (8–10).
- **Poor quality perception** (HV52h): “What are the reasons for not using government health facility?”, sub-item (h): “Poor quality of care.” (1=Yes, 2=No).

Tier 2: ASHA Engagement (Woman Questionnaire, Sections II–III)

- **ASHA motivated ANC** (Q222e): “Who motivated you for ante-natal care?”, sub-item (e): “ASHA / Link worker.” (1=Yes, 2=No).
- **ASHA motivated delivery** (Q227e): “Who motivated you for institutional delivery?”, sub-item (e): “ASHA / Link worker.” (1=Yes, 2=No).
- **ASHA arranged transport** (Q236e): “Who arranged the transport for delivery?”, sub-item (e): “ASHA / Link worker.” (1=Yes, 2=No).
- **ASHA motivated immunization** (Q319e): “Who motivated you for immunization of the child?”, sub-item (e): “ASHA / Link worker.” (1=Yes, 2=No).
- **JSY received** (Q240a): “Did you receive financial assistance for the delivery under any of the following scheme?”, sub-item (a): “Janani Suraksha Yojana (JSY).” Coded 1 if JSY received, 0 otherwise.

Tier 3: Government Scheme Participation (Mixed: Woman Q + Household HV)

- **Pregnancy registered** (Q205): “Was the pregnancy registered?” (1=Yes, 2=No).
- **Health scheme coverage** (HV49): “Is any member of the household covered by a health scheme or health insurance?” (1=Yes, 2=No).
- **RSBY coverage** (HV50b): “What type of health scheme?”, sub-item (b): “Rashtriya Swasthya Bima Yojana (RSBY).” (1=Yes, 2=No).

Tier 4: Placebo, Public vs. Private (Woman Questionnaire, Section II)

- **Public/Private facility delivery** (Q231): “Where did the delivery take place?” Codes 1–7 = public facilities; Codes 8–10 = private facilities; Code 11 = Home.
- **ANC at government/private facility** (Q210): “Where did you receive ante-natal care?” Codes 1–9 = government sources; Codes 10–12 = private sources.

Source: DLHS-4 Household Questionnaire and Woman Questionnaire instruments, International Institute for Population Sciences (IIPS), Mumbai, 2012–13.

F. A6. Census Non-Health Placebo

Table A7

Non-Health Placebo: Sterilization Intensity and Census Outcomes

Outcome	c_{int}^{std}	SE	p -value	N
<i>1961–1971</i>				
Agricultural labor share	0.0082**	(0.0040)	0.042	335
Literacy rate	0.0014	(0.0011)	0.216	336
Matriculation rate	−0.0014*	(0.0007)	0.050	335
Primary education rate	0.0001	(0.0034)	0.971	335
Urbanization rate	0.0002	(0.0013)	0.870	336
Work participation rate	0.0024	(0.0017)	0.150	335
<i>1971–1981</i>				
Agricultural labor share	0.0010	(0.0020)	0.636	328
Female literacy rate	−0.0008	(0.0013)	0.536	328
Literacy rate	0.0003	(0.0012)	0.799	328
Matriculation rate	−0.0015**	(0.0006)	0.019	328
Primary education rate	−0.0056***	(0.0021)	0.009	328
Urbanization rate	0.0030	(0.0020)	0.130	328
Work participation rate	−0.0018	(0.0020)	0.370	328
<i>1971–1991</i>				
Agricultural labor share	0.0009	(0.0025)	0.704	327
Female literacy rate	−0.0017	(0.0029)	0.566	327
Literacy rate	0.0002	(0.0023)	0.927	327
Matriculation rate	−0.0028**	(0.0014)	0.046	302
Primary education rate	−0.0045*	(0.0026)	0.080	302
Urbanization rate	0.0020	(0.0027)	0.463	327
Work participation rate	−0.0024	(0.0022)	0.277	327

Notes: Cross-sectional regressions at the district level. Dependent variable is the change in the outcome rate between the indicated census years. All specifications include state fixed effects and heteroskedasticity-robust standard errors. The 1961–1971 panel serves as a pre-Emergency placebo; 1971–1981 and 1971–1991 test whether Emergency-era sterilization intensity predicts differential changes in non-health outcomes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G. A7. Facility Capacity Interaction

Table A8

Facility Capacity Interaction: Triple-Difference Estimates

Outcome	$c_{int}^{std} \times Post$		$c_{int}^{std} \times Post \times HighCap$		N	Districts
	Coeff	p	Coeff	p		
Institutional delivery	0.0070	0.223	-0.0111	0.148	255,313	375
Any vaccination	0.0219	0.133	-0.0169	0.229	191,977	375
DPT vaccination	0.0074	0.401	-0.0001	0.993	190,128	375
Any ANC	0.0009	0.929	-0.0195	0.187	255,248	375
Tetanus injection	-0.0152	0.242	-0.0256	0.119	255,262	375
Public delivery	-0.0008	0.866	0.0081	0.215	255,313	375

Notes: Triple-difference estimates from the specification $Y_{idt} = \beta_1(c_{int}^{std} \times Post) + \beta_2(c_{int}^{std} \times Post \times HighCapacity) + Controls + \delta_d + \mu_{st} + \epsilon_{idt}$. *HighCapacity* is an indicator for above-median PHC count based on the DLHS-3 facility census (2007–08; median = 15 PHCs per district). A positive β_2 would indicate convergence concentrated in high-capacity districts. The uniformly insignificant β_2 coefficients indicate that utilization convergence occurs similarly in high- and low-capacity districts, consistent with a demand-side trust channel rather than supply-side facility expansion. Standard errors clustered at district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H. A8. HonestDiD Sensitivity Analysis

Figures A12, A13, A14, A15, A16, and A17 report the Rambachan and Roth (2023) sensitivity analysis for the event study coefficients, using the relative magnitudes approach implemented via the HonestDiD package. The sensitivity parameter $\bar{M}$ bounds the ratio of the post-treatment parallel trends violation to the maximal pre-treatment violation. The plots show robust confidence intervals (C-LF method) and, where available, fixed-length confidence intervals (FLCI method) across values of $\bar{M}$ from 0 to 2. With only one pre-period (DLHS-1), the intervals widen rapidly for $\bar{M} > 0.5$. At $\bar{M} = 0$ (exact parallel trends), the robust confidence intervals include zero for all outcomes, but the DPT and any vaccination intervals are tight and concentrated in the positive range, consistent with the convergence finding.

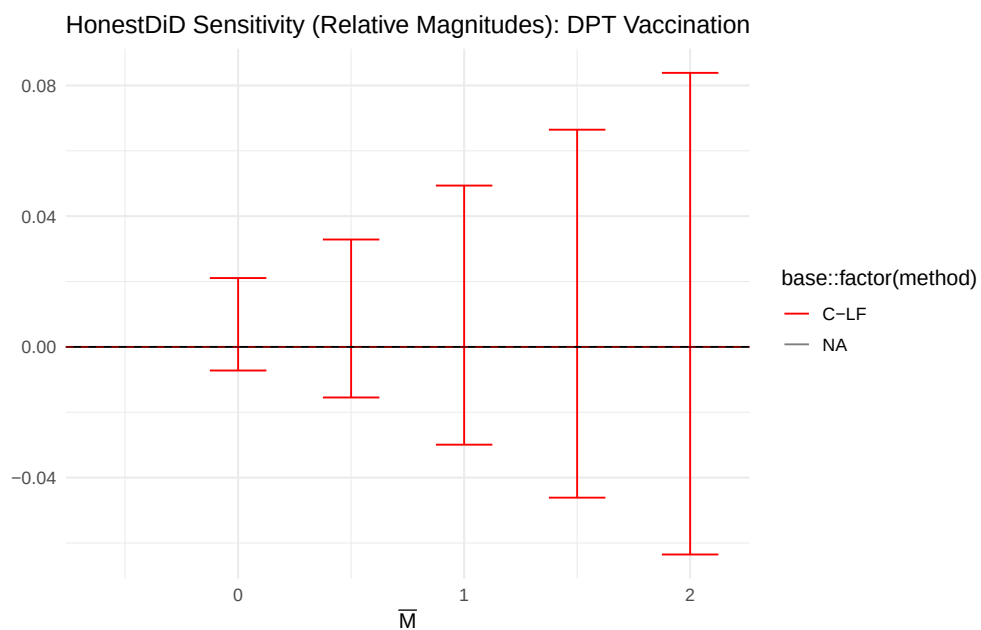


Figure A12

HonestDiD Sensitivity: DPT Vaccination

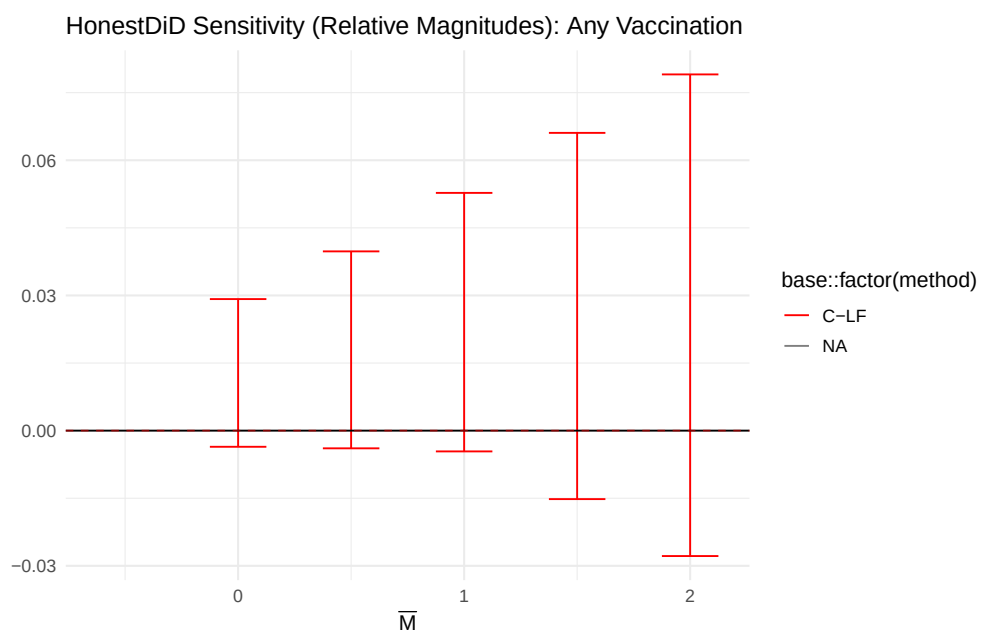


Figure A13

HonestDiD Sensitivity: Any Vaccination



Figure A14
HonestDiD Sensitivity: Institutional Delivery

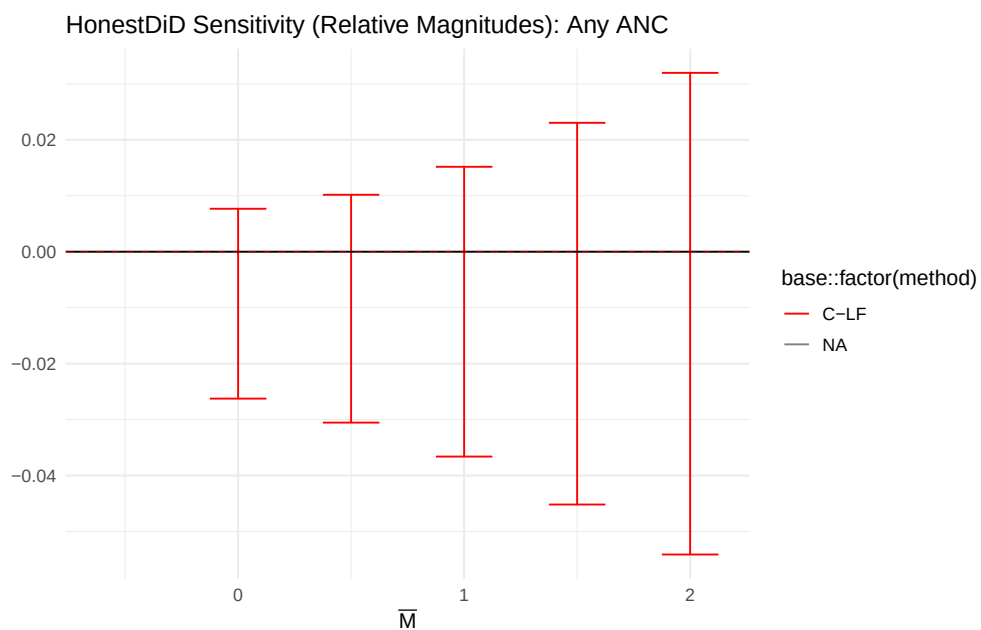


Figure A15

HonestDiD Sensitivity: Any ANC

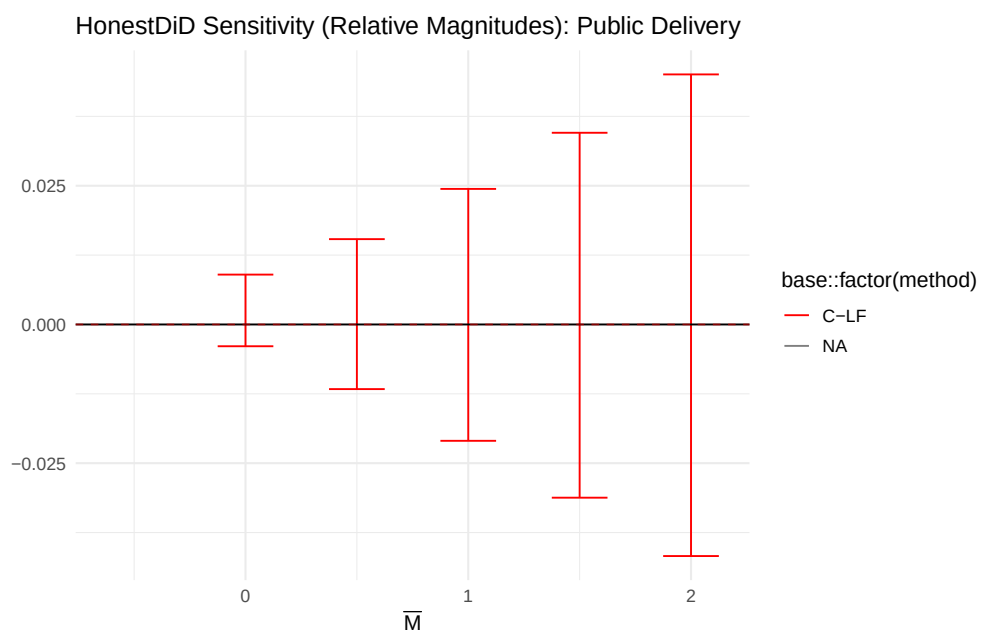


Figure A16

HonestDiD Sensitivity: Public Delivery



Figure A17

HonestDiD Sensitivity: Tetanus Injection

I. A9. Alternative Coercion Measures and the Coercion-Character Modifier

Table A9 reports how the long-run convergence depends on the construction of the coercion-intensity treatment. Panel A re-estimates the event study ((5)) replacing the baseline proportional surge c_{int}^{std} with each alternative measure and reports the NFHS-5 (round 6) coefficient, the round in which the headline convergence is strongest. The result is robust to monotone transformations of the proportional surge (log, quintile rank) but not to constructions that capture administrative over-achievement rather than the coercive shock: the over-target ratio, the preferred measure of Pelras and Renk (2023), yields null convergence at both the within-state district level and at their own state level with region fixed effects, and the per-couple excess is significantly negative. These alternative measures are near-orthogonal to the proportional surge (the over-target/surge correlation is approximately zero); on the *same* 110- and 66-district subsamples for which they are defined, the surge measure retains its positive, significant convergence (institutional delivery +0.030 and +0.036; public delivery +0.047 and +0.048; all $p < 0.05$), so the divergence reflects the measure rather than the sample. Panel B reports the coercion-character modifier: the triple interaction $c_{int}^{std} \times \text{Post} \times \text{male-share}$ is, if anything, negative, indicating that recovery is slower, not faster, where the Emergency campaign was most male and vasectomy-driven, consistent with the weaker recovery found in low-performing states (Section 5.9).

Table A9

Alternative Coercion Measures and the Coercion-Character Modifier (Long-Run Event Study, NFHS-5 Coefficient)

Coercion measure	Inst. delivery	Public delivery	Any ANC	DPT
<i>Panel A: NFHS-5 (round 6) event-study coefficient under alternative treatments</i>				
Proportional surge c_{int}^{std} (baseline)	0.017*** (0.006)	0.025*** (0.009)	0.030** (0.013)	0.022** (0.009)
$\log(1 + c_{int})$	0.021** (0.008)	0.028** (0.011)	0.023 (0.015)	0.021** (0.010)
Quintile rank (z)	0.022** (0.011)	0.027** (0.013)	0.018 (0.015)	0.017 (0.011)
Binary above-median	0.028 (0.022)	0.016 (0.026)	0.013 (0.025)	0.034* (0.020)
Absolute excess	0.006 (0.010)	0.008 (0.013)	−0.021 (0.013)	−0.007 (0.010)
Raw 1976–77 level	−0.004 (0.009)	−0.008 (0.015)	−0.031** (0.013)	−0.017 (0.010)
Over-target ratio, district (P&R measure)	−0.016 (0.012)	−0.017 (0.019)	−0.018** (0.009)	−0.001 (0.010)
Over-target ratio, state + region FE (P&R design)	−0.002 (0.020)	−0.028 (0.036)	−0.011 (0.025)	−0.006 (0.015)
Per-couple excess	−0.071*** (0.021)	−0.082** (0.031)	−0.020 (0.024)	0.032 (0.044)
<i>Panel B: Coercion-character modifier (rounds 2 and 6)</i>				
$c_{int}^{std} \times \text{Post} \times \text{male-share}$	−0.022* (0.013)	−0.060*** (0.020)	−0.010 (0.030)	0.005 (0.019)

Notes: Each cell is the NFHS-5 (round 6) coefficient on the indicated coercion measure interacted with the round dummy (Panel A) or the triple interaction (Panel B), from the long-run event study with district and state×round fixed effects, individual controls, sampling weights, and standard errors (in parentheses) clustered at the district level; DLHS-2 is the reference round. Panel-A rows for the surge, log, quintile, binary, absolute-excess, and raw-level measures use the full long-run sample (≈ 403 districts); the over-target (district) and per-couple rows use the 110- and 66-district subsamples for which the archival targets and by-method counts are available; the state-level over-target uses 23 states with region×round fixed effects, clustered by state. On the 110- and 66-district subsamples the surge measure retains positive, significant NFHS-5 convergence. The baseline proportional surge follows [Singh and Vincent \(2024\)](#); the over-target construction is the preferred measure of [Pelras and Renk \(2023\)](#). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J. A10. A Minimal Trust-Updating Model

This appendix formalizes the conceptual framework sketched in the introduction, trust as an inherited prior updated through repeated positive signals, in the simplest model able to confront the estimated recovery paths, and asks whether the model’s predictions match their shape and timing. The exercise is illustrative, not structural.

Setup. Households in district d hold a belief $T_d \in [0, 1]$ that an encounter with the public health system will be benign. The Emergency shock lowered this belief in proportion to local coercion intensity, opening an initial trust gap $\Delta_0 \propto c_{int,d}$ relative to unexposed districts, transmitted across generations as an inherited prior. Utilization of a public health service is increasing in T , so to first order the cross-district utilization gradient at time τ is proportional to the remaining trust gap Δ_τ .

Updating. Each benign interaction with a trusted intermediary closes a fraction κ of the remaining gap, so after n signals $\Delta = \Delta_0(1 - \kappa)^n$. Signals arrive through ASHA contact at an annual rate ρ_τ that varies over time as the programme scales up. The expected gap after τ years is then $\Delta_\tau = \Delta_0 \exp(-\kappa R(\tau))$, where $R(\tau) = \int_0^\tau \rho_s ds$ is the cumulative number of signals, and the event-study coefficient traces $\gamma(\tau) = \gamma_\infty [1 - e^{-\kappa R(\tau)}]$.

Predictions. The model makes three refutable predictions. First, no discontinuity at the reform date: recovery accrues signal by signal, not at the policy announcement, which matches the null short-run estimates of Section 5.2. Second, recovery is concave in *cumulative contacts* but need not be concave in *calendar time*: if the contact rate ρ_τ rises as ASHA deployment matures, the calendar-time path can accelerate even though updating is saturating in signals. This is exactly the pattern estimated in Section 5.3.3, where the delivery and antenatal coefficients are still rising fifteen years after the reform, and it reconciles that acceleration with a saturating trust mechanism: the exponential-in-time fit fails there not because updating is absent but because the signal arrival rate itself grew. Third, outcomes for which the ASHA herself delivers the benign encounter should converge earliest; immunization, the service with the highest measured ASHA involvement in DLHS-3, is indeed the first outcome to converge.

Calibration. Measured ASHA involvement per recent birth rises from 2–5% in DLHS-3 (2007–08: antenatal 2.4%, delivery 2.1%, immunization 5.4%) to 19% in NFHS-5 (2019–21, antenatal item). Approximating ρ_τ as rising linearly between these endpoints, the cumulative pregnancy-episode signals per mother by 2019–21 are on the order of $R \approx 1$ –1.5. Rationalizing the observed closure of the vaccination gap by NFHS-4, and of the delivery gaps by NFHS-5, then requires a per-signal update κ close to one: a single benign, personally delivered encounter eliminates most of a household’s remaining distrust. Taken literally this is stark, but the pregnancy-episode items undercount total ASHA exposure (home visits, immunization days, village health events), so measured R understates true signal counts and the implied κ is an upper bound. The calibration’s message is accordingly qualitative rather than quantitative: the estimated recovery paths, their lack of a jump at the reform, their acceleration in calendar time alongside rising contact rates, and their ordering across outcomes, are jointly consistent with signal-based updating of an inherited prior, the mechanism the paper’s framework posits.